ENG

gaSteam - UG

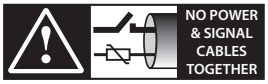
Gas humidifier



USER MANUAL

LEGGI E CONSERVA QUESTE ISTRUZIONI

READ AND SAVE THESE INSTRUCTIONS



READ CAREFULLY IN THE TEXT!

gaSteam - UG

+0300090EN - ENG

Up to date version available on

www.carel.com

IMPORTANT WARNINGS

IF THE INFORMATION IN THIS MANUAL IS NOT FOLLOWED EXACTLY, A FIRE OR EXPLOSION MAY RESULT CAUSING PROPERTY DAMAGE, PERSONAL INJURY OR LOSS OF LIFE

- Do not store or use gasoline or other flammable vapors and liquids in the vicinity of this or any other appliance

WHAT TO DO IF YOU SMELL GAS:

1. **Do not try to light any appliance**
2. **Do not touch any electrical switch; do not use any phone in your building**
3. **Immediately call your gas supplier from a neighbor's phone. Follow the gas supplier's instructions**
4. **If you cannot reach your gas supplier, call the fire department**

- Installation and service must be performed by a qualified installer, service agency or the gas supplier.

CAREL Industries humidifiers are advanced products, whose operation is specified in the technical documentation supplied with the product or can be downloaded, even prior to purchase, from the website www.carel.com. Each CAREL product, in relation to its advanced level of technology, requires setup/configuration/programming to be able to operate in the best possible way for the specific application. Failure to complete such operations, which are required/indicated in the user manual, may cause the final product to malfunction; CAREL accepts no liability in such cases. The customer (manufacturer, developer or installer of the final equipment) accepts all liability and risk relating to the configuration of the product in order to reach the expected results in relation to the specific final installation and/or equipment. CAREL may, based on specific agreements, act as a consultant for the installation/commissioning/use of the unit, however in no case does it accept liability for the correct operation of the humidifier and the final installation if the warnings or suggestions provided in this manual or in other product technical documents are not heeded. In particular, as well as observing the above warnings and suggestions, the following warnings must be observed for correct use of the product:

DANGER OF ELECTRIC SHOCK: the humidifier contains live electrical components. Disconnect the mains power supply before accessing inside parts or during maintenance and installation.

DANGER OF WATER LEAKS: the humidifier automatically and constantly fills/drains certain quantities of water. Malfunctions in the connections or in the humidifier may cause leaks.

DANGER OF BURNS: the humidifier contains high temperature components and delivers steam at 100°C/212°F.

In the event where there is no demand for steam production for a period exceeding 24 hours, the unit will automatically empty the boiler, so as to avoid stagnation of the water inside. Environmental conditions, fuel and power supply voltage must all comply with the specified values. All other uses and modifications made to the appliance that are not authorised by the manufacturer are considered incorrect. Liability for injury or damage caused by the incorrect use of the appliance lies exclusively with the user.

Please note that the appliance is connected to the gas mains, contains live electrical devices and hot surfaces. All service and/or maintenance operations must be performed by specialist and qualified personnel who are aware of the necessary precautions and are capable of performing the operations correctly and in accordance with the safety standards and legislation in force, with specific reference to:

1. Italian law no. 1083/71: "Safety standards relating to the use of gaseous fuel";

2. Italian law no. 46/90: "Safety standards relating to systems in buildings";

3. Italian Presidential Decree no. 447 of 6 December 1991: "Regulations for the enforcement of law no. 46, dated March 5, 1990, on safety relating to systems in buildings";

2. Italian law no. 10/91: "Regulations for the enforcement of the national plan for energy savings and the development of renewable sources of energy"

Adjustment of all units (covered by UL-998) from operation on natural gas (factory setting) to LPG must only be carried out by Carel personnel or Carel service.

IMPORTANT:

The installation of the product must include an earth connection, using the special yellow-green terminal available in the humidifier.

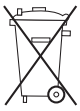
Important:

- Disconnect the appliance from the mains power supply before accessing any internal parts.
- Environmental and power supply conditions must conform to the values specified on the product rating labels.
- The product is designed exclusively to humidify rooms either directly or through distribution systems (ducts).
- Only qualified personnel who are aware of the necessary precautions and able to perform the required operations correctly may install, operate or carry out technical service on the product.
- Only water with the characteristics indicated in this manual must be used for steam production.
- Only water with the characteristics indicated in this manual must be used to produce the spray.
- All operations on the product must be carried out according to the instructions provided in this manual and on the labels applied to the product. Any uses or modifications that are not authorised by the manufacturer are considered improper. CAREL declines all liability for any such unauthorised use.
- Do not attempt to open the appliance in any way other than described in the manual.
- Observe the standards in force in the place where the humidifier is installed.
- The appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.

- Do not install and use the product near objects that may be damaged when in contact with water (or condensate). CAREL declines all liability for direct or indirect damage following water leaks from the humidifier.
- Do not use corrosive chemicals, solvents or aggressive detergents to clean the inside and outside parts of the humidifier, unless specifically indicated in the user manual.
- Do not drop, hit or shake the humidifier, as the inside parts and the linings may be irreparably damaged,

CAREL adopts a policy of continual development. Consequently, CAREL reserves the right to make changes and improvements to any product described in this document without prior warning. The technical specifications shown in the manual may be changed without prior warning. The liability of CAREL in relation to its products is specified in the CAREL general contract conditions, published on the website www.carel.com and/or by specific agreements with customers; specifically, to the extent where allowed by applicable legislation, in no case will CAREL, its employees or subsidiaries/affiliates be liable for any lost earnings or sales, losses of data and information, costs of replacement goods or services, damage to things or people, downtime or any direct, indirect, incidental, actual, punitive, exemplary, special or consequential damage of any kind whatsoever, whether contractual, extra-contractual or due to negligence, or any other liabilities deriving from the installation or use of the product, even if CAREL or its subsidiaries/affiliates are warned of the possibility of such damage.

DISPOSAL



The humidifier is made up of metal parts and plastic parts. In reference to European Union directive 2002/96/EC issued on 27 January 2003 and related national legislation, please note that:

1. WEEE cannot be disposed of as municipal waste and such waste must be collected and disposed of separately;
2. the public or private waste collection systems defined by local legislation must be used. In addition, the equipment can be returned to the distributor at the end of its working life when buying new equipment;
3. the equipment may contain hazardous substances: the improper use or incorrect disposal of such may have negative effects on human health and on the environment;
4. the symbol (crossed-out wheeled bin) shown on the product or on the packaging and on the instruction sheet indicates that the equipment has been introduced onto the market after 13 August 2005 and that it must be disposed of separately;
5. in the event of illegal disposal of electrical and electronic waste, the penalties are specified by local waste disposal legislation.

Warranty on materials: 2 years (from production date, excluding consumables).

Approval: the quality and safety of CAREL products are guaranteed by the ISO 9001 certified design and production system, as well as by the CE and ETL marks.

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1. Introduction and installation

1.1 gaSteam (UG*)

Range of isothermal gas-fired humidifiers for steam production control and distribution, fitted with touch screen graphic display.

Models available (identifiable by the part number shown on the packaging label and on the rating plate): UG045, UG090, UG150, UG180, UG300 and UG450, with steam production capacity up to 450 kg/h (992 lb/h).

gaSteam is available both for indoor and outdoor installation.

Certification:

- 230 Vac models: CE;
- 115 Vac models: CE and ETL.

1.2 Dimensions and weights

Note: models for outdoor installation are equipped with a rain cover (ref. P), consequently the width and depth dimensions vary slightly.

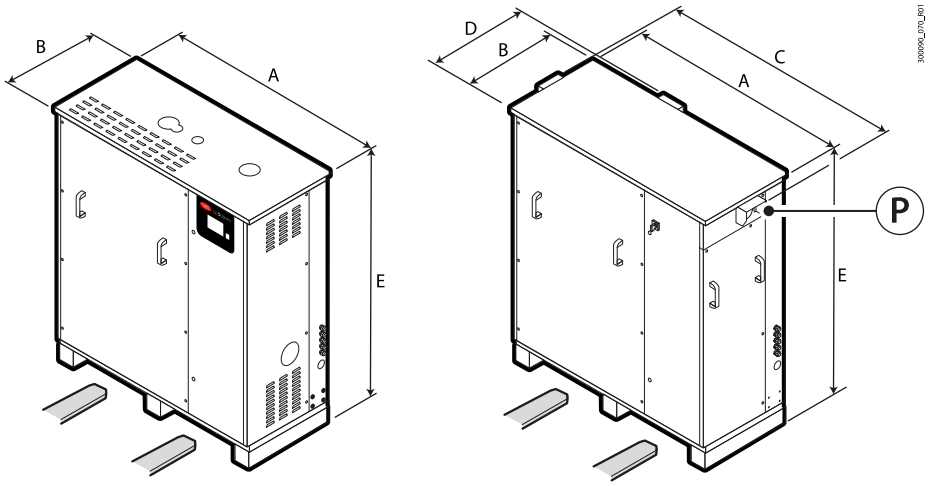


Fig.1.a

						Weight- kg (lb)		
	A - mm (in)	**C - mm (in)**	**B - mm (in)**	**D - mm (in)**	**E - mm (in)**	**Packaged**	**Empty**	**Operating**
UG045	1443 (57)	-	656 (26)	-	1603 (63)	255 (562)	240 (530)	440 (970)
UG045 outdoor	-	1560 (61)	-	800 (31)		270 (595)	255 (562)	455 (1003)
UG090	1443 (57)	-	656 (26)			255 (562)	240 (530)	440 (970)
UG090 outdoor	-	1560 (61)	-	800 (31)		270 (595)	255 (562)	455 (1003)
UG150	1443 (57)	-	656 (26)	-		255 (562)	240 (530)	440 (970)
UG150 outdoor	-	1560 (61)	-	800 (31)		270 (595)	255 (562)	455 (1003)
UG180	1443 (57)	-	993 (39)	-		355 (783)	340 (750)	755 (1664)
UG180 outdoor	-	1560 (61)	-	1107 (44)		370 (816)	355 (783)	770 (1698)
UG300	1443 (57)	-	993 (39)	-		355 (783)	340 (750)	755 (1664)
UG300 outdoor	-	1560 (61)	-	1107 (44)		370 (816)	355 (783)	770 (1698)
UG450 outdoor	-	1620 (64)	-	1668 (66)		550 (1213)	530 (1168)	1150 (2535)

Tab.1.a

1.3 When opening the packaging

- Check that the humidifier is intact upon delivery and notify the carrier immediately, in writing, of any damage that may be due to improper or negligent transport;
- Move the humidifier to the installation site before removing it from the packaging, lifting the box using suitable lifting/handling equipment.

Important: make sure the humidifier does not swing when lifting it.

1.4 Positioning

- For installation choose the most suitable position for steam distribution, that is, the position that minimises the length of the steam pipe. The unit has been designed to be floor standing, on a base that supports its weight in normal operating conditions (see the table of "Dimensions and weights").
- The metal casing of the humidifier heats up during operation, and the top part may reach temperatures > 60°C (140°F); therefore make sure that this does not cause any problems.
- Make sure the humidifier is level, and that the minimum clearances are respected, as shown in the figure below, to allow room for maintenance operations.

⚠ **Important:** make sure that the combustion air intake grille is not blocked or covered.

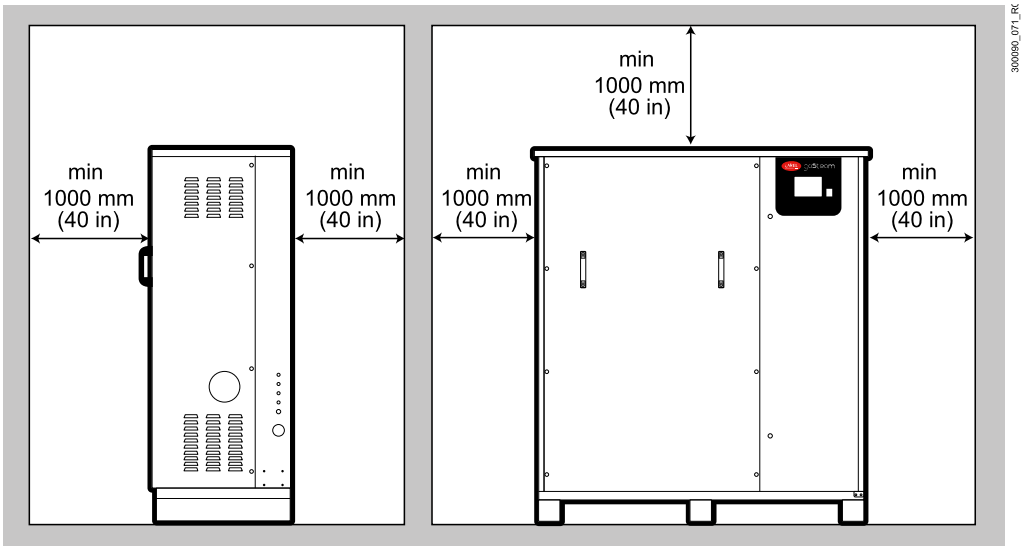


Fig.1.b

1.5 Opening

1.5.1 Front panel

Unscrew the screws using a Phillips-head screwdriver (A), grasp the handles and slide the panel slightly downwards to remove it.

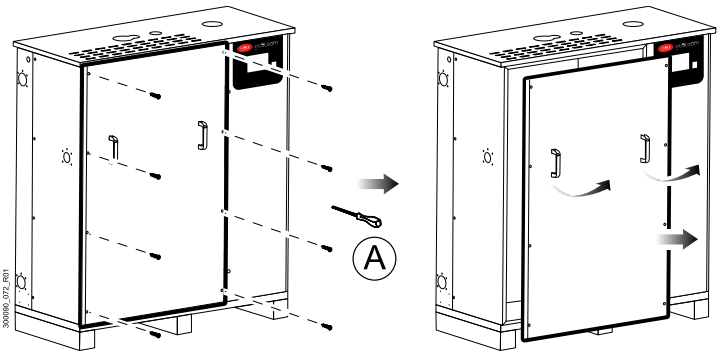


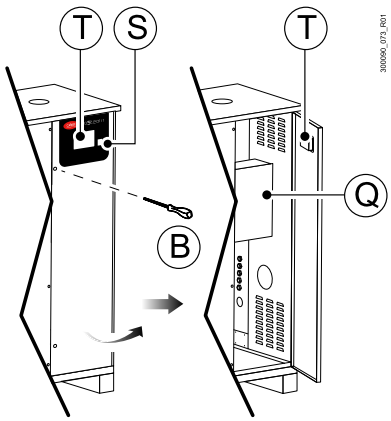
Fig.1.c

1.5.2 Accessing the electrical panel

Note: the user terminal (T) is located:

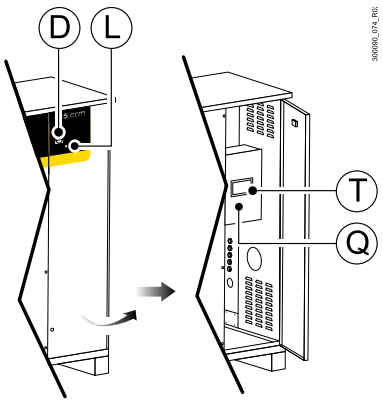
- on the indoor model: mounted on the door and accessible from the outside;
- on the outdoor model: inside the unit, in the electrical panel (Q).

Indoor model



Unscrew the screw using a flat-head screwdriver (B) and open the door to access the electrical panel (Q).

Outdoor model



Procedure:

1. Turn the disconnect switch (D) OFF to electrically disconnect the unit;
2. Turn the key in the lock (L) and open the door to access the electrical panel (Q).

Fig.1.d

1.6 Material supplied

Once the packaging has been opened and the front panel removed, make sure that all of the components listed in the table are present.

	UG45-90-150		UG180-300		UG450
	indoor	outdoor	indoor	outdoor	outdoor
Water circuit					
Water hose with compression fitting	1	1	1	1	2
Screw-band hose clamp	4 (UG45-90); 2 (UG150)	4 (UG45-90); 2 (UG150)	4 (UG180); 2 (UG300)	4 (UG180); 2 (UG300)	3
Threaded hose coupling	1	1	1	1	2
Spring hose clamp	2	2	2	2	4
Gas shut-off valve	1	1	1	1	1
Water shut-off valve with gasket	1	1	1	1	2
Probe flanges with gasket	1	1	2	2	3
Cable gland PG13.5	1	1	1	1	1

	UG45-90-150		UG180-300		UG450
Cable gland PG9	4	4	4	4	4
Weatherproof cover	-	6	-	6	12
Flue gas exhaust:					
straight pipe	1	2	2	4	6
probe section	1	1	2	2	3
reducer	1	1	2	2	3
flange with screws	1	1	2	2	3
elbow	-	1	-	2	3

Tab.1.b

1.7 Structure

The figure shows the structure of model UG45; for the other models, see par. "Dimensions by model".

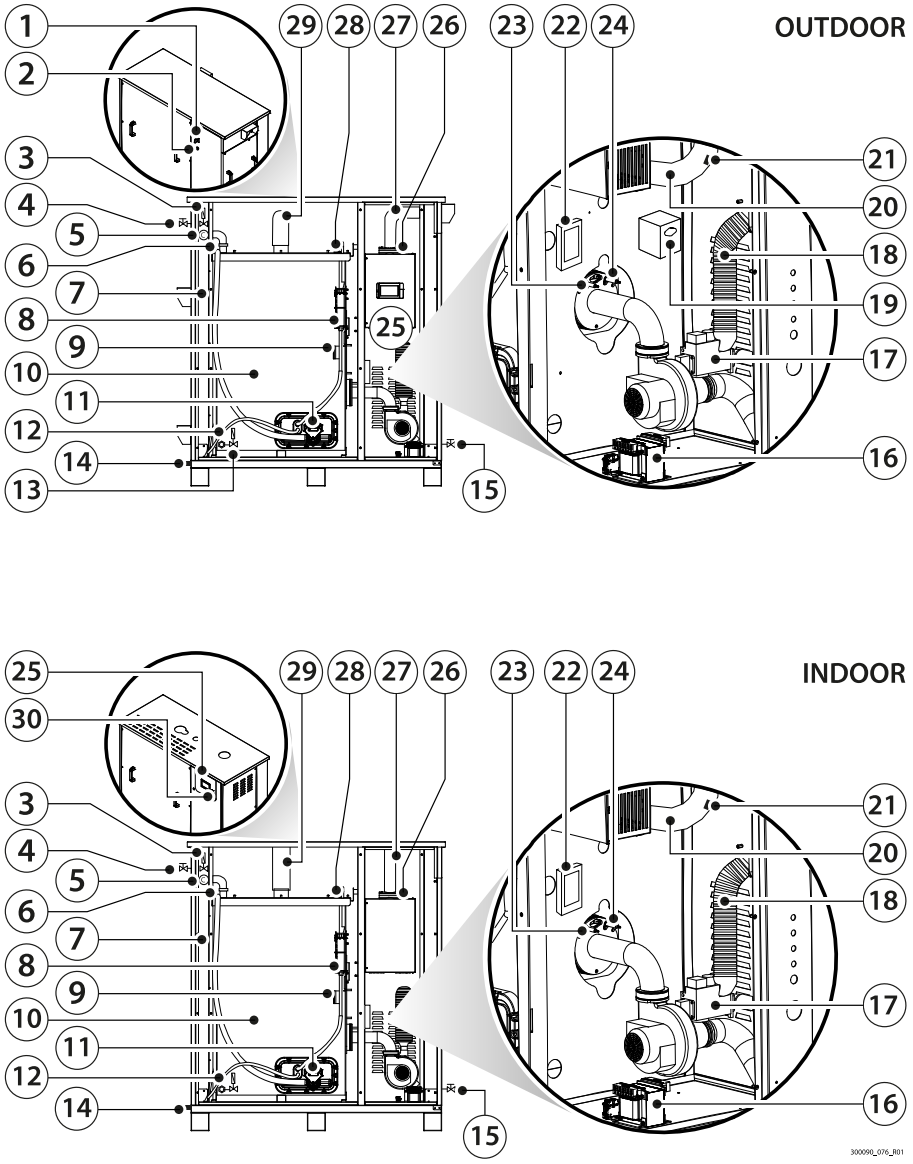


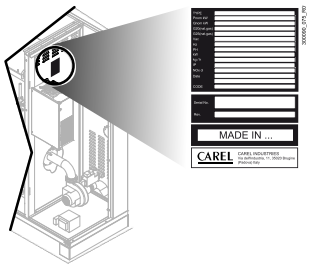
Fig.1.e

Ref.	Description	Ref.	Description
1	Disconnect switch	16	115-230 V transformer (US market)
2	Lock	17	Gas valve
3	Fill solenoid valve	18	Combustion air intake pipe
4	Fill shut-off valve	19	Heater (accessory)
5	Conductivity meter	20	Flue gas exhaust pipe
6	Drain column	21	Safety thermostat
7	Overflow/drain pipe	22	Flame control board
8	Level sensor	23	Ignition electrode
9	Preheating temperature sensor	24	Detection electrode
10	Boiler	25	User terminal
11	Drain pump	26	Fan electrical panel
12	Fill pipe	27	Flue gas exhaust pipe
13	Total boiler drain valve	28	Foam sensor
14	Tank drain	29	Steam outlet
15	Gas shut-off valve	30	ON/OFF switch

Tab.1.c

1.8 Rating plate

The humidifiers can be identified by the rating plate placed on the partition in the electrical cabinet. There is also a label showing the types of gas allowed for each country and the corresponding supply pressures (classification according to EN437).



Note: tampering, removal or absence of the rating plate or anything else that does not allow certain identification of the product will make any installation or maintenance operations difficult.

1.9 Dimensions by model - mm(in)

1.9.1 Indoor models

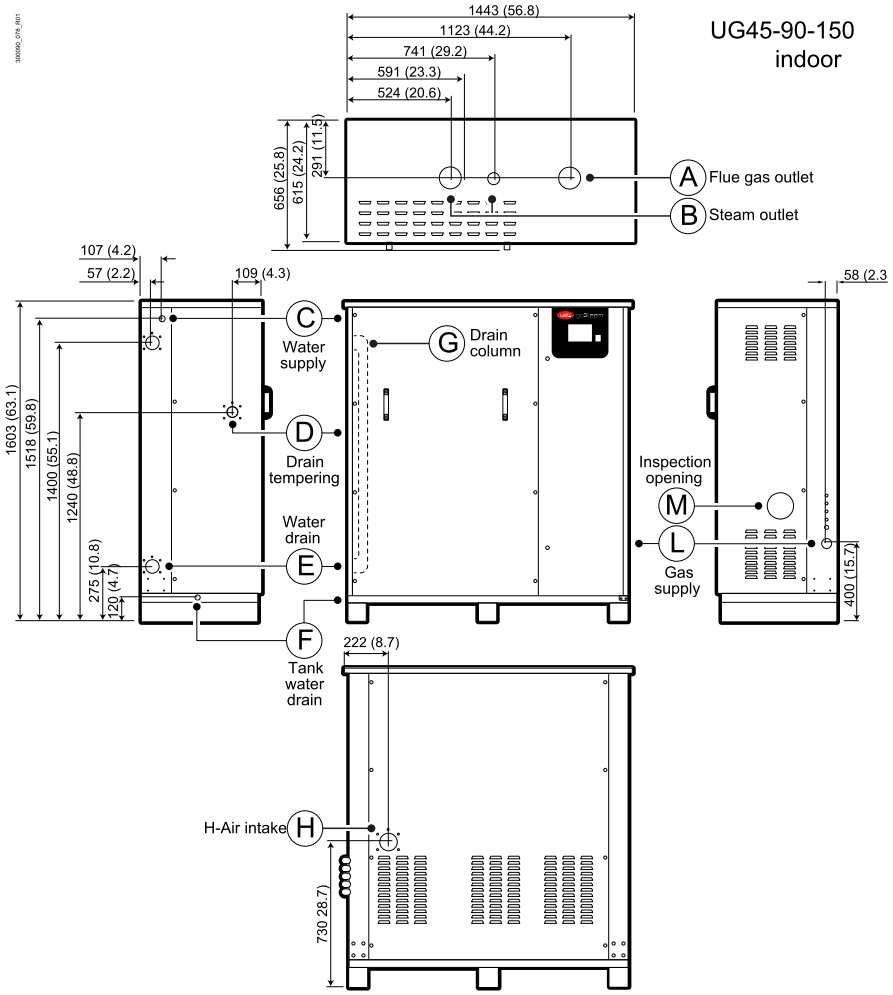


Fig.1.f

Key

Ref.	Description
A	Flue gas outlet
B	Steam outlet
C	Water inlet
D	Drain tempering (accessory)
E	Water drain

Ref.	Description
F	Tank drain
G	Drain column
H	Combustion air intake
L	Gas inlet
M	Flame inspection opening

Tab.1.d

Model	No. burners	Flue gas outlets	Steam outlets	Water inlet	Gas inlet	Air intake	Tank drain	Drain tempering (accessory)
UG45	1	1 X Ø 80(3.1)	2 X Ø 40(1.6)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG90	1	1 X Ø 80(3.1)	2 X Ø 40(1.6)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG150	1	1 X Ø 80(3.1)	1 X Ø 80(3.1)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"

Tab.1.e

UG180-300
indoor

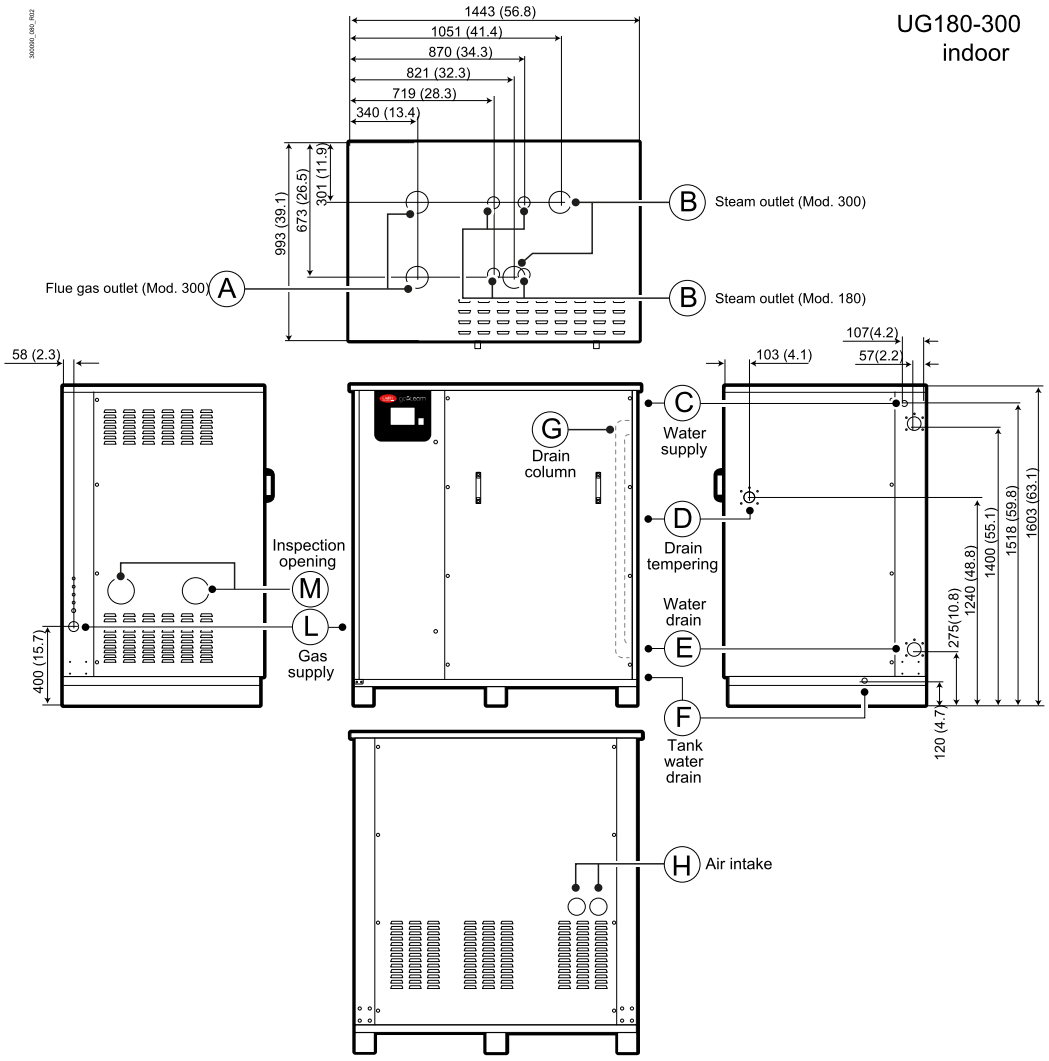


Fig.1.g

Key

Ref.	Description
A	Flue gas outlet
B	Steam outlet
C	Water inlet
D	Drain tempering (accessory)
E	Water drain

Ref.	Description
F	Tank drain
G	Drain column
H	Combustion air intake
L	Gas inlet
M	Flame inspection opening

Tab.1.f

Model	No. burners	Flue gas outlets	Steam outlets	Water inlet	Gas inlet	Air intake	Tank drain	Drain tempering (accessory)
UG180	2	2 X Ø 80(3.1)	4 X Ø 40(1.6)	1 X 3/4" M	1 X 1"1/4 G	2 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG300	2	2 X Ø 80(3.1)	2 X Ø 80(3.1)	1 X 3/4" M	1 X 1"1/4 G	2 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"

1.9.2 Outdoor models

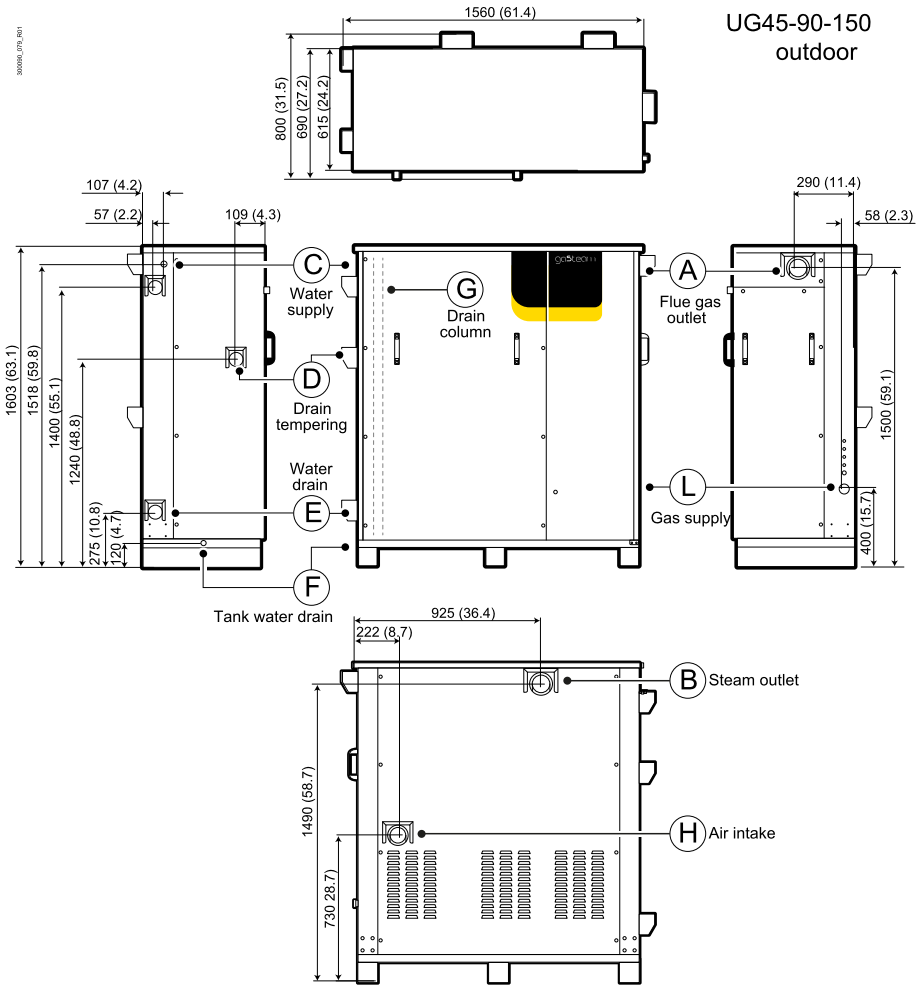


Fig.1.h

Key	
Ref.	**Description**
A	Flue gas outlet
B	Steam outlet
C	Water inlet
D	Drain tempering (accessory)
E	Water drain

Ref.	**Description**
F	Tank drain
G	Drain column
H	Combustion air intake
L	Gas inlet

Tab.1.g

Model	No. burners	Flue gas outlets	Steam outlets	Water inlet	Gas inlet	Air intake	Tank drain	Drain tempering (accessory)
UG45	1	1 X Ø 80(3.1)	2 X Ø 40(1.6)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG90	1	1 X Ø 80(3.1)	2 X Ø 40(1.6)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG150	1	1 X Ø 80(3.1)	1 X Ø 80(3.1)	1 X 3/4" M	1 X 1"G	1 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"

Tab.1.h

UG180-300
outdoor

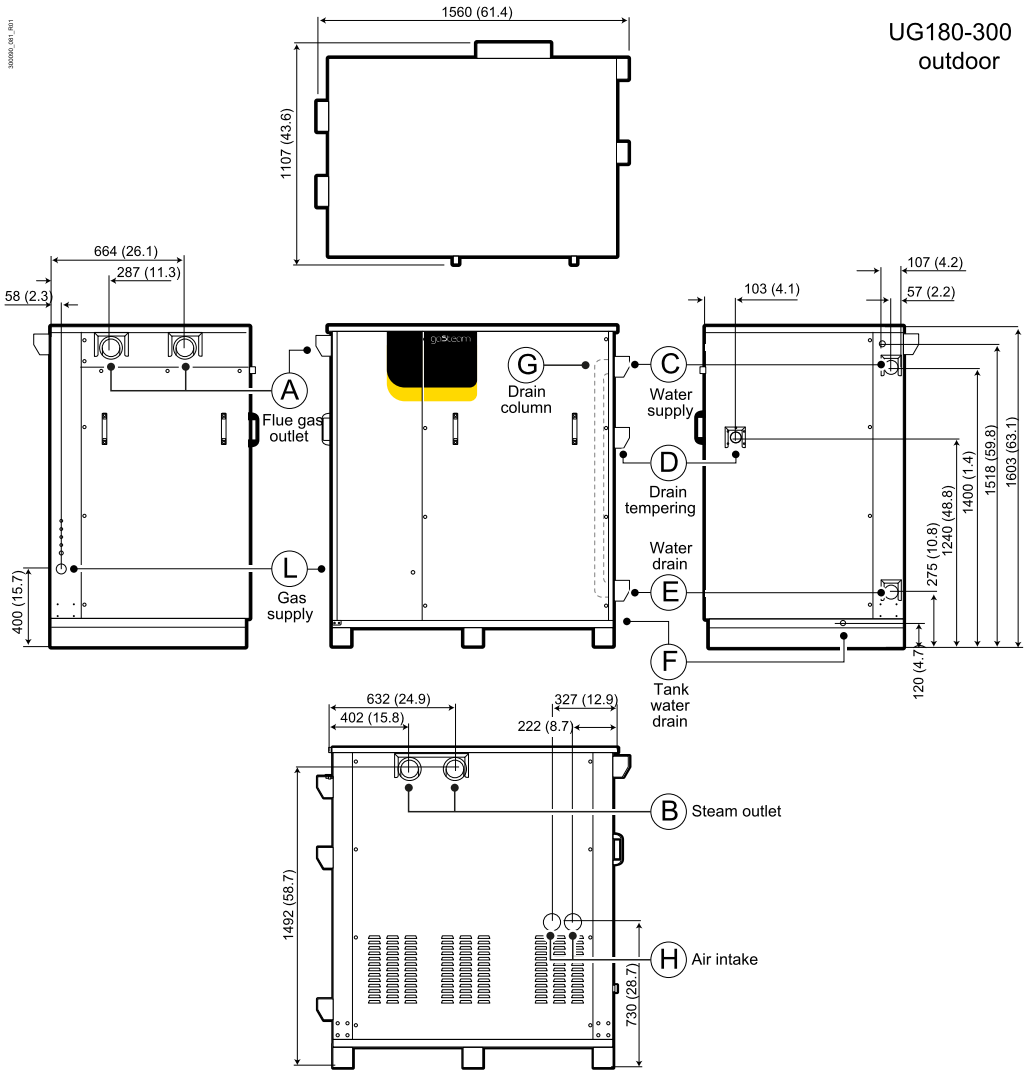


Fig.1.i

Key

Ref.	Description
A	Flue gas outlet
B	Steam outlet
C	Water inlet
D	Drain tempering (accessory)
E	Water drain

Ref.	Description
F	Tank drain
G	Drain column
H	Combustion air intake
L	Gas inlet

Tab.1.i

Model	No. burners	Flue gas outlets	Steam outlets	Water inlet	Gas inlet	Air intake	Tank drain	Drain tempering (accessory)
UG180	2	2 X Ø 80(3.1)	4 X Ø 40(1.6)	1 X 3/4" M	1 X 1"1/4 G	2 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"
UG300	2	2 X Ø 80(3.1)	2 X Ø 80(3.1)	1 X 3/4" M	1 X 1"1/4 G	2 X Ø 80(3.1)	1 X Ø 20(0.8)	1 X 3/4"

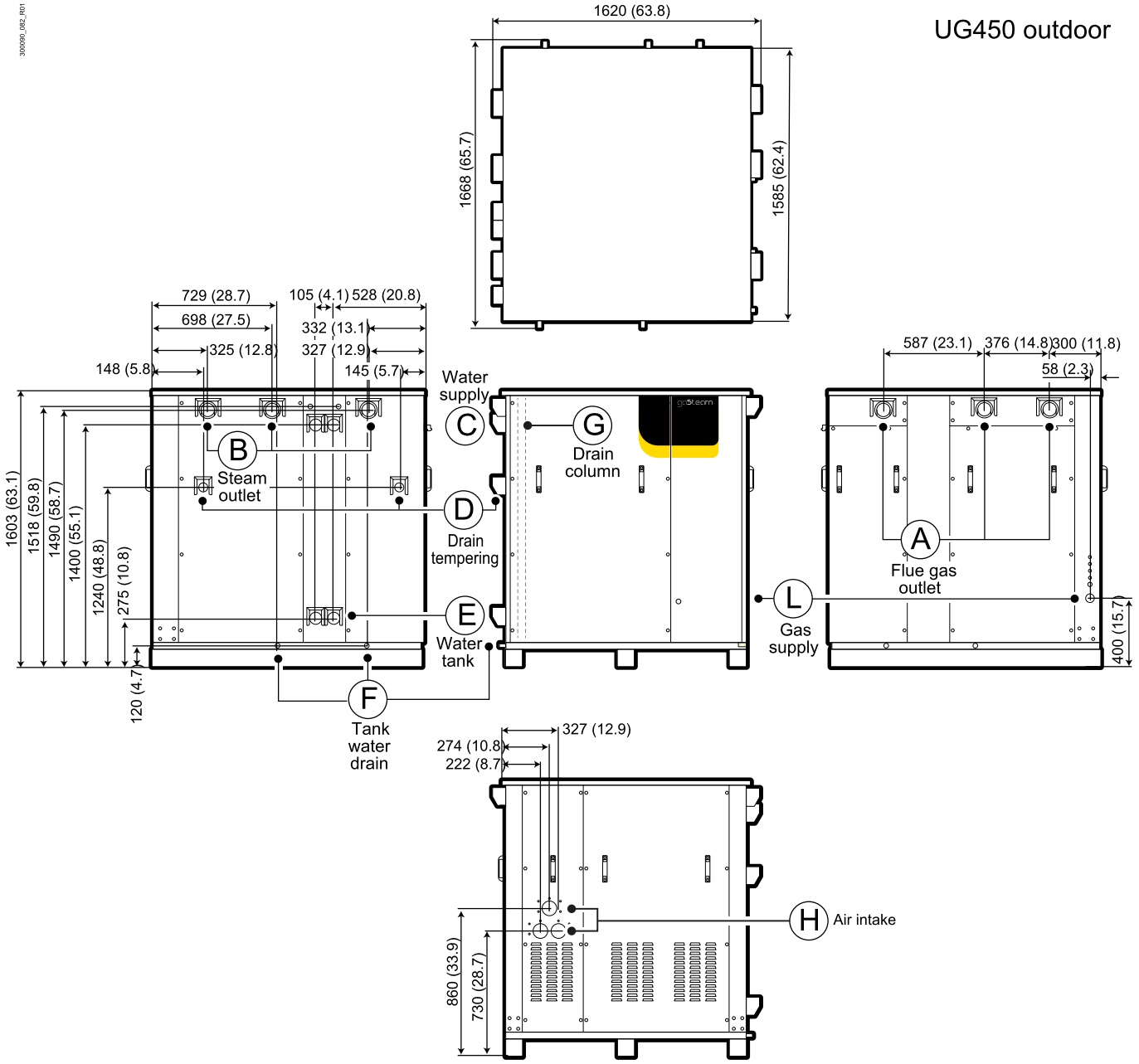


Fig.1.j

Key	
Ref.	**Description**
A	Flue gas outlet
B	Steam outlet
C	Water inlet
D	Drain tempering (accessory)
E	Water drain

Ref.	**Description**
F	Tank drain
G	Drain column
H	Combustion air intake
L	Gas inlet

Tab.1.j

Model	No. burners	Flue gas outlets	Steam outlets	Water inlet	Gas inlet	Air intake	Tank drain	Drain tempering (accessory)
UG450	3	3 X Ø 80(3.1)	3 X Ø 80(3.1)	2 X 3/4" M	1 X 1"1/2 G	3 X Ø 80(3.1)	2 X Ø 20(0.8)	1 X 3/4"

2. Connections

Important: before making the connections, disconnect the power supply by turning the main switch to OFF.

2.1 Pipe connections

gaSteam requires a connection to the gas and water supply and water drain pipes. The feedwater supply can be connected using a hose with a minimum inside diameter (recommended) of 6 mm. Install a shut-off valve to allow the appliance to be disconnected during maintenance. We recommend the use of CAREL hoses (P/N FWH3415003), installing a mechanical filter to trap any solid impurities.

To make the connection:

- drain water: use non-conductive plastic tubing resistant up to 100°C (212°F), with an outside diameter of 50 mm (ref. 3 in the figure);
- gas supply: use a flexible metal pipe (with vibration isolating joint) connected to a manual shut-off valve.

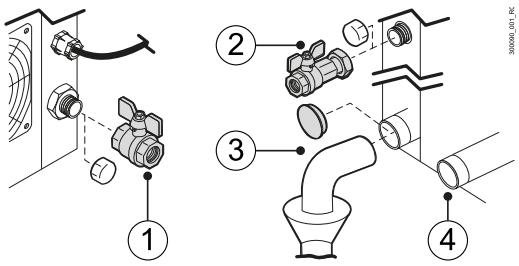


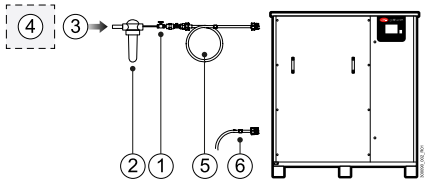
Fig.2.a

Ref.	Description
1	gas shut-off valve
2	water inlet shut-off valve
3	water drain connection
4	tank drain connection

Tab.2.a

2.2 Water connection diagram

Important: the drain pipe must not be obstructed, must not have back pressure and must be fitted with a drain trap immediately downstream of the connection to the humidifier.



Ref.	Description
1	shut-off valve
2	filter
3	feedwater
4	possible reverse osmosis water treatment system
5	hose (P/N FWH3415003, not supplied)
6	hose (drain tempering option, not supplied)

2.3 Feed water

The feedwater supply to the humidifier must not be corrosive, must not emit bad odours, and must not have an excessive lime content to avoid scale build-up. The water supply must be potable or demineralised water, and must comply with the limits shown in the following table. For proper operation, it is recommended to use demineralised water; for this purpose, the use of a reverse osmosis demineralisation system is suggested.

Important: water treatment with polyphosphates or water softeners is allowed, guaranteeing at least 60% of the initial hardness value and no less than 5°f. Softened water reduces maintenance however favours the concentration of dissolved salts in the water inside the boiler, with the consequence of foam formation.

Note that:

1. no relationship can be proven between water hardness and conductivity; nonetheless, as an indication only, water with a hardness of 40°f should have a conductivity of approximately 900-1000 µS/cm at 20 °C.
2. if supplied with water coming from an external reverse osmosis system, an instant flow-rate of 20 l/min (5.28 Gal/min) must be guaranteed. For UG450, there are two inlets, therefore the instant flow-rate must be 2 x 20 l/min (2 x 5.28 Gal/min).

		Min	Max
Specific conductivity at 20°C	σ20 - µS/cm	0	1500
Total dissolved solids	TDS – mg/l	(1)	(1)
Dry residue at 180°C	R180 – mg/l	(1)	(1)
Hydrogen ion activity	pH	6	8.5
Total hardness	TH– mg/l CaCO3	0 (2)	400
Temporary hardness	mg/l CaCO3	0 (3)	300
Chlorides	ppm Cl	-	50 (4)
Iron + Manganese	mg/l Fe+Mn	-	0.2
Silica	mg/l SIO2	-	20
Residual chlorine	mg/l Cl-	-	0.2
Calcium sulphate	mg/l CaSO4	-	100
Metallic impurities	mg/l	0	0
Solvents, thinners, detergents, lubricants	mg/l	0	0

Tab.2.b

(1) Values dependent on the specific conductivity; in general: TDS ≈0.93* σR, 20°C; R180 ≈0.65* σR, 20°C;

(2) Not lower than 200% the chloride content in mg/l Cl-;

(3) Not lower than 300% the chloride content in mg/l Cl-;

It may be necessary to adjust the drain rate to avoid a concentration in the boiling water higher than 300 mg/l Cl-;

2.4 Drain water characteristics

The drain water:

- contains the same substances that are dissolved in the supply water, yet in greater quantities;
- can reach a temperature of 100°C (212°F);
- is not toxic and thus may be drained into the sewage system;
- for drain tempering versions, water temperature will be 60°C (140°F), with a ma inlet temperature of 25°C (77°F).

⚠ **Important:** completely demineralised water is aggressive. For connection to a feedwater system with completely demineralised water, only use components made from chemically-resistant plastic (e.g. polypropylene) or stainless steel.

2.5 Air intake and flue connections

The gaSteam humidifier is an approved type C appliance (sealed appliance). Regarding the installation type, the following configurations are possible: B23, C13, C33, C43, C53.

⚠ **Important:** the air intake pipes and flues must be installed and positioned according to the corresponding legislation and standards in force, as well as to the any national or local provisions, and therefore the validity of the diagrams below should always be verified.

The air intake/flue gas exhaust pipes must be connected as shown in the following figure.

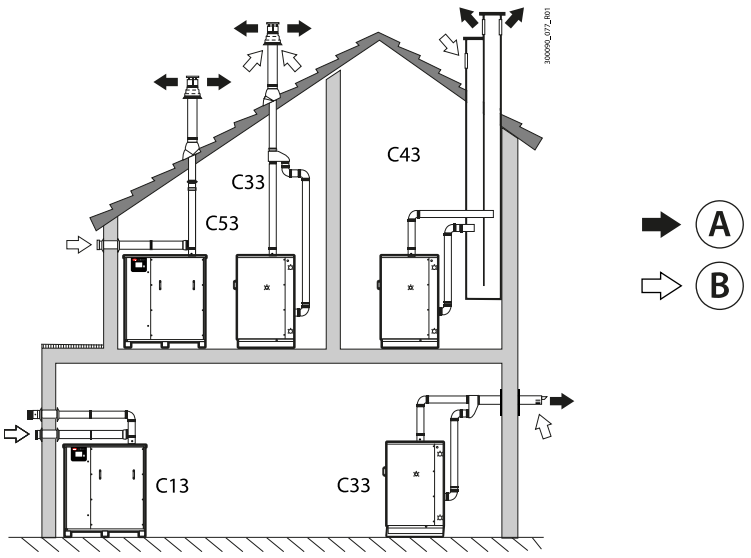


Fig.2.b

Ref.	Description
A	Flue gas outlet
B	Combustion air intake

For further requirements, use flue gas accessories available on the market. The maximum lengths specified in the following installation diagrams have been verified using CAREL pipes.

For the position of the combustion air intake and flue gas outlet, see the paragraph "Dimensions by model".

The indoor version is factory-set as follows:

- flue gas outlet from the top of the humidifier;
- air intake from the rear;
- supplied with a 135 mm long inspectable section of piping (two sections for UG180/UG300) and a 135 mm section for condensate collection.

⚠ **Note:** assemble the condensate collection section and connect a pipe to the hole on the side to drain the condensate and prevent it from causing the burner to malfunction.

Both the flue gas outlet and the combustion air intake can be moved using the accessories available, based on installation requirements, with reference to the pressure drop values shown in the table.

		pipe diameter (aluminium or stainless steel)		
UG45	**DP**	**d=80mm**	**d=100mm**	**d=120mm**
linear section of pipe l= 1m	Pa	2.5	1	0.4
90° bend r/d=0.5	Pa	6.4	2.6	1.2
90° bend r/d=1	Pa	2.6	1	0.5
45° bend r/d=0.5	Pa	3.2	1.3	0.6

		pipe diameter (aluminium or stainless steel)		
flue gas exhaust terminal	Pa	7	2.8	1.4
"T" section for condensate collection	Pa	7.7	3.1	1.5
Expansion in cross-section	Pa	1	0.4	0.2

Tab.2.c

		pipe diameter (aluminium or stainless steel)		
UG90	**DP**	**d=80mm**	**d=100mm**	**d=120mm**
UG180	**DP**	**2 xd = 80mm**	**2 xd = 100mm**	**2 xd = 120mm**
linear section of pipe l= 1m	Pa	7.7	2.6	1.1
90° bend r/d=0.5	Pa	19.1	7.6	3.6
90°bend r/d=1	Pa	7.6	3	1.5
45° bend r/d=0.5	Pa	9.5	3.8	1.8
flue gas exhaust terminal	Pa	21	8.3	4
"T" section for condensate collection	Pa	23	9	4.3
Expansion in cross-section	Pa	3	1.2	0.6

Tab.2.d

		pipe diameter (aluminium or stainless steel)			
UG150	**DP**	**d=80mm**	**d=100mm**	**d=120mm**	**d=150mm**
UG300	**DP**	**2 xd = 80mm**	**2 xd = 100mm**	**2 xd = 120mm**	**2 xd = 150mm**
UG450	**DP**	**3 xd = 80mm**	**3 xd = 100mm**	**3 xd = 120mm**	**3 xd = 150mm**
linear section of pipe l= 1m	Pa	22.5	7.6	4	1
90° bend r/d=0.5	Pa	66	26	12.3	5
90° bend r/d=1	Pa	27	11	5	2
45° bend r/d=0.5	Pa	33	13	6.2	2.5
flue gas exhaust terminal	Pa	73	29	13.6	5.5
"T" section for condensate collection	Pa	79	32	15	6
Expansion in cross-section	Pa	10	4	1.9	0.8

Tab.2.e

Flue gas accessories (to be purchased separately):

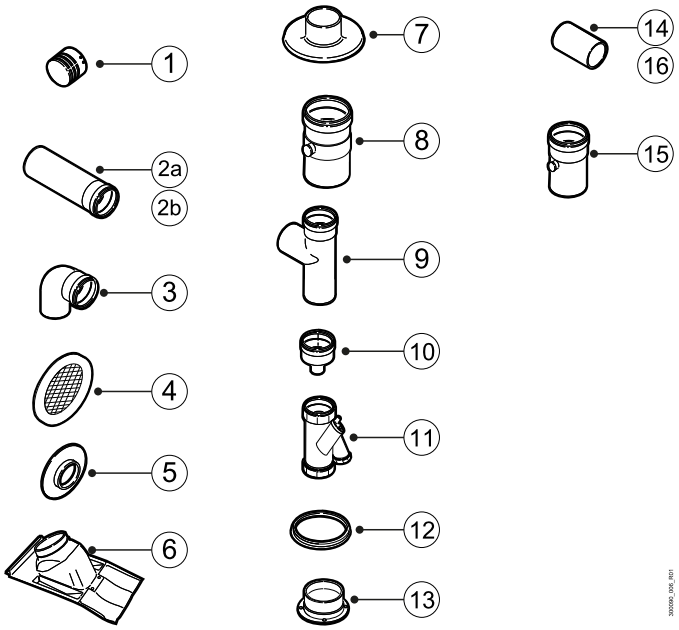


Fig.2.c

Ref.	Description	Code
1	Stainless steel flue grille, Ø 80 mm	EXHZ080000
2a	Ø 80 mm extension, L = 1000 mm	EXHP080100
2b	Ø 80 mm extension, L = 500 mm	EXHP080500
3	Ø 80 mm 90° bend	EXHC080080
4	Stainless steel intake grille, Ø 80 mm	EXHX080000
5	Internal/external flange, Ø 80 mm	EXHQ080000
6	Tile for sloping roof, Ø 80 mm	EXHNI00125
7	Flat tile Ø 80 mm	EXHN080000
8	Condensate drain section, Ø 80 mm L=135mm	EXHS0A0011
9	"T" connector Ø 80 mm	EXHD080000
10	Cap	EXHG000000
11	Condensate drain trap with drain pipe	EXHDS00000
12	Silicone flue gasket Ø 80mm	EXHW000080
13	Flanged section, Ø 80 mm, L=56 mm	EXHL080056
14	Pipe section, Ø 80 mm, L=120 mm	EXHT080120
15	Inspection section, Ø 80 mm, L=135 mm	EXHI080025
16	Pipe section, Ø 80 mm, L=100 mm	EXHT080100

Tab.2.f

2.6 Installation of the appliance with air intake from the room

gaSteam humidifiers can be also installed as type B appliances, i.e. with air intake from the room where the appliance is installed.

The maximum allowable pressure drop in the ø 80 mm air intake/flue gas exhaust pipes is:

- for gaSteam 45/90/180: 0-70 Pa (0-0.70 mbars / 0-0.0101 PSI);
- for gaSteam 150/300/450: 0-260 Pa (0-2.6 mbars / 0-3.771 PSI);

⚠ **Important:** for flues > 2 m long, a condensate collection section (part 6) or an appropriate drain (part 9) must be installed.

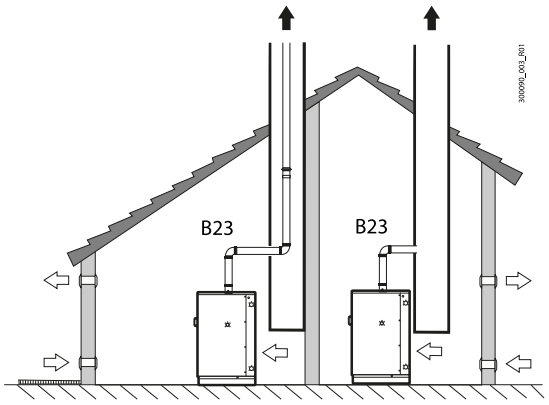


Fig.2.d

2.7 Pressure switch

The pressure switch (P/N UGKPRES180, not supplied) is used to verify that there is no return of flue gases in the event of installation with a common flue/chimney.

2.8 Checks

Verity that the water connections are correct, as follows:

- the feedwater supply line can be closed using a shut-off valve;
- a mechanical filter is installed on the feedwater line;
- water temperature and pressure are within the permitted values;
- the drain pipe is resistant to operating temperatures of 100°C (212°F);
- minimum inside diameter of the drain pipe is 40 mm (1"1/2);
- minimum slope of the drain pipe is greater than or equal to 5°;
- an inspection section is fitted with connection to a condensate drain.

⚠ **Important:** when installation is completed, flush the feedwater line for around 30 minutes by piping the water directly into the drain, without sending it into the humidifier. This will eliminate any scale or processing residues that may cause foam when boiling.

3. Steam distribution

To achieve optimal humidifier efficiency, the steam produced must be introduced into the room uniformly, in order to prevent the formation of droplets and condensation. This is achieved using linear steam distributors. The right steam distributor must be chosen according to the place where the steam is to be introduced.

3.1 Steam nozzles

Compact steam distributors for small ducts or applications such as steam baths. These can be mounted either horizontally or vertically (steam outlet upwards).

Mounting instructions (see the figure):

- drill a series of holes in the wall of the pipe, using the drilling template supplied with the distributor;
- insert the distributor;
- fasten the distributor flange using four screws.

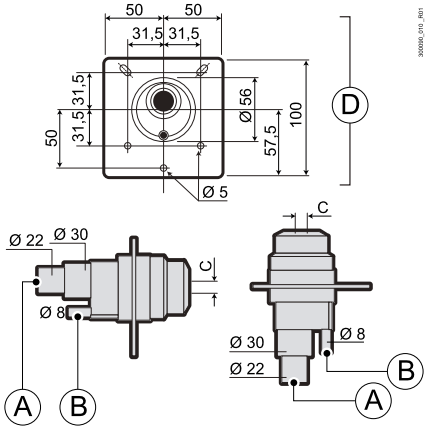


Fig.3.a

Key:

A.	steam inlet
B.	condensate drain
C.	steam outlet The hole dimensions vary depending on the distributor model: ○ model SDPOEM0000: hole to be drilled manually, up to 30 mm (1.2 in) diameter; ○ model SDPOEM0012: hole diameter 12 mm (0.5 in); ○ model SDPOEM0022: hole diameter 22 mm (0.9 in).
D.	drilling template

Tab.3.a

Note: if using steam pipes with an inside diameter of 30 mm (1.2 in), remove the 22 mm (0.9 in) inlet section from the steam nozzle.

3.2 Linear distributors for air ducts or AHUs

The steam distributors for AHU (air handling unit) or duct applications (sold separately) deliver steam along their entire length in order to ensure the shortest not-wetting distance. The distributor must be chosen based on:

- maximum capacity required;
- dimensions of the AHU/duct;
- outlet diameter on the humidifier it is connected to.

Note: install away from any obstacles (curves, branches, changes in cross-section, grilles, filters, fans).

Minimum distance between distributor and obstacle: 1-1.5 m (3.3-4.9 ft).

Increase the distance in the event of:

1. increased air speed in the duct;
2. less turbulence.

Mounting instructions (see the figure):

- drill a series of holes in the wall of the pipe, using the drilling jig of the distributor (found in the packaging of the distributor);
- insert the distributor with the steam outlet holes facing upwards;
- fasten the distributor flange using four screws.

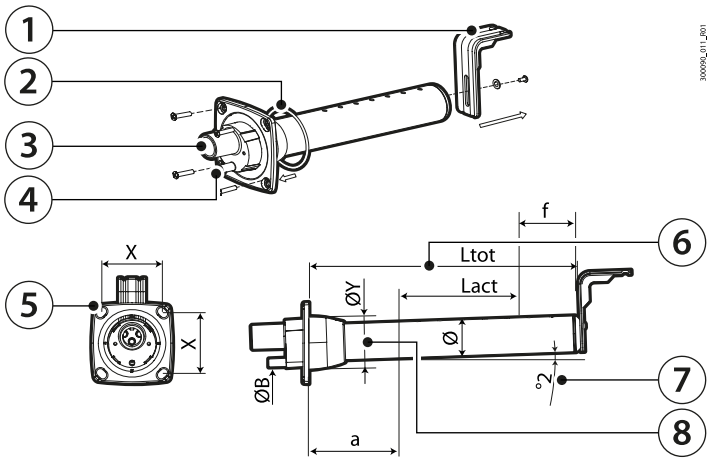


Fig.3.b

Key:

1. L-shaped mounting bracket (where featured)
2. flange gasket
3. steam inlet (ØA)
4. condensate drain (ØB)
5. flange screw (see technical leaflet supplied)
6. Ltot = total length from the support to the end of the pipe, including the cover (depending on the model, see the paragraph "Linear distributors");
   - Lact = actual length used (from the first to the last hole);
   - a = distance from the support to the first hole;
   - f = distance from the last hole to the end of the pipe;
1. slope (about 2°) for condensate drainage
2. diameter of the hole in the wall (ØY)

Dimensions - mm (in)

	DP***D22R0	DP***D30R0	DP***D40R0
ØA	22 (0.9)	30 (1.2)	40 (1.6)
ØB	10 (0.4)	10 (0.4)	10 (0.4)
ØY	58 (2.3)	68 (2.7)	89 (3.5)
Ø	35 (1.4)	45 (1.8)	60 (2.4)
X	68 (2.7)	77 (3.0)	99 (3.9)

Tab.3.b

Length L

P/N	L tot	max kg/h	lbs/hr
DP035D22R0	332	4	8.81
DP045D22R0	438	6	13.22
DP060D22R0	597	9	19.82
DP085D22R0	835	9	19.82
DP035D30R0	343	5	11.01

P/N	L tot	max kg/h	lbs/hr
DP045D30R0	427	8	17.62
DP060D30R0	596	12	26.43
DP085D30R0	850	18	39.65
DP105D30R0	1048	18	39.65
DP125D30R0	1245	18	39.65
DP165D30R0	1640	18	39.65
DP085D40R0	834	25	55.07
DP105D40R0	1015	35	77.09
DP125D40R0	1222	45	99.12
DP165D40R0	1636	45	99.12
DP205D40R0	2025	45	99.12
DP035D30RH	343	5	11.01
DP045D30RH	427	8	17.62
DP060D30RH	596	12	26.43
DP085D30RH	850	18	39.65
DP105D30RH	1048	18	39.65
DP125D30RH	1245	18	39.65
DP165D30RH	1640	18	39.65
DP085D40RH	834	25	55.07
DP105D40RH	1015	35	77.09
DP125D40RH	1222	45	99.12
DP165D40RH	1636	45	99.12
DP205D40RH	2025	45	99.12

⚠ Important:

1. mount the distributor with a slight slope (at least 2°, to facilitate condensate drainage);
2. the L-shaped support is supplied with steam distributor models from DP085* to DP205*. For shorter lengths, the support can be supplied as an option (P/N 18C478A088).

3.3 High-efficiency linear distributors for air ducts or AHUs

High-efficiency linear distributors (sold separately), can be used in all cases where the formation of condensate needs to be reduced inside the distributor itself. All the information provided in the paragraph on linear distributors still applies; the dimensions and diameters are shown below.

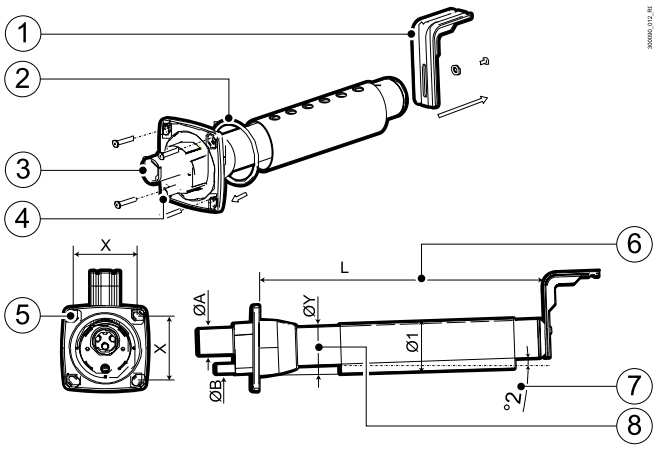


Fig.3.c

Key:

1. L-shaped mounting bracket (where featured)
2. flange gasket
3. steam inlet (ØA)
4. condensate drain (ØB)
5. flange screw (see technical leaflet supplied)

6. L = length (depending on the model)
7. slope (about 2°) for condensate drainage
8. diameter of the hole in the wall (ØY)

Dimensions - mm (in)

	DP***D30RH	DP***D40RH
ØA	30 (1.2)	40 (1.6)
ØB	10 (0.4)	10 (0.4)
ØY	68 (2.7)	89 (3.5)
Ø	45 (1.8)	60 (2.4)
Ø1	60 (2.4)	80 (3.1)
X	77 (3.0)	99 (3.9)

Tab.3.c

3.4 Steam blower for room installation

Steam distributor for installation inside the room, sold separately, equipped with a fan to improve the absorption of steam into the air. The VRDXL00001 blower with 230 Vac power supply (mains) can be used for humidifiers with a capacity up to 45 kg/h (99.2 lb/h), and therefore with models UG045 and UG090 (two units). The blower can be mounted separately from the humidifier. See instruction manual +030222195.

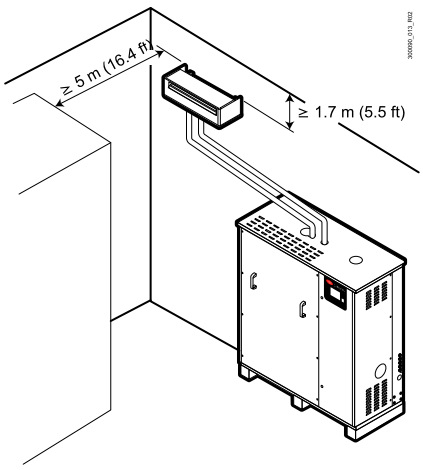


Fig.3.d

⚠ **Important:** observe the distances shown in the figure for correct steam distribution.

3.5 Steam hose

Warnings:

- use CAREL hoses (max length = 4 m, see par. "Steam hose models");
- avoid the formation of pockets or traps (these cause condensate accumulation)
- avoid choking the hose with sharp bends or twists.
- use metal clamps to secure the ends of the hose to the connections on the humidifier and the steam distributor so that they do not come loose due to the high temperatures.
- avoid strain, including mechanical stress on the boiler steam outlet spigot.

3.6 Condensate drain pipe

During operation of the humidifier, part of the steam may condense, causing a decline in efficiency and noise (gurgling). To drain the condensate, connect to the base of the distributor a drain pipe with a trap and a minimum slope of 5°.

CAREL condensate drain pipes: P/N 1312368AXX (ø 10mm) - (CHOSE00516 (5/16") for the North American market) for DP* series linear steam distributors; P/N 1312353APG (ø 7mm) - (CHOSE0038 (3/8") for the North American market) for steam blowers and steam nozzles.

⚠ **Important:** the condensate drain trap must be filled with water prior to turning on the humidifier.

Below are some requirements and installation examples of the steam hoses and condensate drain pipes:

- instructions provided in this chapter, the steam outlet hoses are aimed upwards and the distributor has an upwards slope of at least 2°;
- any curves in the pipes or hoses are sufficiently wide (radius > 300 mm - 12 in) so as to not cause bending or choking;
- the paths of the steam hose and condensate pipe comply with the instructions provided in this chapter;
- the slope of the steam hose is sufficient to allow correct condensate drainage (> 20° for upward sections,> 5° for downward sections);
- the slope of the condensate drain pipe is at least 5° at every point;
- the condensate drain pipe is fitted with a drain trap (filled with water before starting) to prevent steam from escaping.

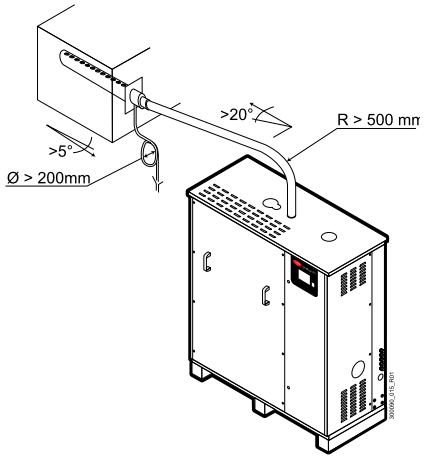


Fig.3.e

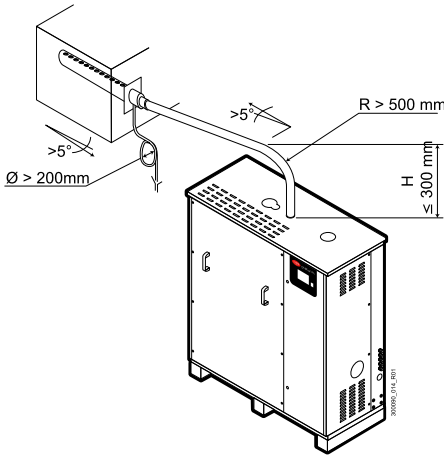


Fig.3.f

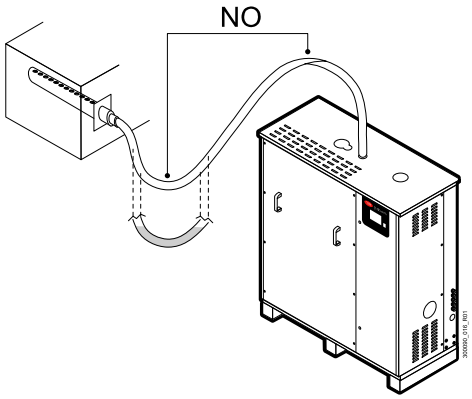


Fig.3.g

3.7 Outlet pressure limits

The back-pressure at the boiler outlet, either positive or negative, depends both on the relative pressure in the duct/AHU and the pressure drop in the steam hose, due to bends or adapters, and the steam distributor. Carel steam hoses have a pressure drop of around 150 Pa/m (0.021 psi) (in compliance with the maximum flow-rate recommended by Carel). Carel DP* series linear distributors have a pressure drop of around 25 Pa (0.003psi) (in compliance with the maximum flow-rate recommended by Carel). Considering these values, the boiler outlet pressure must be:

- boiler outlet pressure > 150 [Pa/m]* steam hose length [m] + 25 [Pa] DP + pressure in duct/AHU [Pa]

Boiler outlet pressure values for gaSteam:

	UG045	UG090	UG150	UG180	UG300	UG450
pressure limits at boiler outlet Pa (PSI)	0 to 2000 (0 to 0.290)					

Tab.3.d

Note: a pressure of around -200 Pa (-0.029 psi) at the boiler outlet corresponds to around 20 mm

(0.78") of water inside the boiler (for all sizes).

4. Electrical connections

4.1 Wiring requirements

The cables coming from the outside must pass through the side cable glands (see the figure).

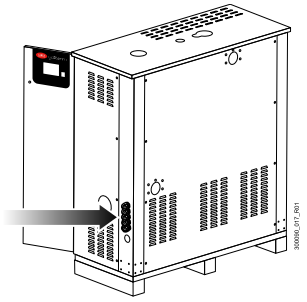


Fig.4.a

4.2 Power cable

Before making the electrical connections, disconnect the unit from the mains power supply.

Check that the unit's power supply voltage corresponds to the rated value shown inside the electrical panel.

230 V/50 Hz version

Run the power and earth wires into the electrical panel through the tear-resistant cable gland supplied, and connect the ends to terminal AL (L1, L2, GND), as shown in the figure.

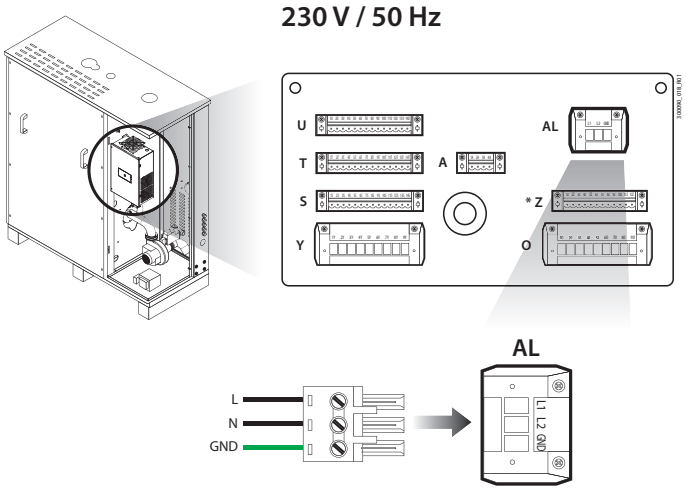


Fig.4.b

115 V/60 Hz version

Run the power and earth wires into the electrical panel through the tear-resistant cable gland supplied, and connect the ends to the terminals on the DIN rail, connected directly to the main transformer (primary 115 V, secondary 230 V).

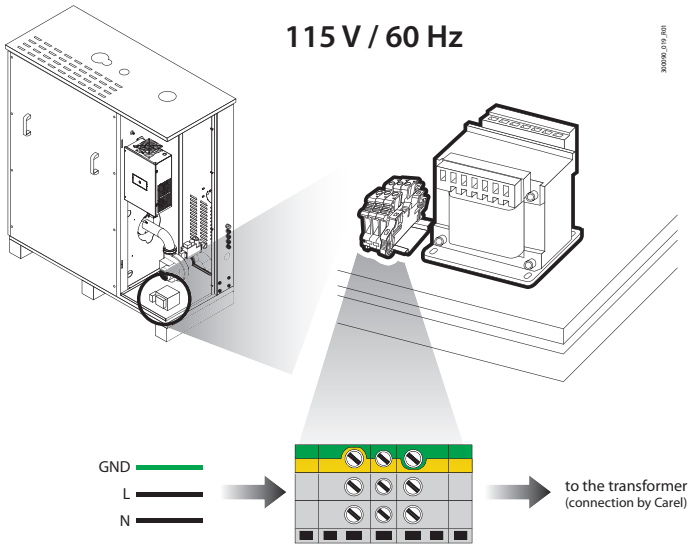


Fig.4.c

Note: to avoid unwanted interference, the power cables should be kept separate from the probe signal cables.

Important: connect the yellow-green wire to the earth terminal (GND).

4.3 Power supply

The humidifier power line must be fitted by the installer with a disconnect switch and protection fuses. The table shows the recommended size of the power cable and fuse ratings. These data are purely indicative; local regulations must always prevail.

	power supply		rated characteristics				
mod.	code	voltage (V - type) (1)	current (A) (2)	power (W) (3)	production (kg/h (lbs/h)) (2, 4)	cable size (mm2) (AWG14) (3)	line fuses (A - type) (3)
UG045*D004	D0	230V 50Hz - 1~	0.34	180	45 (100)	2.5	16 F
UG045*1104	11	115V 60Hz - 1~	0.68	180	45 (100)	2.5	16 F
UG090*D004	D0	230V 50Hz - 1~	0.67	250	90 (200)	2.5	16 F
UG090*1104	11	115V 60Hz - 1~	1.34	250	90 (200)	2.5	16 F
UG150*D004	D0	230V 50Hz - 1~	0.71	260	150 (330)	2.5	16 F
UG150*1104	11	115V 60Hz - 1~	1.41	260	150 (330)	2.5	16 F
UG180*D004	D0	230V 50Hz - 1~	1,246	385	180 (400)	2.5	16 F
UG180*1104	11	115V 60Hz - 1~	2.5	385	180 (400)	2.5	16 F
UG300*D004	D0	230V 50Hz - 1~	1.29	400	300 (660)	2.5	16 F
UG300*1104	11	115V 60Hz - 1~	2.6	400	300 (660)	2.5	16 F
UG450*D004	D0	230V 50Hz - 1~	2	660	450 (990)	2.5	16 F
UG450*1104	11	115V 60Hz - 1~	4	660	450 (990)	2.5	16 F

Tab.4.a

(1) Tolerance allowed on the nominal mains voltage: -15%, ± 10%.

(2) Tolerance on nominal values: + 5%, -10% (EN 60335-1).

(3) Recommended values; referred to cables laid in closed PVC or rubber conduits 20 m long; the standards in force must always be followed.

(4) Instant steam production: average steam production may be affected by external factors, such as: ambient temperature, water quality, steam distribution system.

4.4 Auxiliary circuit transformer

The multi-voltage auxiliary circuit transformer has a primary winding for 230 V (protected by fuses, 10.3x38 mm) and a secondary winding (24 V). The transformer is connected and checked in the factory, according to the rated voltage.

4.5 Main control board

The auxiliary connections must be made by running the cables from the outside into the electrical panel compartment through the cable gland located on the side of the humidifier until reaching the plug-in screw terminal block located on the bottom of the electrical panel (as shown in the figure). It is recommended to secure the probe, remote ON/OFF contact connection cables etc. with appropriate cable ties, in order to prevent disconnection.

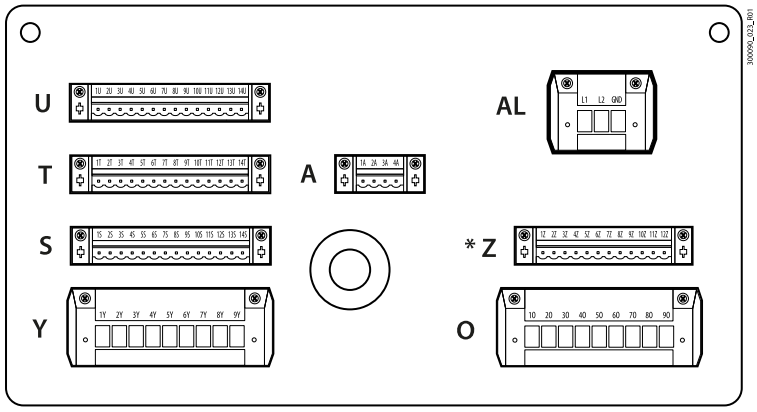


Fig.4.d

Note: terminal Z only available on UG180.

c.pHC control board

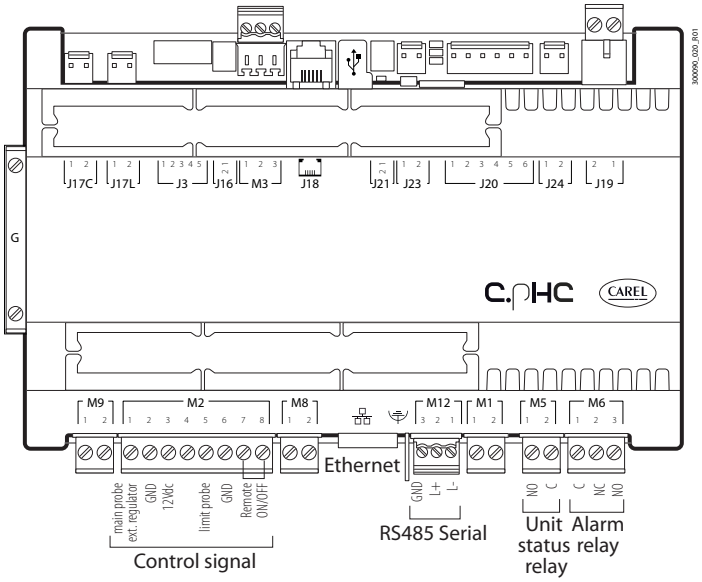


Fig.4.e

Terminal		Function	Electrical characteristics
M1	M1.1	GND (G0)	
	M1.2	Controller power supply	24Vac +10%/-15% 50/60Hz
M2	M2.1	Main room probe input or signal from external controller	0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA, NTC 10 kΩ at 25 °C
	M2.2	GND	
	M2.3	Probe power supply (+G)	+12 Vdc, maximum current that can be delivered 50 mA; protection against short-circuits
	G	Probe power supply (+G)	CAREL 0-10 V
	M2.4	Digital input for backup/rotation function	Maximum output current: 5 mA; maximum voltage with open contact: 13 Vdc
	M2.5	Limit humidity probe signal input	0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA, NTC
	M2.6	GND	
	M2.7	GND	
	M2.8	Remote on/off contact digital input	Maximum output current: 5 mA; maximum voltage with open contact: 13 Vdc
M3	M3.1	Fieldbus	Tx/Rx-
	M3.2		Tx/Rx+
	M3.3		GND
M5	M5.1	Unit status contact (NO)	250 Vac; 2 A with resistive load; 2 A with inductive load
	M5.2	Unit status contact common	
M6	M6.1	Alarm common	250 Vac; 2 A with resistive load; 2 A with inductive load
	M6.2	NC alarm contact	
	M6.3	NO alarm contact	
M8	M8.1	Production request analogue output	Output voltage range: 0-10V, max 10 mA
	M8.2	GND	
M9	M9.1	Backup and rotation contact common	250 Vac; 2 A with resistive load; 2 A with inductive load
	M9.2	Backup and rotation NO contact	
M11		Ethernet port	
M12	M12.1	BMS 485	Tx/Rx-
	M12.2		Tx/Rx+
	M12.3		GND
⏚		Earth connection	
J18		pLAN/display terminal	
J19	J19.1	Auxiliary/blower contact (NO)	250 Vac; 2 A with resistive load; 2 A with inductive load
	J19.2	Auxiliary/blower contact common	
⇕		USB port (type A)	
⊞		Ethernet RJ45 port	

Tab.4.b

4.6 Operation and control

Before describing the electrical connections to the terminals in detail, below is an introduction to the humidifier control principles.

4.6.1 Operating principle of a gas-fired humidifier

In a gas-fired humidifier, the production of steam is obtained inside a boiler containing water that is heated to and then held at boiling temperature. The heat required to boil the water is provided by a heat exchanger, heated by a modulating premixed gas burner. Burner operation is completely automatic and there is no pilot flame. All stages of burner operation are managed by an electronic board that, through ionisation control, constantly checks the flame. The continuously-delivered heat output reflects heat demand, across a wide modulation ratio (1: 4). The variable-speed fan (driven by the control board), combined with a proportional gas valve, ensures modulation of heat output (the gas flow-rate is proportional to the air needed for combustion). The water that evaporates over time is automatically replenished from the mains supply. When fully operational, the required level of production is achieved automatically by adjusting the heat output of the burner. The salts introduced by automatically refilling the water partly deposit in the form of scale inside the boiler, causing a progressive decline in performance, and partly remain dissolved in the water. To avoid excessive accumulation of salts, a certain amount of water is periodically and automatically drained, and then replaced with fresh water.

Control principles

The humidifier can be configured to ensure the desired humidity or temperature value. It can be controlled in the following ways:

4.6.2 ON/OFF control

Operation is all or nothing, and is activated by an external contact; the maximum percentage of unit production can be set.

4.6.3 Proportional to an external signal (modulating control)

Steam production is proportional to the value of an external signal Y (selectable by programming one of the following standards: 0-1Vdc; 0-10Vdc; 2-10Vdc; 0-20mA; 4-20mA). The maximum production Pmax corresponds to the maximum value of the external signal Y and will be the humidifier's rated output. The activation hysteresis hy is settable by the user and refers to the external signal Y.

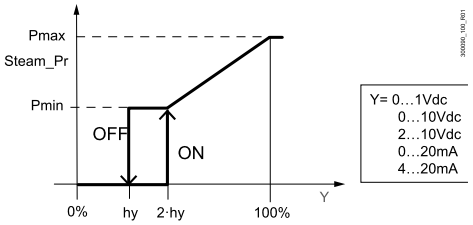


Fig.4.f

Key:

Steam_pr	Steam production	Y	External signal
Pmax	Max production	hy	Activation hysteresis
Pmin	Min production		

Note: the above graph applies if the preheating function is disabled.

4.6.4 Autonomous control with humidity probes

The production of steam is related to the % rH reading made by the relative humidity probe, and increases as the value read deviates from the set point St. The maximum production Pmax is delivered when the relative humidity read by the probe is equal to the value of BP (proportional band) from the set point. The activation hysteresis hy is settable by the user.

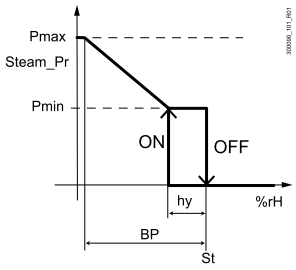


Fig.4.g

Key:

Steam_pr	Steam production	%rH	Relative humidity measured
Pmax	Max production	hy	Activation hysteresis
Pmin	Min production		

Tab.4.c

To verify that the relative humidity measured by the probe is within certain predefined values, two alarm thresholds can be set in autonomous control:

- high relative humidity;
- low relative humidity.

When these thresholds are exceeded, an alarm is activated and the corresponding relay contact is closed.

4.6.5 Autonomous control with relative humidity probe and outlet limit probe

In this case too, the controller modulates steam production based on the % rH measured by the main relative humidity probe, while limiting production if the humidity measured by a second limit probe, located in the air duct downstream of the steam distributor, approaches the maximum desired value. Consequently, to prevent the relative humidity of the outlet air from exceeding a value that is considered excessive, autonomous control with a limit probe can be set with a high relative humidity alarm threshold. When this threshold is exceeded, an alarm is activated and the corresponding relay contact is closed. The limit probe allows steam production to be modulated depending on the specific set limit differential.

4.6.6 Application for steam baths

In applications for steam baths, in which the control probe measures temperature rather than humidity, the same rules apply as for autonomous control with probe. By setting control based on temperature, the humidifier will continue to produce steam until reaching the desired set point temperature inside the steam bath, and consequently the desired saturation of the air (fog effect). Recommended CAREL transducer: ASET030000 or ASET030001 or NTC probe UEKNTC0*.

4.7 Steam production control signals

Depending on the type of signal used, steam production can be enabled and/or managed in different ways (ON/OFF or modulating).

Important: the probe inputs are protected against short-circuits, and the maximum current delivered (3U) is 50 mA. Despite this, it is recommended to configure the "Control type" before connecting the probes to the terminals.

To help users configure the unit, a screen index is shown at the top right of the display. The screen index corresponds to the sequence in each menu in order to reach the specific page.

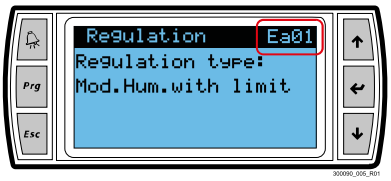


Fig.4.h

Steam production can be started by:

HUMIDISTAT (ON/OFF operation):

- connect terminals 1U and 2U (production request) to a humidistat;
- short-circuit terminals 7U and 8U (jumper) to enable production;
- to enable ON/OFF operation, set:

Index	Parameter	Description
Ea01	Type of control	External ON/OFF signal
Ea04	Maximum production	25-100% of rated production

Tab.4.d

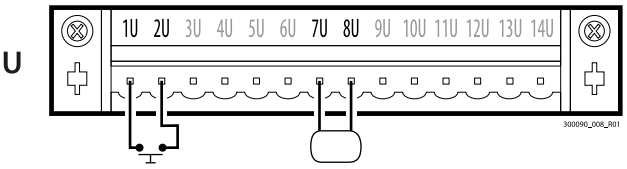


Fig.4.i

HUMIDISTAT and REMOTE CONTACT (ON/OFF operation)

- connect terminals 1U and 2U (production request) to a humidistat;
- connect inputs 7U and 8U (enable) to a remote contact (e.g.: switch, timer,...);
- to enable ON/OFF operation, set:

Index	Parameter	Description
Ea01	Type of control	External ON/OFF signal
Ea04	Maximum production	25-100% of rated production

Tab.4.e

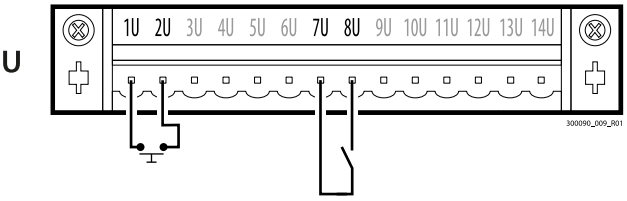


Fig.4.j

Steam production can be enabled and controlled by:

EXTERNAL PROPORTIONAL CONTROLLER (modulating operation)

- short-circuit terminals 7U and 8U (jumper) to enable production;
- connect terminals 1U and 2U (production request) to an external controller;
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	proportional to external signal
Ea03	Proportional band	Set: Hysteresis (0-100%) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.f

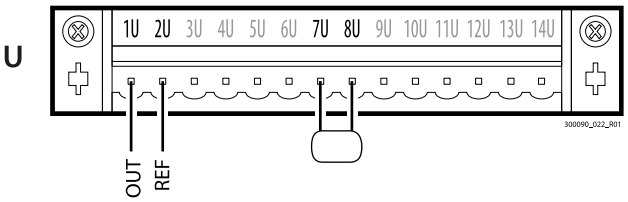


Fig.4.k

EXTERNAL PROPORTIONAL CONTROLLER and REMOTE CONTACT (modulating operation)

- connect terminals 1U and 2U (production request) to a humidistat;
- connect inputs 7U and 8U (enable) to a remote contact (e.g.: switch, timer,...);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	proportional to external signal
Ea03	Proportional band	Set: Hysteresis (0-100%) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.g

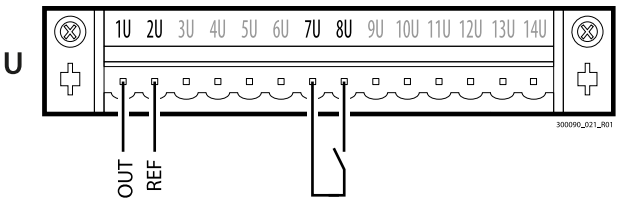


Fig.4.l

EXTERNAL PROPORTIONAL CONTROLLER and REMOTE CONTACT (modulating operation) with LIMIT PROBE

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect terminals 1U and 2U (production request) to an external controller;
- connect the active limit probe to terminals 5U, 3U (+12Vdc), 6U (GND);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Proportional to external signal with limit probe
Ea03	Proportional band	Set: Hysteresis (0-100%) Minimum production (25-100%) Maximum production (25-100%)
Ea06	Limit probe	Set: Set point (0-100 %rH)

Index	Parameter	Description
		Differential (0-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA
Ec02	Limit probe type	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.h

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "G" on the board rather than terminal 3U.

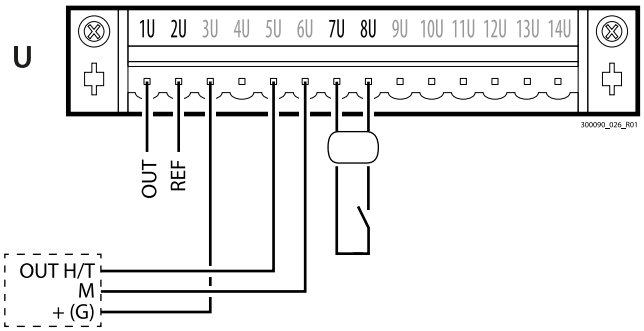


Fig.4.m

Note: in industrial environments (IEC EN61000-6-2) the signal cables leaving the unit must not exceed 30 m (98') in length: steam production signal cable (terminals 1U, 2U), digital remote on/off input (terminals 7U, 8U) and shielded cable for RS485 communication.

4.8 Control with humidity probes

The main control board, connected to a room humidity probe, manages steam production based on the humidity measured. A second outlet humidity limit probe can also be connected: with this configuration, typical of air handling units, the controller continues managing steam production according to the humidity requirement, however production is limited according to the relative humidity measured in the outlet duct.

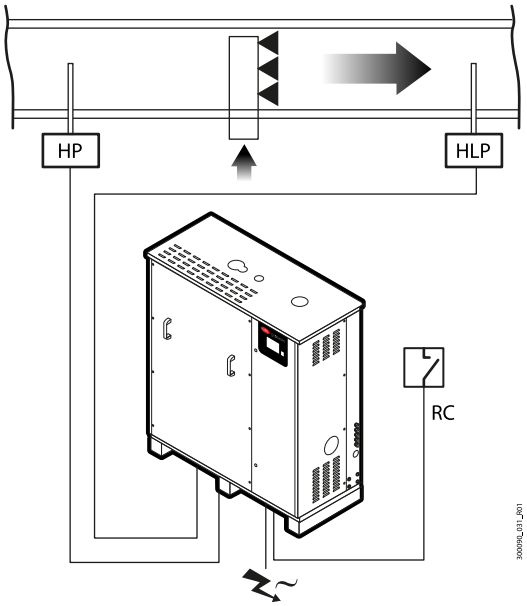


Fig.4.n

Key:

HP	Humidity control probe (intake/room humidity probe)
HLP	Limit humidity probe (outlet humidity probe)
RC	Remote contact

Tab.4.i

CONTROL WITH ONE HUMIDITY PROBE

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	one humidity probe
Ea05	Modulating control	Set: humidity set point (0-100 %rH) differential (2-20 %rh) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.j

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the terminal block "T" inside the electrical panel, rather than terminal 3U.

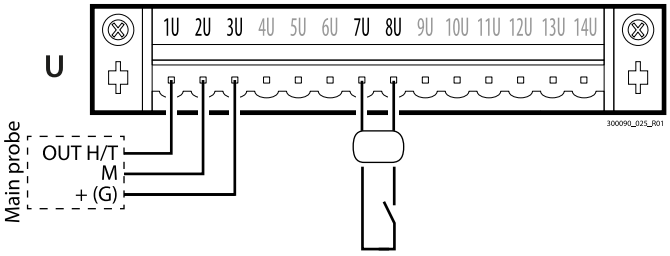


Fig.4.o

CONTROL WITH ONE HUMIDITY PROBE AND LIMIT PROBE

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- connect the active limit probe to terminals 5U, 3U (+12Vdc), 6U (GND);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Humidity with limit probe
Ea05	Modulating control	Set: Humidity set point (0-100 %rH) Differential (2-20 %rh) Minimum production (25-100%) Maximum production (25-100%)
Ea06	Limit probe	Set point (0-100 %rH) Differential (0-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA
Ec02	Limit probe type	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.k

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the

terminal block "T" inside the electrical panel, rather than terminal 3U.

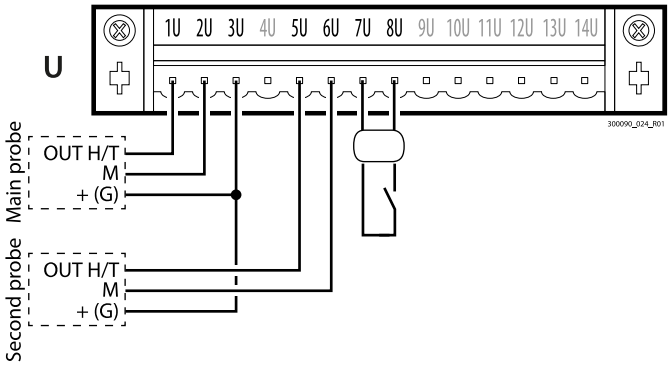


Fig.4.p

The following probes can be connected:

CAREL probes for rooms	DPWC111000
for air ducts	DPDC110000, DPDC210000
for industrial applications	DPPC210000

Tab.4.l

Note: third-party active probes can also be connected to the controller.

CONTROL WITH TWO HUMIDITY PROBES

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- connect the second active probe to terminals 5U, 3U (+12Vdc) and 6U (GND);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Modulation with two humidity probes
Ea02	Control with two probes	Set the weight of the two probes (0-100%)
Ea05	Modulating control	Set: humidity set point (0-100 %rH) differential (2-20 %rh) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA
Ec02	Second probe type	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.m

The controller will calculate the weighted average between the two probes. The weight of the two probes can also be set.

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the terminal block "T" inside the electrical panel, rather than terminal 3U.

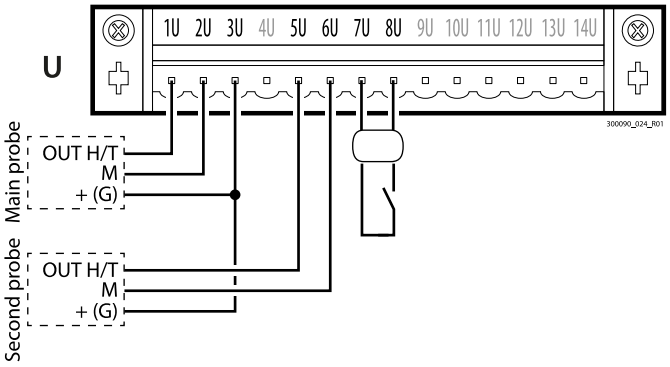


Fig.4.q

The following probes can be connected:

CAREL probes for rooms	DPWC111000
for air ducts	DPDC110000, DPDC210000
for industrial applications	DPPC210000

Tab.4.n

Note: third-party active probes can also be connected to the controller.

4.9 Control with temperature probes

The controller also allows autonomous temperature control, and can be connected to a temperature probe TT. It performs complete control based on the temperature measured inside the controlled environment.

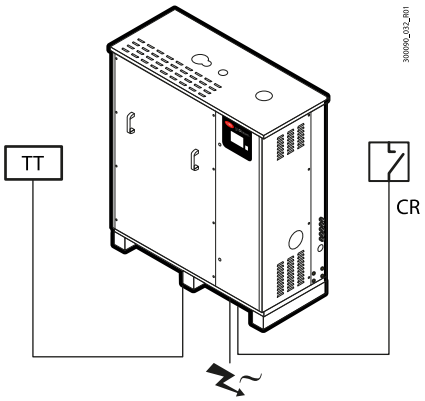


Fig.4.r

Key:

TT	Active temperature probe
RC	Remote contact

Tab.4.o

CONTROL WITH ONE ACTIVE TEMPERATURE PROBE

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	One temperature probe
Ea05	Modulating control	Set: temperature set point (0-100 °C) (32-212°F) differential (2-20°C) (3.6-36°F) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.p

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the terminal block "T" inside the electrical panel, rather than terminal 3U.

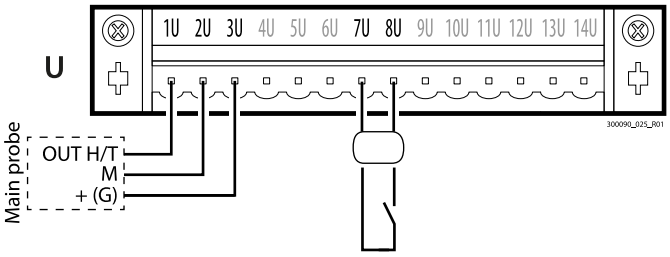


Fig.4.s

CONTROL WITH ONE TEMPERATURE PROBE AND LIMIT PROBE

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- connect the active limit probe to terminals 5U, 3U (+12Vdc), 6U (GND);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Temperature with limit
Ea05	Modulating control	Set: temperature set point (0-100 °C) (32-212°F) differential (2-20°C) (3.6-36°F) Minimum production (25-100%) Maximum production (25-100%)
Ea06	Limit probe	Set point (0-100°C/°F) Differential (0-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA
Ec02	Limit probe type	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.q

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the terminal block "T" inside the electrical panel, rather than terminal 3U.

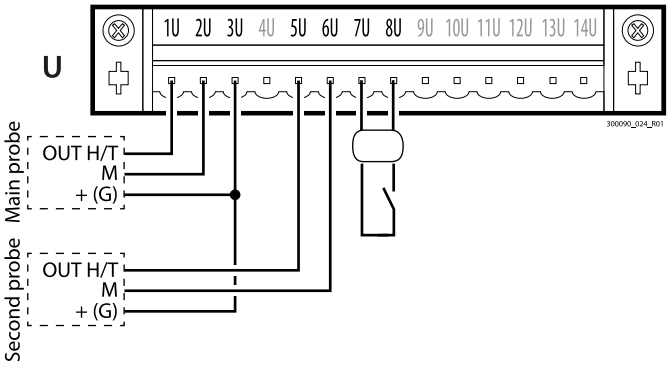


Fig.4.t

The following probes can be connected:

CAREL probes for rooms	DPWC111000
for air ducts	DPDC110000, DPDC210000
for industrial applications	DPPC210000

Tab.4.r

Note: third-party active probes can also be connected to the controller.

CONTROL WITH TWO ACTIVE TEMPERATURE PROBES

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main active room probe to terminals 1U, 2U (GND) and 3U (+12Vdc);
- connect the active limit probe to terminals 5U, 3U (+12Vdc) and 6U (GND);
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Temperature (two probes)
Ea02	Control with two probes	Set the weight of the two probes (0-100%)
Ea05	Modulating control	Set: temperature set point (0-100°C) (32-212°F) differential (2-20°C) (3.6-36°F) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA
Ec02	Second probe type	Select from: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA

Tab.4.s

The controller will calculate the weighted average between the two probes. The weight of the two probes can also be set.

Note: when using Carel 0-10V probes, connect the probe power supply +(G) to terminal "14" on the terminal block "T" inside the electrical panel, rather than terminal 3U.

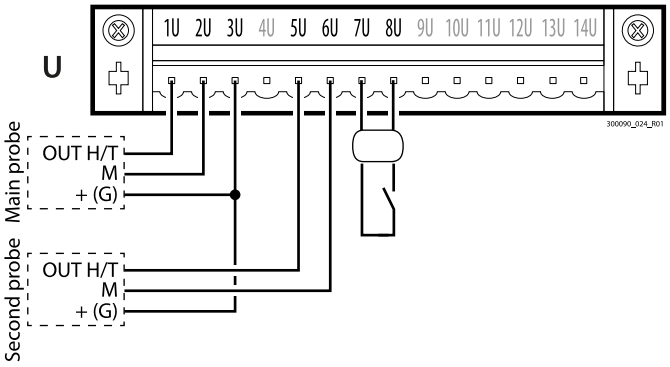


Fig.4.u

The following probes can be connected:

CAREL probes for rooms	DPWC111000
for air ducts	DPDC110000, DPDC210000
for industrial applications	DPPC210000

Tab.4.t

Note: third-party active probes can also be connected to the controller.

CONTROL WITH ONE NTC TEMPERATURE PROBE (passive)

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);
- connect the main NTC room probe to terminals 1U and 2U;
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	one temperature probe
Ea05	Modulating control	Set: temperature set point (0-100 °C) (32-212°F) differential (2-20°C) (3.6-36°F) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Set probe type: NTC

Tab.4.u

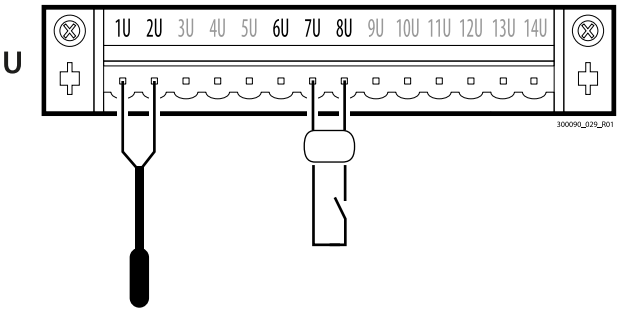


Fig.4.v

CONTROL WITH TWO NTC TEMPERATURE PROBES (passive)

- short-circuit terminals 7U and 8U (jumper) to enable production; alternatively connect terminals 7U – 8U to a remote contact (e.g.: switch, timer,...);

- connect the first NTC probe to terminals 1U and 2U;
- connect the second NTC probe to terminals 5U and 6U;
- to enable control, set:

Index	Parameter	Description
Ea01	Type of control	Modulation with two temperature probes
Ea02	Control with two probes	Set the weight of the two probes (0-100%)
Ea05	Modulating control	Set: temperature set point (0-100 °C) (32-212°F) differential (2-20°C) (3.6-36°F) Minimum production (25-100%) Maximum production (25-100%)
Ec01	Type of main probe	Set probe type: NTC
Ec02	Second probe type	Set probe type: NTC

Tab.4.v

The controller will calculate the weighted average between the two probes. The weight of the two probes can also be set.

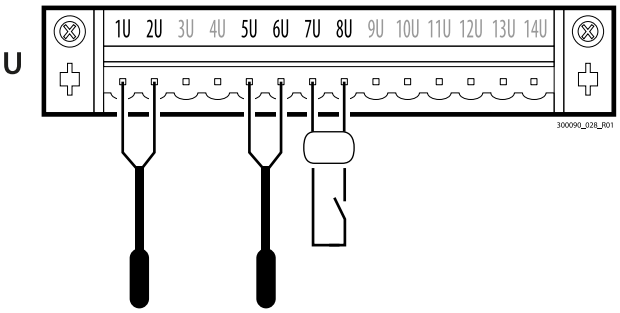


Fig.4.w

4.10 Alarm contact

The humidifier controller is fitted with a relay contact for remote signalling of one or more faults or alarms. The alarm contact (250 Vac; max. rating: 2 A resistive - 2 A inductive) is connected using terminals 7O, 8O and 9O.

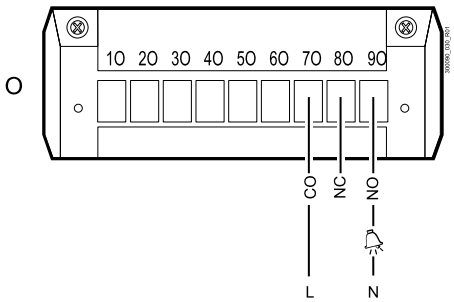


Fig.4.x

7O	CO - Common
8O	NC - Normally closed
9O	NO - Normally open

Tab.4.w

4.11 Production request analogue output

The humidifier controller is fitted with an analogue output (0-10 V signal) that reflects the production request. The production request output (0-10 V max 10 mA) is connected to terminals 11U, 12U.

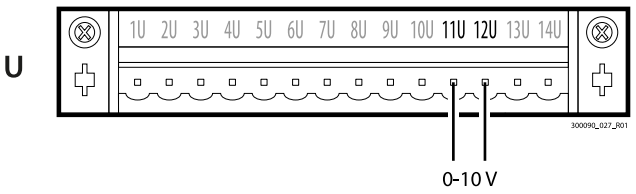


Fig.4.y

11U	Production request analogue output
12U	GND

Tab.4.x

⚠ **Important:** to avoid unbalanced control, the earth of the probes or external controllers must be connected to the unit controller's earth.

4.12 Remote installation of the touch display

To install the 4.3" touch screen in a position away from the unit, use the kit HCTXRC0000. The kit comprises a touch display, a 24 Vdc power supply, a telephone cable and a telephone splitter for simultaneous connection of the two displays (one in the remote position and the other installed on the humidifier).

4.13 Final checks

When installation is complete, verify that:

1. mains power to the humidifier corresponds to the voltage shown on the rating plate;
2. the fuses installed are suitable for the line and power voltage;
3. a mains disconnect switch has been installed so as to be able to disconnect power to the humidifier;
4. the humidifier has been correctly earthed;
5. the power cable is secured to the tear-resistant cable gland;
6. terminals 7U, 8U are jumpered or connected to an enabling contact;
7. if the humidifier is controlled by an external controller, the signal earth is electrically connected to the controller earth.

5. Preparing for operation

5.1 Preliminary checks

Before starting the humidifier, check that:

- the water and electrical connections have been completed and the steam distribution system configured according to the instructions contained in this manual;
- the humidifier water shut-off valve is open;
- the power fuses are installed and intact;
- terminals 7U and 8U are jumpered or connected to the remote ON/OFF contact, and that the latter is closed;
- the probes or the external control device are correctly connected (and that the earth of these devices is electrically connected to the earth of the main control board);
- the steam hose is not choked;
- in the event of ducted humidification, operation of the humidifier is slaved to the operation of the air fan (replacing or in series with the remote ON/OFF contact);
- the condensate return pipe from the blower has been installed and is not obstructed;
- the drain pipe is connected correctly and unobstructed.

⚠ **Important:** before starting, check that the humidifier is in perfect condition, that there are no water leaks and that the electrical parts are dry. Do not connect power if the appliance is damaged or even partially wet!

5.2 Different types of gas supply

⚠ **Important:** the appliances are delivered calibrated and tested to operate on natural gas (methane).

The humidifier can be supplied by the following types of gas:

- Natural gas (G20-G25 methane - factory setting);
- Propane-Butane (LPG G30-G31);

A number of parameters on the electronic controller need to be set for correct operation, calibrating combustion by checking the values of CO2 (% vol) and CO (ppm) in the flue gas:

NATURAL GAS/LPG CALIBRATION

	no. turns											
	UG45		**UG90**		**UG150**		**UG180**		**UG300**		**UG450**	
Gas	**min**	**max**	**min**	**max**	**min**	**max**	**min**	**max**	**min**	**max**	**min**	**max**
G20-G25	1450	4700	1700	5050	1900	6750	2000	5300	1900	6750	1900	6750
G30-G31	1650	4200	1850	4450	2000	5150	2000	5150	-	-	-	-

Tab.5.a

5.2.1 Gas burner calibration

The burner is pre-calibrated in the testing phase by the manufacturer; nonetheless, combustion should be checked and adjusted if necessary.

5.2.2 Preparing to analyse combustion

If the flue gas exhaust pipe is positioned horizontally or vertically:

1. remove the cap from the inspection section of the humidifier flue gas exhaust pipe;
2. insert the flue gas analyser probe;
3. perform the flue gas analysis.

Once analysis has been completed, reposition the cap on the inspection section.

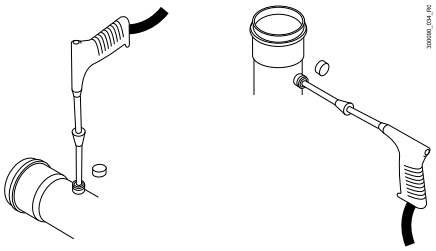


Fig.5.a

5.2.3 Type of calibration

Two different types of burner calibration are available:

A. Guided;

B. Manual.

The guided procedure automatically provides the user, step by step, with all the information necessary to complete the calibration process. For the manual procedure, follow the instructions below.

5.2.4 Burner calibration at maximum output

Force the burner to operate at maximum output, as shown in the table, by setting the fan to the maximum speed and verifying by flue gas analysis that the CO2 value corresponds to the table.

If the values measured are different, proceed as follows:

1. open the front door;
2. adjust the gas flow regulator B (figure) using a flat-head screwdriver or a 4 mm spanner: turn anticlockwise to increase the CO2 value, clockwise to decrease it;
3. wait until the CO2 value stabilises, then check again and, if necessary, repeat the operations.

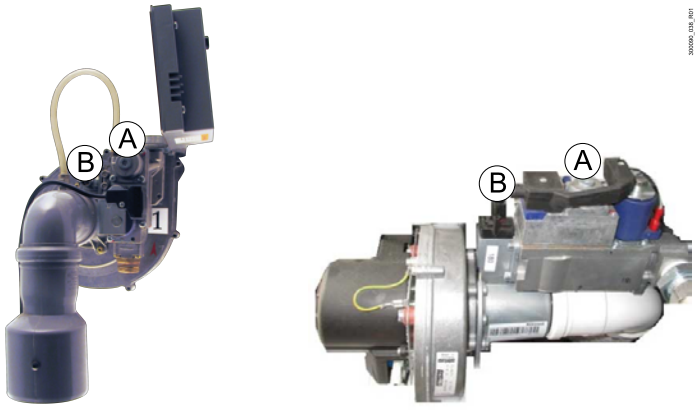


Fig.5.b

	CO2 (%)	
Gas	**min**	**max**
G20-G25	8.5 ±0.3	9.4 ±0.2
G30-G31	9.9 ±0.3	10.5 ±0.3

Tab.5.b

5.2.5 Burner calibration at minimum output

Force the burner to operate at minimum output, by setting the fan to the minimum speed as described previously and verifying by flue gas analysis that the CO2 value corresponds to the table.

If the values measured are different:

1. open the front door;
2. remove the gas valve pressure regulator cap and adjust the pressure regulator A (see the figure) using a no. 40 Torx screwdriver: turn clockwise to increase the CO2 value, anticlockwise to decrease it (apply small adjustments, as the adjustment screw is very sensitive);
3. wait until the CO2 value stabilises: if necessary, repeat the operations.
4. replace the cap on the regulator;

⚠ **Important:** once the calibration at maximum output has been performed, check the minimum calibration again, as it may have been affected by the maximum calibration; if necessary, repeat the operations described in Burner calibration at MAXIMUM output. Then reset the burner to automatic operation.

To check the correct position of the burner electrodes, see "Cleaning the burner".

⚠ **Important:** make sure that the CO values in steady operation do not exceed 100 ppm.

PROPANE/BUTANE/LPG CALIBRATION

The following units only can also work on propane, butane and LPG: UG45, UG90 and UG180 (UG150, UG300 and UG450 only work on natural gas). Do not ignite the burner (factory calibrated for natural gas!) with propane or LPG until turning the screw (B) approximately 2.5-3 turns clockwise to adjust maximum gas flow. In fact, the heat power of propane is about three times that of natural gas (methane), therefore it is essential to first reduce the maximum gas flow before ignition. After having reduced the gas flow rate, ignite the burner and subsequently calibrate the gas flow rate by analysing the flue gas, as reported in the previous paragraphs.

6. Start-up and user interface

6.1 Start-up

After having connected power using the main disconnect switch (ON), switch the appliance on by moving the switch on the front panel to "ON". The activation sequence will start, which includes an initial phase, an autotest phase and finally actual operation. Each step in the activation sequence is identified by a different display.

6.2 Activation sequence

1. STARTING THE APPLICATION PROGRAM THE FIRST TIME

The "gaSteam" humidifier logo is displayed. When starting the first time, the menu language needs to be selected:

English

Italiano

Deutsch

Français

Español

Select the desired language and confirm. This screen is displayed for 60 seconds.

2. FIRST START-UP (WIZARD)

When starting the first time, a wizard is provided to quickly set the main unit parameters. The wizard comprises ten steps (some of which may not be displayed, if not necessary):

1/10 - model (only if the replacement controller is not configured): select and set the parameters relating to the model (size, voltage,…);

2/10 - water hardness: auto or user-defined. If choosing "auto", the controller automatically sets the water hardness, based on feedwater conductivity;

3/10 - manual water hardness setting. The possible options are:

- Demineralised, maintenance at 3000h (hours)
- 0-10°f, maintenance at 3000h (hours)
- 10-20°f, maintenance at 1500h (hours)
- 20-30°f, maintenance at 1000h (hours)
- 30-40°f, maintenance at 800h (hours);

4/10 - control type: external On/Off signal, proportional to external signal with limit probe, proportional to external signal, one humidity probe, one temperature probe, one humidity probe and limit probe, one temperature probe and limit probe, two temperature probes (average), two humidity probes (average);

5/10 – select the main room probe type: 0-1V (active), 0-10V (active), 2-10V (active), 0-20 mA (active), 4-20 (active), NTC (passive);

6/10 – select the limit probe type: 0-1V (active), 0-10V (active), 2-10V (active), 0-20 mA (active), 4-20 (active), NTC (passive);

7/10 - set the limits for active probes.

Room min (%): set the minimum relative humidity limit rH% for the main probe;

Room max (%): set the maximum relative humidity limit rH% for the main probe;

Limit min (%): set the minimum relative humidity limit rH% for the limit probe;

Limit max (%): set the maximum relative humidity limit rH% for the limit probe;

8/10 - set the drain to dilute cycles: auto or user-defined. If choosing "auto", the controller automatically sets the number of evaporation cycles that must occur between two consecutive drain to dilute cycles. This setting is based on the feedwater conductivity read by the conductivity meter, thus reducing the use of water, reducing maintenance and extending boiler life;

9/10 - manual drain to dilute cycle setting. Enter the number of evaporation cycles before forcing a dilution cycle.

10/10 - date and time setting.

At the end of the guided procedure, a message prompts whether to show the wizard again when next starting? Yes/No

3. AUTOTEST PROCEDURE

Indicated on the display by the unit status shown as "AUTOTEST".

Whenever the humidifier is started (switch moved from OFF to ON), an autotest procedure is run by default to check operation of the level sensor and the appliance as a whole. The autotest procedure involves a water fill cycle to above the high level (green LED), followed by a drain cycle until below the minimum level (red LED). The procedure then refills the unit with water in order to restart production (if required).

Note: all of the screens in the wizard (except for the language selection screen) remain on the display until the user enters the settings.

4. OPERATION

The humidifier starts operating and the standard display is shown. If an alarm is active, the corresponding icon (bell) comes on red, see the chapter "Alarms" for the complete list and description.

Note: model selection and configuration (only for non-configured replacement controllers). If needing to replace the controller only (controller replacement part number) UGKA0D0040), the first time the controller is powered on, the model will need to be set (capacity, power supply and version).

6.3 Shutdown

- To avoid stagnation, drain the water from the boiler using the manual procedure.
- move the switch to 0, "OFF".

6.4 Graphic terminal

The 4.3" touch graphic terminal has a graphic interface with coloured and animated icons. The contents of the display can be scrolled up and down simply and intuitively.



Fig.6.a- home page

6.5 Touch Display

"HOME" menu

The "HOME" menu comprises information on the probes, the set point and external request signal.

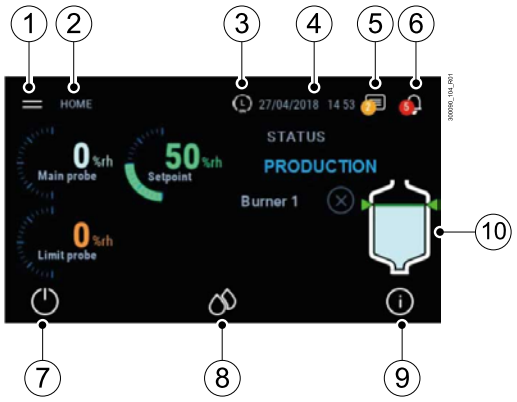


Fig.6.b

Position	Function
1	System menu

Position	Function
2	home
3	Scheduler settings
4	System date and time
5	Notification centre
6	Alarm list
7	ON/OFF
8	Set point setting
9	System information
10	Descriptive icon of humidifier status

Tab.6.a

6.5.1 System menu

The System menu provides access to settings that are available without entering a password.

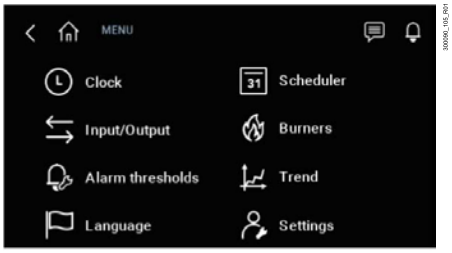


Fig.6.c- system menu

Menu	Description
Clock	Date and time setting
Input/Output	Display the analogue and digital inputs/outputs
Graphs	Display historical and real-time operation of the humidifier
Functions	Special and manual functions
Scheduler	Manage scheduling of working time bands
Alarm threshold	Set the alarm thresholds
Language	Set the menu language
Settings	Access advanced humidifier configuration (Installer password 77). Menu: E. Settings. Change unit of measure (Imperial/International)

Tab.6.b

6.5.2 Scheduler settings

Activate and set the time bands for humidifier activation. After enabling the scheduler, six on/off time bands can be set for each day. Use the copy button to copy the set time bands from one day to the next.

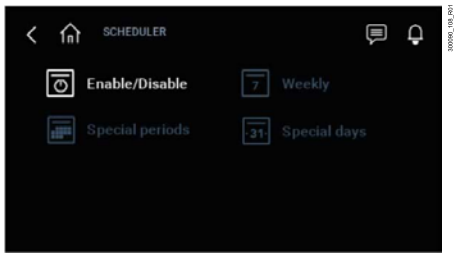


Fig.6.d

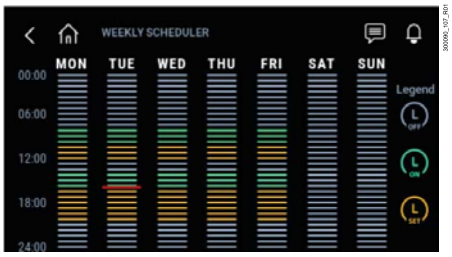


Fig.6.e

6.5.3 Notification centre

The main messages regarding the activities performed by the humidifier can be quickly displayed in the notification centre. The main screen shows whether any notifications are present, by indicating the number of active notifications. The list of messages that can be viewed in the notification centre is shown below:

Message	Description	Type of notification
Maintenance required in xxx hours. boiler lifetime hours: yyy hours	The unit will require routine maintenance	System
The unit has performed x drain cycles due to the presence of foam	The unit has performed a drain cycle due to foam detection	System
The maximum production of the Master/Slave system is lower than the value set. Set: xxxx Current: yyyy	The maximum production of the Master/Slave system is lower than the value set.	System
The unit has restarted automatically after a shutdown. Was there a blackout?	The unit has restarted automatically after a shutdown. Was there a blackout?	User
Periodical drain cycle performed	The periodical drain was performed correctly	User
Factory reset	The factory parameters have been reset correctly	User

Tab.6.c

The system notifications cannot be reset by the user and will be automatically cancelled, user notifications however can be deleted.

6.5.4 ON/OFF

Switch the humidifier on and off

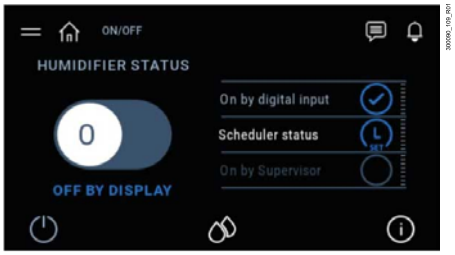


Fig.6.f

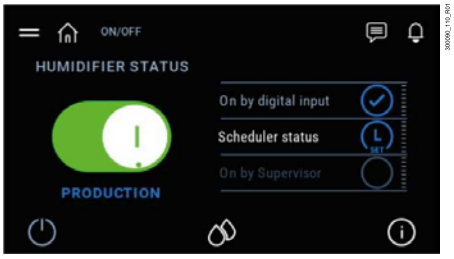


Fig.6.g

6.5.5 Set point setting

Set the set point, proportional band and maximum production



Fig.6.h

6.5.6 System information

Menu providing humidifier status, software and hardware information.

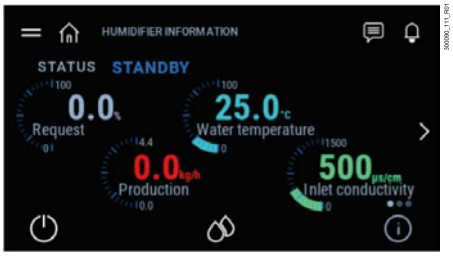
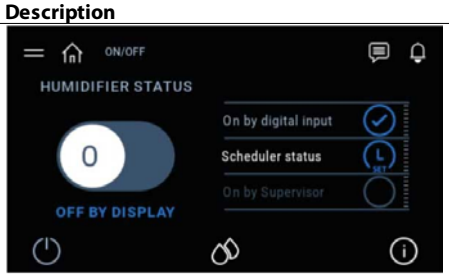


Fig.6.i

Unit status may be as described below:

Unit status	Description
Standby	unit in standby and ready to operate
Production	the unit is producing steam
Alarm	at least one alarm is active. Display the alarm by pressing the button
Off from BMS	steam production disabled by the supervisor
Off by scheduler	steam production disabled due to set time band
Off by Din	production of steam stopped due to opening of the "remote ON/OFF" contact;
Off by keyboard	the unit has been switched off from the keypad

Unit status	Description
	
Ready for backup	the unit is ready and awaiting to start operation if there is a fault on the main unit.
Manual mode	test mode for commissioning and to check functions (for example: activate drain pump, activate fill valve...)
Warning	warning notification
Preheating	the unit is preheating the water in the boiler
Startup preheat.	at start-up, the unit heats the water to reach the production request
Autotest	the unit is running the autotest
Inactivity drain	water drain due to inactivity or periodically, the drain pump is operating

Tab.6.d

6.5.7 Descriptive icon of humidifier status

Graphic display of unit status via animated icons. The status may be:

Fill (fill valve active)
Drain (drain pump active)
Steam production
Minimum water level in the boiler (yellow and red LEDs on);
Water level above the maximum in the boiler (green LED on);

6.6 Complete programming tree

Below is the complete tree of the settings menu. The screen index at the top right corresponds to the sequence in each menu in order to reach the specific page. There are two levels of access: Installer and Service. To login to the settings menu, use the following passwords:

- service: 0044;
- installer: 0077.

Menu A. Clock

Index	Description	Level
A01	Set the time, date and time zone	Installer Service

Menu B. Time bands (scheduler)

Index	Description	Level
B01	Enable time bands	Installer Service
B02	(visible if time bands are enabled) Set the time bands: day, on time, off time	Installer Service

Menu C. Alarm thresholds

Index	Description	Level
C01	Set alarm thresholds Low humidity/temperature alarm threshold High humidity/temperature alarm threshold Limit humidity/temperature alarm threshold	Installer Service

Menu D. Inputs/outputs

Menu	Index	Description	Level
Analogue inputs	D01	Values read by main probe, limit probe, boiler water temperature, feedwater conductivity	Installer, Service
Digital inputs	D02	Read status of remote On/Off, motor protector, level sensor float position, foam sensor	Installer, Service
Analogue outputs	D03	Read current production	Installer, Service
Digital outputs	D04	Read status of unit On/Off, contactor, drain pump, fill valve, SSR/fan	Installer, Service
In/Out - WPrb 1 (In/Out wireless probe 1)	D05	Read temperature and/or humidity values Read signal level and battery level on probe 1	Installer, Service
In/Out - WPrb 2 (In/Out wireless probe 2)	D06	Read temperature and/or humidity values Read signal level and battery level on probe 2	Installer, Service
In/Out - WPrb 3 (In/Out wireless probe 3)	D07	Read temperature and/or humidity values Read signal level and battery level on probe 3	Installer, Service
In/Out - WPrb 4 (In/Out wireless probe 4)	D08	Read temperature and/or humidity values Read signal level and battery level on probe 4	Installer, Service

Menu E. Settings(password)

Menu		Index	Description	Level
a. Control	Control type	Ea01	Set the type of control	Installer
	Weights	Ea02	Set the weight of the two probes	Installer
	Control proportional to external signal	Ea03	Set the hysteresis, minimum production and maximum production	Installer
	External ON/OFF	Ea04	Set the maximum production with external ON/OFF control	Installer
	Modulating control	Ea05	Set the set point, differential, min production, max production	Installer
	Integral function	Ea05a	Set the integral time and the neutral zone	Installer
	Limit probe	Ea06	Set the set point and differential	Installer
	Boiler operating hours	Ea07	Display boiler operating hours Reset hour counter and set maintenance pre-alert threshold	Service
	Burners rotation	Ea10	Set the burner activation sequence	Installer
b. Functions	Enable preheating	Eb02	Enable preheating Set the water temperature to maintain Set the offset	Installer
	Part fill cycles	Eb03	Enable part fill cycles Set part fill time	Installer
	Drain do to inactivity	Eb04	Enable drain due to inactivity Set the inactivity threshold	Installer
	Periodical drain	Eb05	Enable periodical drain cycles Set the threshold for the periodical drain cycles	Installer
	M5 Signal. M6 Signal	Eb07	Enable the output relay for unit status (production) or maintenance pre-alert	Installer
	Blower unit	Eb08	Set on and off delay for blower unit	Installer
	Export logs	Eb09	Save the log of main variables to USB pen drive	Service Installer
	Export Alarms	Eb10	Save alarm log to USB pen drive	Service Installer

Menu		Index	Description	Level
	Frost protection	Eb11	Set the frost protection function	Installer
c. Configuration	Main probe	Ec01	Set the main probe: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA, NTC	Installer
	Limit/second probe	Ec02	Set the limit probe: 0-1V, 0-10V, 2-10V, 0-20mA, 4-20mA, NTC	Installer
	Wireless probes	Ec03	Set the wireless probes (main/limit)	Installer
	Wireless probe 1	Ec04	Set the weight of probe 1 Set the communication times	Installer
	Wireless probe 2	Ec05	Set the weight of probe 2 Set the communication times	Installer
	Wireless probe 3	Ec06	Set the weight of probe 3 Set the communication times	Installer
	Wireless probe 4	Ec07	Set the weight of probe 4 Set the communication times	Installer
	Number of evaporation cycles before drain	Ec11	Set the number of evaporation cycles between two drain cycles	Installer
	Variation in fill and drain time	Ec12	Set the fill and drain time in relation to factory default	Installer
	Water hardness	Ec13	Hardness value	Installer
	Missing water warning retry time	Ec14	Set time to check for no water	Installer
	High conductivity	Ec15	Enable the high conductivity alarm Set the alarm delay	Installer
	High conductivity	Ec16	Set conductivity thresholds: warning, alarm and hysteresis	Installer
	Remote ON input logic	Ec22	Operating logic for remote ON/OFF	Installer
	Burner 1 settings	Ec23	Set pre-purge speed, minimum speed and maximum speed for production (Burner 1)	Installer
	Burner 2 settings	Ec24	Set pre-purge speed, minimum speed and maximum speed for production (Burner 2)	Installer
	Burner 3 settings	Ec25	Set pre-purge speed, minimum speed and maximum speed for production (Burner 3)	Installer
	Burners calibration	Ec28	Start the procedure to calibrate the burners.	Installer
d. Master/Slave	Master/Slave	Ed01	Initial Master/Slave system configuration Press "PRG" to configure the Master/Slave network	Installer
	Unit 1 Unit 2 . . . Unit 20	Ed02	Add unit to the Master/Slave system	Installer
	Maximum production of Master/Slave system. Unit rotation	Ed03	Set maximum production of the Master/Slave system Set the unit rotation logic	Installer
	Advanced preheating. Rotation time	Ed04	Enable the advanced preheating function for Master/Slave systems Set the rotation time between one unit and the next	Installer
	Offline timeout	Ed05	Set the unit offline time in the Master/Slave system	Installer
	Disconnect unit from Master/Slave system	Ed06	Disable/disconnect the current unit from the Master/Slave system	Installer
	Master/Slave production	Ed07	Display the request (%) and the production (kg/h / lbs/h) of the Master/Slave system Press "PRG" to configure the Master/Slave network	Installer

Menu		Index	Description	Level
	Display Master/ Slave system	Ed08	Unit status with relative production as a %. Press "PRG" to configure the Master/Slave network	Installer
	Display info on individual unit. Unit 1, Unit 2, ...,Unit 20	Ed09	Display unit operating hours, current production and any alarms To move between the various units use the UP and DOWN arrows	Installer
	Unit switched-off for maintenance	Ed10	Unit switched off to carry out maintenance	Installer
e. Backup	Enable backup	Ee01	Enable automatic unit backup in the event of shutdown, using a second independent unit	Installer
	Priority when starting	Ee02	Define the priority of the units when in backup mode	Installer
f. Manual mode	Manual mode	Ef01	Enable manual mode: Disabled, manual management of outputs and manual request	Service, Installer
	Manual management of outputs	Ef02	Test the outputs: fill valve, drain pump, ...	Service, Installer
	Manual request management	Ef03	Manually set the production request	Service, Installer
g. Initialisation	Wizard	Eg01	Start the wizard for initial unit configuration Set whether to display the wizard when next restarting	Service, Installer
	Language	Eg02	Set the language	Service, Installer
	Unit of measure and language when starting	Eg03	Set unit of measure (International or Imperial). Select the language when starting	Installer
	Change password	Eg04	Change the password (Service, Installer).	Service, Installer
	Inst. default	Eg06	Unit factory reset. Warning, when changing the model all the control settings will be lost and the values will return to the factory defaults	Service, Installer
	Software update	Eg07	Update the unit software from USB pen drive	Service, Installer
	Export parameters	Eg08	Export the unit configuration parameters to a USB pen drive	Service, Installer
	Import parameters	Eg09	Import configuration parameters from USB pen drive to the unit	Service, Installer
h. Supervisor	Unit supervision address on BMS port	Eh01	Set the unit address for supervision. Enable and set type of supervisor protocol	Installer
	BMS port configuration settings	Eh02	Set the communication parameters for supervision via BMS: Baud rate, stop bits and parity	Installer
	BACnet MS/TP configuration	Eh03	Set the address, maximum number of masters and maximum number of frames	Installer
	Device Inst.	Eh06	Device inst. configuration for BACnet protocol	Installer
	Ethernet network settings	Eh04	Set DHCP, IP address, mask, gateway, DNS for the Ethernet network. Important: these values must be provided by the local network administrator	Installer
	Supervsion settings on Ethernet port	Eh05	Select the protocol on the Ethernet port: ModBus or BACnet	Installer

Menu		Index	Description	Level
	Regulation from SV	Eh07	Select the port that the supervisor is connected to	Installer
i. Logout	Logout	Ei01	Information on the type of login performed. Possibility to logout	Service, Installer

Tab.6.e

7. User menu and unit configuration

The following paragraphs describe the gaSteam programming menus. The screen index at the top right corresponds to the sequence in each menu in order to reach the specific page.

7.1 Main menu

7.1.1 Clock menu

The Clock menu is used to set the time, date and time zone. Setting the time zone will automatically update standard/daylight saving time. To access, enter the system menu and then click the clock icon:

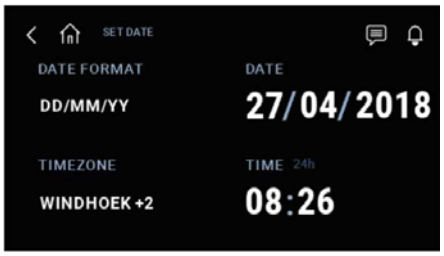


Fig.7.a

7.1.2 Inputs/outputs

The Inputs/outputs menu is used to read the status of the inputs and outputs and check unit operation and status.

From the system menu, enter the "Input/Output" menu:

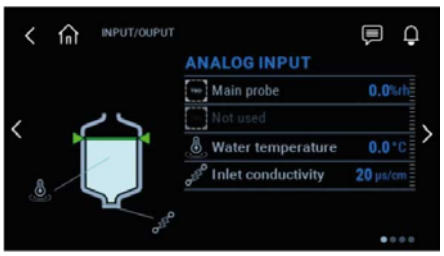


Fig.7.b

Parameter	Description
Analogue inputs	Values read by main probe, limit probe, boiler water temperature, feedwater conductivity.
Digital inputs	Read status of remote On/Off, level sensor float position, foam sensor.
Analogue outputs	Read current production
Digital outputs	Read status of unit On/Off, contactor, drain pump, fill valve, SSR/fan
In/Out - WPrb 1 In/Out wireless probe 1	Read temperature and/or humidity values Read value of level signal and battery level on probe 1
In/Out - WPrb 2 In/Out wireless probe 2	Read temperature and/or humidity values Read value of level signal and battery level on probe 2
In/Out - WPrb 3 In/Out wireless probe 3	Read temperature and/or humidity values Read value of level signal and battery level on probe 3
In/Out - WPrb 4 In/Out wireless probe 4	Read temperature and/or humidity values Read value of level signal and battery level on probe 4

7.1.3 Alarm thresholds

To check that the relative humidity measured by the probe transducer is within certain predefined values, two alarm thresholds can be configured:

- high relative humidity alarm threshold, for both the main probe and limit probe;
- low relative humidity alarm threshold for the main probe.

On exceeding these thresholds, an alarm is activated and the corresponding relay contact on the main control board is closed. Temperature thresholds can also be set.

From the system menu, enter the "alarm thresholds" menu:



Fig.7.c

Parameter	Description
Alarm thresholds	Set alarm thresholds
	Low humidity/temperature alarm threshold
	High humidity/temperature alarm threshold
	Limit humidity/temperature alarm threshold
	Default:
	low humidity/temperature: 0% rH / 0°C (32°F);
	high humidity/temperature: 100% rH / 100°C (° F);
	humidity/temperature limit: 100% rH / 100°C (° F);
	Range: 0-100

7.1.4 Scheduler menu

Parameter	Description
Scheduler	Enable time bands
	Default: time bands disabled

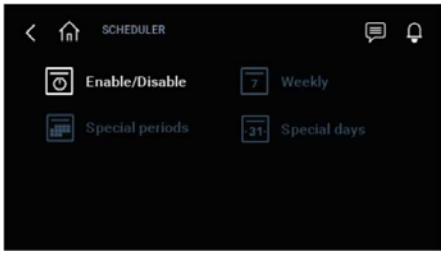


Fig.7.d

After having enabled the scheduler in the corresponding menu, the humidifier operating intervals can be set for one day (24h) and for the whole week. A time band can be set with production disabled (OFF), enabled (ON), or enabled with s specific set point (ON+SET). Setting the ON time band, the unit uses the main set point configured. If setting ON+SET, the specific set point for the time band can be configured.

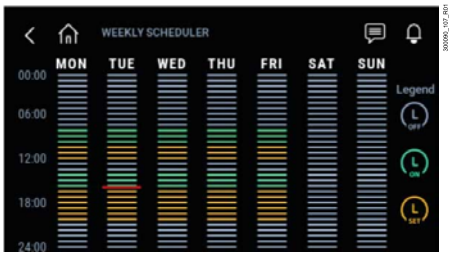


Fig.7.e

Symbol	Unit of measure
%	%rH
°C	Degrees Celsius
°F	Degrees Fahrenheit

Tab.7.a

Note: if control is by external signal, the humidifier can only be switched ON or OFF.

The "Special period" function can be used to set operation (or switch off) for a specific period of time (from day x to day y).



Fig.7.f

The "Special day" function can be used to set operation (or switch off) on a specific day.

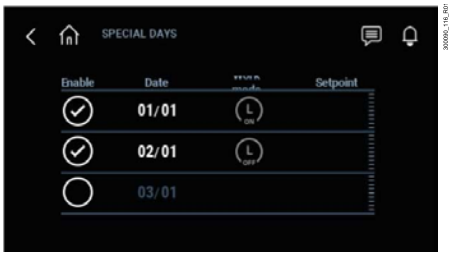


Fig.7.g

7.2 Menu E. Settings - a. Control

Login with password is required to access the settings menus:

- service: 0044;
- installer: 0077.

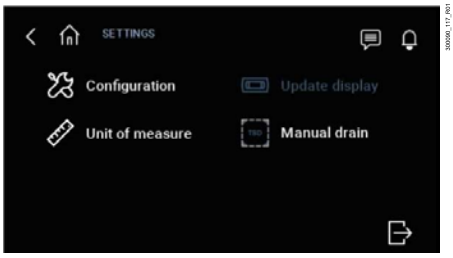


Fig.7.h

from the system menu, enter the "Settings" menu and enter the password. Then click the "Configuration" icon:

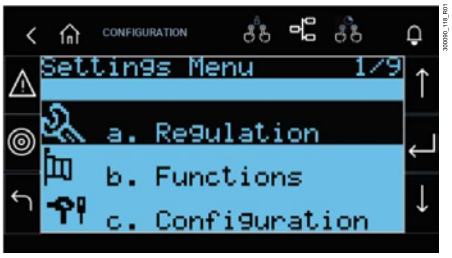


Fig.7.i

The "Settings" menu also has functions to:

- set the units of measure;
- manually drain the water in the boiler.

7.2.1 Control type (Installer menu)

Index	Parameter	Description
Ea01	Control type	Set the type of control Default: humidity (one probe) Possible settings: proportional to external signal, proportional to external signal with limit probe, On/Off signal, humidity (one probe), temperature (one probe), humidity with limit, temperature with limit, humidity (two probes), temperature (two probes)

The possible settings are:

- **proportional to external signal**: proportional control with signal from an external controller;
- **proportional to external signal with limit probe**: proportional control with signal from an external controller plus limit probe;
- **On/Off signal**: humidity control with humidistat;
- **humidity (one probe)**: humidity control with main probe;
- **temperature (one probe)**: temperature control with main probe;
- **temperature with limit**: temperature control with main probe and limit probe;
- **temperature with limit**: temperature control with main probe and limit probe;
- **humidity (two probes)**: humidity control with two probes, the controller calculates the weighted average of the two values read;
- **temperature (two probes)**: temperature control with two probes, the controller calculates the weighted average of the two values read.

For "humidity (one probe)" or "temperature (one probe)" control, one single main probe can be connected and configured, either wired or wireless.

For "humidity with limit" or "temperature with limit" control, a wired probe can be connected as the main probe and a wired probe as the limit. If using wireless probes (maximum of four), two groups of probes can be defined: the group of main probes and the group of limit probes. In this case, the average

will be calculated between the main probes, depending on the defined weight, and the limit probes will also have their own average, again depending on the defined weight.

For "humidity (two probes)" or "temperature (two probes)" control, only a group of main probes can be defined. Wired probes can be connected to the main probe input (1U) and the limit probe input (5U), which will be used as a second probe, with the average calculated. If using wireless probes (maximum of four), only a group of main probes can be defined, with the average calculated, depending on the defined weight.

For the connections of the signals and/or probes, see chap. "Electrical connections".

7.2.2 Weighted average of the probes (Installer menu)

If using two temperature probes or two humidity probes, the humidifier controller will calculate the weighted average of the probe readings. In this way, two probes can be used, for example humidity probes, at opposite ends of the room, calculating the average.

Index	Parameter	Description
Ea02	Weights	Set the weight of the probes
		Default: 100
		Range: 0-100
		Step: 1

The weight of each probe is expressed with a value from 0 to 100.

The weighted average is calculated as follows:

Weighted average = ((S1×p1)+(S2×p2)) / (p1+p2)

where "Si" is the probe reading, and "pi" the relative weight.

To calculate the arithmetic average, equal weights should be set (for example: p1 = p2 = 100).

If using wireless probes, the weights of each device can also be defined; in this case a maximum of four wireless probes can be connected, and consequently the formula shown previously for calculating the weighted average is extended to four devices.

7.2.3 Proportional control configuration

If using control proportional to external signal or control proportional to external signal with limit probe, the hysteresis, minimum and maximum production need to be set.

Index	Parameter	Description
Ea03	Control proportional to external signal	Set the hysteresis, min production and max production
		Default:
		Hysteresis = 5%
		Minimum production = 25%
		Maximum production = 100%
		Possible settings:
		Hysteresis = 0.5-100%
		Minimum production = 0-10%
		Maximum production = 0-100%

7.2.4 Control from external ON/OFF signal configuration

If using control with On/Off signal, the maximum production needs to be set.

Index	Parameter	Description
Ea04	External ON/OFF	Set the maximum production with external ON/OFF control
		Default: 100%
		Range: 0-100%

7.2.5 Modulation configuration (Installer menu)

If using modulating control, the related parameters need to be set:

Index	Parameter	Description
Ea05	Modulating control	Set the set point, differential, min production, max production
		Default: Set point = 50%rH (42°C) (107.6°F) Differential = 5%rH (5°C) (9°F) Minimum production = 25% Maximum production = 100%
		Range: 0-100

7.2.6 Integral function in probe control

If using a probe that is connected directly to the humidifier (control: humidity probe), the Integral (I) control function can be selected. This means the humidity level over time can be considered, bringing the value to the set point even when the proportional action (P) alone is null.

To activate the Integral function, set humidity control (single probe) on screen [Ea01]; also adjust the proportional band on screen [Ea05] (for example, by setting it to a value of 50%). The proportional band has to be at least 10% or higher, so that screen Ea05a will be visible. On screen [Ea05a], two parameters can be set, "integral time" and "neutral zone".

Index	Parameter	Description
Ea05a	Integral time	Integral time setting
		Default: 120 s
		Minimum: 0 sec (integral function disabled)
		Maximum: 300 s
	Neutral zone	Integral neutral zone setting, inside which gain remains constant
		Default: 2.5%
		Minimum: 0%
		Maximum: 20%

7.2.7 Limit probe configuration (Installer menu)

A second probe can be connected as a limit probe at the outlet. This probe has the purpose of preventing the relative humidity downstream of the steam distributor from exceeding a set value, configured by the user. As this probe has a modulating action, the differential can also be set. The limit probe, within its range of activation, has priority over the main probe (as the limit probe set point is higher than the main probe set point).

Index	Parameter	Description
Ea06	Limit probe	Limit probe set point and differential
		Default: Set point = 100% rH Differential = 5%
		Range: 0-100

7.2.8 Boiler operating hours (Service menu)

The "Boiler op. hours" screen displays the effective number of boiler working hours.

Index	Parameter	Description
Ea07	Boiler operating hours	Display boiler operating hours, reset hour counter and set maintenance pre-alert threshold
		Default: Pre-alert = 240 hours
		Range: 0-999

If the boiler needs to be replaced, the hour counter must be reset using the "Reset" parameter; the hour counter will then start again from zero.

The "Pre-alert" item sets the maintenance pre-alert, which is activated "x" hours before the maintenance alarm, and where "x" is the value set for the "Pre-alert" parameter. This allows time to plan the

maintenance operation. The "maintenance alarm" time is the number of boiler operating hours before it needs to be cleaned. During the start-up wizard, the feedwater hardness is entered, and the "maintenance alarm" depends on this value, as shown in the table below:

Water hardness	Maintenance alarm
Demineralised	Cleaning/maintenance 3000 hours (NO STOP)
0-10°f	STOP for cleaning/maintenance 3000 hours
10-20°f	STOP for cleaning/maintenance 1500 hours
20-30°f	STOP for cleaning/maintenance 1000 hours
30-40°f	STOP for cleaning/maintenance 800 hours

If during the wizard the value is set to "automatic", rather than entering the hardness value, the maintenance alarm is automatically associated to the feedwater conductivity read by the conductivity meter. The following table shows the dependency of the maintenance alarm on water conductivity.

Water conductivity	Maintenance alarm
1-50 µS/cm	Warning at 3000 hours without STOP for cleaning-maintenance (it is assumed that the water comes from a reverse osmosis system).
50-100 µS/cm	STOP for cleaning/maintenance 3000 hours
> 100 µS	STOP for cleaning/maintenance 1500 hours

The conductivity is read periodically, and the maintenance alarm times is thus updated as a consequence.

Thus, for example, if a water hardness of 15°f is set, the "maintenance warning" time will correspond to 1500 actual boiler operating hours; if the "maintenance pre-alert" time is set to 240 hours (default value), the maintenance pre-alert will be activated after 1260 hours.

The humidifier will signal an alarm (and STOP the unit) for maintenance once the pre-set hours have been reached, plus an additional 120 hours (in the above example, 1500 + 120 = 1620 hours).

It must be stressed that if demineralised water (manual mode) or 1-50µS/cm (automatic mode) are selected, the unit will signal maintenance and cleaning via the warning only, without ever shutting down the humidifier.

7.2.9 Burner rotation/activation

Burner rotation and activation can be configured.

gaSteam 180/300/450 manages burner operation according to the setting of the "Burners Sequence" parameter.

In normal operation, depending on the humidity request and the selected control mode (series or parallel), the control logic automatically calculates the production for each unit. Rotation of the request involves periodically changing these production references. In the long term, operating hours of the burners are balanced, with consequently more uniform wear of the components and scale build-up in the heat exchangers.

There are two types of rotation:

- IN PRODUCTION: timed rotation, i.e. whenever the gap between the operating hours of the two burners reaches 10 hours, the production references are reversed.
- ON IGNITION: the burner with the lowest number of operating hours is ignited first, as long as the difference between the operating hours is greater than 10.

There are also two types of activation:

- "PARALLEL": the steam request is divided equally between the burners so that steam production is the same.
- "SERIES" (also called "sequential"): if the steam request is less than 50% of the rated production, only one burner is operated. Only if the request exceeds 50% does the other burner start.

Index	Parameter	Description
Ea10	Burners rotation	Set the burner sequence (series or parallel). Burner rotation (in production, on ignition, no rotation) Operating hours before rotation.

7.3 Menu E. Settings - b. Functions

7.3.1 Preheat boiler water (Installer menu)

To ensure production starts quickly, the preheat function can be enabled. In this way, even when there is no steam request, the water temperature in the boiler is kept at a value specified by the user. When

production is next needed, the water is thus warmer than ambient temperature and consequently production will start faster.

Index	Parameter	Description
Eb02	Enable preheating	Enable preheat Set the water temperature to maintain Set water temperature offset
		Default: Preheating disabled Water temp. set point in the boiler = 70°C (176°F) Offset = 3% rH (3°C/5.4°F)
		Set point range: 50-80°C (122-194°F) Offset range: 2-20%rH (0-20°C/32-68°F)

The water temperature in the boiler is read by the passive NTC temperature probe in contact with the boiler. The operating principle of the preheat function is described in the following graphs, the first for control with probes, the second for control by external signal.

Preheat with autonomous modulating control with probes

The preheating function, if active, overlaps the control diagram and modulates the power delivered to the heaters in relation to the water temperature and the preheating set point.

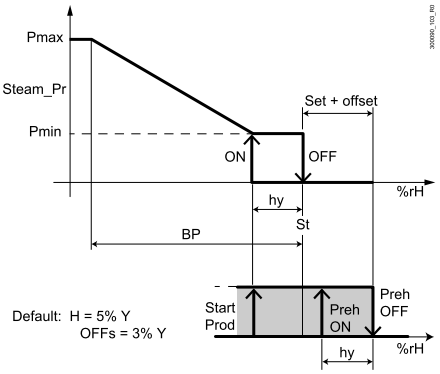


Fig.7.j

Key

Steam_pr	Steam production	%rH	Humidity measurement
Pmax	Max production	hy	Activation hysteresis
Pmin	Min production	BP	Proportional band
St	Set point		

Tab.7.b

Preheat with control by external signal

The preheating function, if active, shifts the control diagram by a value equal to the "offset". The preheat function modulates the power delivered to the heaters in relation to the water temperature and the preheating set point.

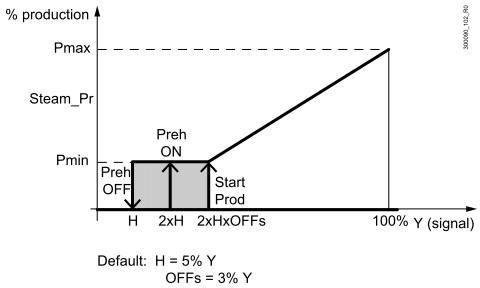


Fig.7.k

Key

Steam_pr	Steam production	Pmin	Min production
Pmax	Max production	y	External signal

Tab.7.c

7.3.2 Fill cycles in PWM mode after drain to dilute cycles and high level/foam (Installer menu).

After a drain to dilute cycle or high level/foam, the fill valve is opened to replenish water up to the maximum level of the float. The addition of fresh water affects steam production, as it decreases the average water temperature inside the boiler: to reduce the negative impact of fresh water on steam production, the user can activate PWM mode for the fill cycles after drain to dilute cycles and due to high level/foam.

PWM mode for fill cycles works as follows:

1. the mass of fresh water needed to replenish the water level is divided into smaller amounts of fresh water;
2. by keeping the partial fill cycles as far apart as possible, each small amount of fresh water has more time to heat up before more fresh water is filled. This reduces the negative impact of the fresh water on steam production.

The duration of each fill cycle can be defined in seconds:

Index	Parameter	Description
Eb03	Part fill cycles	Enable part fill cycles. Set the part fill duration.
		Default: part fill: enabled fill time: 5 seconds
		Range 1-199 seconds

7.3.3 Total drain due to inactivity (Installer menu)

For reasons of hygiene, it is recommended to empty the boiler so as to prevent water from stagnating inside when there is no humidification request for an extended period. The user can set the automatic total drain due to inactivity time in hours:

Index	Parameter	Description
Eb04	Drain do to inactivity	Enable drain due to inactivity; set the hours of inactivity without production request that must expire before draining
		Default: drain due to inactivity: enabled; hour threshold: 72 hours
		Range: 1-999 hours
		Remarks: gaSteam must remain on in order to empty the boiler. The display shows “DRAIN TO DILUTE” during the drain due to inactivity

Index	Parameter	Description
		cycle.

Automatic total drain due to inactivity cycle is enabled by default and the maximum inactivity time is three days (72 hours). The boiler will be automatically emptied when gaSteam remains on for at least three days without humidification request.

7.3.4 Periodical drain (Installer menu)

If the feedwater is turbid or has a high mineral content, the boiler can be periodically drained in order to clean and dilute the water as much as possible. For this operation to be effective, it is recommended to carry it out at least once every two-three days. The user can set the automatic periodical total drain cycle as follows:

Index	Parameter	Description
Eb05	Periodic drain	Enable periodical drain; Set the time between one periodical drain cycle and the next (in hours);
		Default: periodical drain: disabled; hour threshold: 10 hours
		Range: 0-999 hours
		Remarks: gaSteam must remain on in order to empty the boiler. The display shows "PERIODICAL DRAIN" during the periodical drain cycle.

Periodical drain is disabled by default. The periodical drain hour counter considers the effective production time.

7.3.5 Unit status signal or maintenance pre-alert (Installer menu)

The humidifier controller features two relay contacts for remote warning of the maintenance pre-alert or unit status (production). The terminals for these functions are 13U, 14U and 7O, 8O, 9O.

Index	Parameter	Description
Eb07	M5 Signal	Alarm type on relay output M5 Default: maintenance pre-alert Normally-closed
	M6 signal	Alarm type on relay output M6 Default: alarm (general) Normally-closed

Digital outputs M5 and M5 can be configured as follows:

- production
- failed autotest alarm
- high conductivity warning
- level sensor malfunction alarm
- motor protector alarm (overtemperature)
- high boiler temperature alarm
- model not selected
- no water alarm
- low production warning
- main probe broken or disconnected alarm
- limit probe broken or disconnected alarm
- pre-heating probe broken or disconnected alarm
- main wireless probe group not working
- limit wireless probe group not working
- maintenance pre-alert (warning)
- maintenance alarm
- offline

Note: the alarm shuts the unit down, the warning is an alert only

M5 (13U, 14U) = pre-alert for maintenance	maintenance early warning signal, helps plan maintenance in advance. See the paragraph "Boiler operating hours" for more details on pre-alert configuration

M6 (7O, 8O, 9O) = alarm (general)	

The general alarm (associated, for example, with output M6 - 7O, 8O, 9O) considers all the alarms with shutdown in the list, unless the alarm in question is selected for the other digital output (in this example M5 - 13U, 14U).

7.3.6 Blower unit configuration (Installer menu)

To best manage the blower units in applications where steam is delivered directly into the room, the blower on and off times can be set. Delaying activation of the blower unit allows the system to reach the operating temperature before the fan is activated. Delaying deactivation of the blower unit guarantees that parts in contact with the steam will be completely dried, and that when next starting, there is no condensate in the blower (no droplets delivered into the room).

Index	Parameter	Description
Eb08	Blower unit	Set the delay time (seconds) for activation and deactivation of blower units
		Default: start delay: 0 seconds stop delay: 90 s
		Range: 0-600 s

7.3.7 Save logs (Service and Installer menu)

The log of the main variables can be saved to USB pen drive; these are:

- Unit On/Off
- Unit status
- Active alarms
- Request (%)
- Production (kg/h)
- Level sensor status
- Fill
- Drain

The file is saved in .csv format. The logs are constantly saved to internal memory. To export them to the memory key in the USB Host port, enter menu screen Eb09. Set "Export to USB" to "YES". Progress of the operation will be shown under "progress".

Index	Parameter	Description
Eb09	Export logs	Save the log of main variables to USB pen drive
		Export to USB = Yes, to save the log

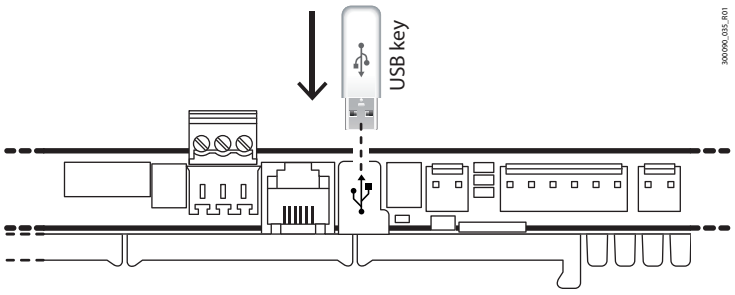


Fig.7.I

7.3.8 Save alarm log

The alarm log can be saved to USB pen drive. To export the file, plug the USB pen drive into the c.pHC and go to screen Eb10, then set "YES" for the "Export?" parameter.

Index	Parameter	Description
Eb10	Export alarms	Save alarm log to USB pen drive
		Default: No

The file will be saved in the pen drive's root directory and will be called "AlrmLog.txt".

7.3.9 Frost protection

The temperature of the unit is controlled using the burners. When the internal temperature measured is lower than the value of the "Forced preheating threshold" parameter (default value = 10°C (50°F)), the unit switches on the burners to keep the water and therefore the humidifier warm. If the temperature inside the unit is lower than the temperature set for the "Pump drain threshold" parameter (default value = 5°C (41°F)), the unit completely drains the water inside the boiler.

Index	Parameter	Description
Eb11	Frost protection	Set the frost protection function with burner management. Sett the safety function to completely drain the water in the boiler.

In addition to these functions, on outdoor models there is also a normally-open valve connected to a temperature probe (independent), which completely drains the boiler if it measures a temperature of 3°C (37.4°F) (default value, can be set manually). It is also possible to install heaters inside the unit, which work independently (kit to be purchased separately, P/N: UGKHEAT115 for models powered at 115 Vac, UGKHEAT230 for models powered at 230 Vac). The recommended calibration temperature for the heaters is 8°C (46.4°F).

7.4 Menu E. Settings - c. Configuration

7.4.1 Main probe (Installer menu)

The main probe can be configured on the screen with index Ec01.

Index	Parameter	Description
Ec01	Main probe	Main probe configuration; Default: Enable: Enabled (depending on the type of control) Type: 0-10V Minimum: 0% rH Maximum: 100% rH Offset: 0 Enable Al (enable alarms): YES Delay: 120 seconds Possible settings: Type: 0-10V/0-1V/NTC/4-20mA/0-20mA/2-10V Minimum: 0-100% rH Maximum: 0-100% rH Offset: 0 Enable Al (enable alarms): YES Delay: 0-999 seconds

For each probe, after having specified the type, the minimum and maximum values readable can be defined, as well as an "offset" to compensate for any imprecisions in the value read (example: offset = 3 % rH corresponds to 3 percent increase of the humidity value read by the probe). Parameter "En. Al." activates the alarms corresponding to any probe faults. If a malfunction is detected for a time greater than the value of "Delay" (seconds), the "main probe broken or disconnected" alarm is activated.

7.4.2 Limit probe (or second probe - Installer menu)

The limit probe (if present) can be configured on the screen with index Ec02.

Index	Parameter	Description
Ec02	Limit/second probe	Limit probe configuration; Default: Enable: Enabled (depending on the type of control) Type: 0-10V Minimum: 0% rH Maximum: 100% rH Offset: 0 Enable AI (enable alarms): YES Delay: 120 seconds
		Possible settings: Type: 0-10V/0-1V/NTC/4-20mA/0-20mA/2-10V Minimum: 0-100% rH Maximum: 0-100% rH Offset: 0 Enable AI (enable alarms): YES Delay: 120 seconds

The same explanations provided in the previous paragraph also apply here.

7.4.3 Wireless probes (Installer menu)

Configuration of the wireless probes involves defining the weight of each probe; for further details on the averages see paragraph 7.2.2 "Weighted average of the probes". Screen Ec03 can be used to deactivate, activate as main probe or limit probe each of the four possible wireless probes.

Index	Parameter	Description
Ec03	Wireless probes	Wireless probe configuration (main and limit) Default: Probe 1: disabled Probe 2: disabled Probe 3: disabled Probe 4: disabled
		Possible settings: disabled, main probe, limit probe

To configure the wireless probes, weights and communication times, use screens Ec04, Ec05, Ec06 and Ec07, depending on the number of probes connected.

Index	Parameter	Description
Ec04 Ec05 Ec06 Ec07	Wireless probe 1 Wireless probe 2 Wireless probe 3 Wireless probe 4	Configure probe weight and communication times; Default: Weight: 100 Transmission time: 10 s Disconnection delay: ms
		Range: Weight: 0-100 Transmission time: 5-3600 seconds Disconnection delay: ms

Note:

- if wanting to use two wireless probes, simply set "humidity (one probe)" control (Ea01) and then activate the two wireless probes on screen Ec03;
- "humidity+limit" control (Ea01) can also be used, in this case on screen Ec03 choose which of the four wireless probes is the limit, and set the others as main;
- "humidity+limit" control (Ea01) can also be used, in this case on screen Ec03 choose which of the four wireless probes is the limit, and set the others as main;

7.4.4 Maximum number of evaporation cycles between drain to dilute cycles settable by the user (Installer menu)

The "Evaporation cycles before drain" parameter is used to manually set the maximum number of evaporation cycles allowed between two drain to dilute cycles. The number of evaporation cycles between two successive drain to dilute cycles can also be calculated internally, based on feedwater conductivity. To use the automatic formula, set "Evaporation cycles before drain" = "Auto". The number of evaporation cycles used by the controller will be the lower of the value set manually by the user and calculated automatically.

Index	Parameter	Description
Ec11	Number of evaporation cycles before drain	Set the number of evaporation cycles between two drain to dilute cycles.
		Default: number of cycles: Auto;
		Possible settings: Auto (automatic management according to feedwater conductivity); 1 to 40

7.4.5 Duration of the fill and drain to dilute cycles settable by the user (Installer menu)

On screen Ec12, the duration of the fill time after the evaporation cycle and the duration of the drain to dilute cycle can be set in relation to the default parameter values.

Index	Parameter	Description
Ec12	Variation in fill and drain time	Set the fill after evaporation and drain to dilute times in relation to the default values
		Default: variation in fill time: 100% variation in drain time: 100%
		Possible settings: variation in fill time: 20-100% variation in drain time: 0-190%

The duration of the drain to dilute cycle is set using the "variation in drain time" parameter, which defines the time as a% of the default value:

new drain time = (default drain time × variation in drain time)/100

Example: if the default time is 10 s and "variation in drain time" = 50%, then the new time will be = 10 s x 50/100 = 5 s.

The same also applies to the fill time after the evaporation cycle, in this case using the "variation in fill time" parameter:

new fill time = (default fill time × variation in fill time)/100

Note: if the drain to dilute time is very short, there may be the **RISK OF FOAM/CORROSION** due to increased internal conductivity. Low values should be set for the "variation in drain time" parameter only after having carefully evaluated the water quality and the consequences.

Warning: RISK OF FOAM FORMING WHEN "variation in drain time" IS TOO LOW FOR THE WATER QUALITY.

"variation in drain time" = 100 % means that the default value will be used.

"variation in drain time"< 100 % means that the drain time will be less than the default, consequently introducing less fresh water when next refilling, reducing the negative impact on steam production (important in high-precision applications).

Setting the "variation in drain time" for precise %RH control: the "variation in drain time" must be set as low as possible, without however causing formation of foam/corrosion. Proceed by trial and error so as to find the most suitable value.

Note: the duration of the water fill cycle can be used to offset water pressure upstream of the humidifier. Decrease the fill time in the case of high pressure and vice-versa. Nonetheless, feedwater pressure must be within the range of pressure values specified in "General characteristics and models" (1-8 bars)

7.4.6 Set feedwater hardness (Installer menu)

To read feedwater hardness, Carel provides an analysis kit (P/N: KITTH00000). The "Water hardness" parameter defines the maintenance alarm for cleaning the boiler and the heat exchanger.

Index	Parameter	Description
Ec13	Water hardness	Set the feedwater hardness
		Possible settings: Hardness: Auto, Manual Manual setting: demineralised, 0-10°f, 10-20°f, 20-30°f, 30-40°f.
		Remarks: if the value is set to Auto, the water hardness is estimated by the controller based on the conductivity value.

If water hardness is not set, the controller independently manages the type of water based on the conductivity reading. Although there is no reliable relationship between water hardness and conductivity, a hardness of 40°f is typically equivalent to a conductivity of approximately 900-1000 µS/cm at 20°C. See "Boiler operating hours" for information on the maintenance alarm times in relation to water hardness or conductivity.

7.4.7 No feedwater setting (Installer menu)

In the event of no feedwater, the corresponding "No water" alarm is displayed. After the alarm has been displayed and the "no feedwater time" has elapsed, the controller attempts a new fill cycle. At each attempt, the "no feedwater time" is multiplied by the number of times it has already been carried out. The controller therefore performs a series of attempts to refill with water, for a maximum time of five hours after the alarm occurs. The alarm is reset only if the water level effectively reaches the intermediate position (yellow LED).

Index	Parameter	Description
Ec14	Missing water warning retry time	Set the time to check for no feedwater.
		Default: 10 minutes
		Range: 0-20 minutes
		Step: 1 minute
		Remarks: if the value is set to 0, no more attempts will be made to refill the water and the alarm will not be reset

7.4.8 Enable and set high conductivity alarm (Installer menu)

The controller allows conductivity thresholds to be set for the activation of alarms when the limits are exceeded. Excessive conductivity and consequently high concentration of salts in the feedwater can be signalled. These alarm thresholds are programmable using the following parameters:

Index	Parameter	Description
Ec15	High conductivity	Enable the high conductivity alarm and set the alarm delay.
		Default: enabled: YES alarm delay: 60 seconds
		Possible settings: enabled: YES/NO; alarm delay: 0-300 seconds
		Remarks: once the threshold has been exceeded (if the alarm is enabled), the controller waits for the "alarm delay" to elapse before displaying the alarm. If during this waiting time the conductivity falls back below the threshold, the alarm is not activated. This helps avoid false alarms due to variations in the conductivity reading.

7.4.9 Set high conductivity alarm thresholds (Installer menu)

Two high conductivity thresholds can be set. If exceeding:

- the first (lower) threshold, a warning is shown without stopping operation;
- the second (higher) threshold, an alarm is activated and the unit shuts down as a precaution.

Index	Parameter	Description
Ec16	High conductivity	Set the high conductivity thresholds. Default: pre-alert: 1250 µS/cm alarm: 1500 µS/cm Hysteresis: 25 µS/cm Possible settings: pre-alert: 0-1500 µS/cm alarm: 0-1500 µS/cm Hysteresis: 0-100 µS/cm Remarks: the hysteresis is used to reset the alarm if the conductivity falls below the threshold minus the "Hysteresis".

7.4.10 Remote ON/OFF operating logic (Installer menu)

Screen Ec22 is used to set the operating logic of the remote ON/OFF contact (normally open or normally closed).

Index	Parameter	Description
Ec22	Remote ON input logic	Set the operating logic of the remote ON/OFF input Default: Remote ON input logic = N.O Possible settings: Remote ON input logic: N.O., N.C.

7.4.11 Fan speed setting

Setting the fan **speed for minimum and rated production**. These values are set by default to ensure the correct air/gas mix with the natural gas

Pre-purge speed. The pre-purge speed is indicated in "RPM" and is normally set by the manufacturer based on the characteristics of the combustion system and the gas.

Note: these parameters must only be modified by qualified personnel.

Display	Value and notes	Default	Unit of measure
Pre-Purge	Pre- purge fan speed set points	UG045 = 2500 UG090 = 2500 UG150 = 2500 UG180 = 2500 UG300 = 2500 UG450 = 2500	RPM
Min speed	Fan speed set point at minimum production	UG045 = 1450 UG090 = 1700 UG150 = 1900 UG180 = 2000 UG300 = 1900 UG450 = 1900	RPM
Max speed	Fan speed set point at rated production	UG045 = 4700 UG090 = 5050 UG150 = 6750 UG180 = 5300 UG300 = 6750 UG450 = 6750	RPM

Index	Parameter	Description
Ec23	Burner 1 settings	Set pre-purge speed, minimum speed and maximum speed for rated steam production

Index	Parameter	Description
Ec24	Burner 2 settings	Set pre-purge speed, minimum speed and maximum speed for rated steam production
Ec25	Burner 3 settings	Set pre-purge speed, minimum speed and maximum speed for rated steam production

7.4.12 Burner calibration

Also see the chapter "Preparing for operation".

The Burner Calibration submenu is used to access the procedure for configuring and calibrating the burners on the humidifier.

Two types of gas can be selected:

- Natural gas (G20-G25);
- LPG (G30-G31).

Note: on selecting the type of gas, the humidifier automatically sets all the control parameters for correct operation with that specific family of gas.

Index	Parameter	Description
Ec28	Burners calibration	Start the procedure to calibrate the burners. Follow the instructions shown on the display

Guided calibration is a step-by-step procedure that helps the installer correctly set up the burner.

It is divided into three different steps:

1. Calibration at maximum fan speed (maximum output delivered by the humidifier);
2. Calibration at minimum fan speed (minimum output delivered by the humidifier=;
3. Verification of calibration at maximum output.

The first step once the type of gas has been selected; the humidifier will be switched on, if off, and operated at the maximum output available. The screen will show info on the unit operating status, such as pre-purge, fan speed and waiting. Wait for the humidifier to reach maximum output, then calibrate the gas valve (see chap. "Type of calibration"), adjusting it so that the percentage of CO2 measured is within the allowed range.

Repeat the operations for the other two steps, following the instructions shown on the screen, to complete the calibration procedure.

Note:

- if an alarm is activated during calibration, the procedure stops and a warning message is shown;
- If during calibration no button is pressed for 5 minutes, the procedure will be interrupted and the unit will return to normal operation.

7.5 Menu E. Settings – d.Master/Slave

7.5.1 Master/Slave system network settings (Installer menu)

To increase total production capacity when one single unit is not sufficient, the Master/Slave function can be used to connect up to 20 units together in just one system. To set and enable the individual units in the system, go to the "Network" menu, specifically starting from screen Ed01.

Initial Master/Slave system configuration:

Index	Parameter	Description
Ed01	Master/Slave	Initial Master/Slave system configuration
		Press "PRG" to configure the Master/Slave network
Ed02	Unit 1 Unit 2 - - Unit 20	Add unit to the Master/Slave system
		To add a unit to the network enter the unit's IP address. Scroll between the units using the UP/DOWN buttons

Master/Slave system function configuration: from screen Ed07 (Master/Slave production) press "PRG" for the following settings:

Index	Parameter	Description
Ed02	Unit 1 Unit 2 - - Unit 20	Add unit to the Master/Slave system. To add a unit to the network enter the unit's IP address. Scroll between the units using the UP/DOWN buttons
Ed03	Maximum production of Master/Slave system. Unit rotation	Set maximum production of the Master/Slave system. Load: this is the maximum capacity effectively required for the Master/Slave system, settable by the user. Maximum: this is the sum of the maximum capacities of the units added to the Master/Slave system
		Set the unit rotation logic. Rotation: Grouped (Default), Balanced
Ed04	Advanced preheating Rotation time	Enable the advanced preheating function for Master/Slave systems. Possible settings: YES/NO Set the auto-rotation time between units Possible settings: 0-65535 hours (Default 3 hours). If auto-rotation time = 0 the function is disabled.
Ed05	Offline timeout	Set the offline time for units in the Master/Slave system Range: 500-1000 ms
Ed06	Disconnect unit from Master/Slave system	Disconnect the current unit from the Master/Slave system Possible settings: YES/NO Remarks: this disconnects and removes the current unit from the Master/Slave system.

Display Master/Slave system production.

Index	Parameter	Description
Ed07	Master/Slave production	Display the request (%) and the production (kg/h / lbs/h) of the Master/Slave system

Display unit status and the percentage of production for each unit in the Master/Slave system:

Index	Parameter	Description
Ed08	Display Master/ Slave system	Unit status with corresponding % of production
Ed09	Display info on individual unit. Unit 1, Unit 2, ...,Unit 20	Display unit operating hours, current production and any alarms To move between the various units use the UP/DOWN arrows The display is available for each unit

Maintenance on the units in the Master/Slave system:

Index	Parameter	Description
Ed10	Unit switched-off for maintenance	Unit switched off to carry out maintenance.

For detailed information on the Master/Slave system, see the chapter "Master/Slave system".

7.6 Menu E. Settings - e. Backup

7.6.1 Enable backup unit (Installer menu)

In some critical applications where relative humidity control is very important, it may be essential have a backup unit available in the event of malfunctions on the main unit. To enable the backup unit, go to screen Ee01:

Index	Parameter	Description
Ee01	Enable backup	Enable automatic unit backup in the event of shutdown, using a second independent unit
		Default: Disabled
		Possible settings: Enabled/Disabled

After having enabled the backup function, the unit that will start first in the event of simultaneous activation can be selected.

Index	Parameter	Description
Ee02	Priority when starting	Define the priority of the units when in backup mode Default: Disabled Possible settings: Enabled/Disabled

The priority must be set to "YES" only on one of the two units; the "priority" parameter on the second unit must be set to "NO".

7.7 Menu E. Settings - f. Manual mode

7.7.1 Manual mode (Service and Installer menu)

During first start-up or maintenance, "manual mode" may be useful in order to check the operation of the main devices on the humidifier. Manual mode is only available when the unit is OFF and without any active alarms. In addition, to safeguard the unit, the heaters can only be activated when there is water in the boiler (high level, corresponding to green LED on)

⚠ **Important:** manual mode should only be activated by qualified personnel; improper use may cause serious damage to the unit.

Index	Description	Parameter
Ef01	Manual mode	Enable "Manual mode" to check operation of the individual components. Default: Disabled Possible settings: Disabled, manual outputs, manual production Remarks: when exiting "Manual mode", the settings are restored and normal operation resumes automatically. If the user does not set any parameters in the "Manual mode" menu for 30 minutes, manual mode is automatically disabled.
Ef02	Manual management of outputs	Enable "Manual mode" to check operation of the individual components. Test fill valve Test drain pump Activate the blower relay
Ef03	Manual request management	Manually set the production request.

7.8 Menu E. Settings - g. Initialisation

7.8.1 Initial wizard (Service and Installer menu)

The Wizard menu, screen index Eg01, is used to initiate step-by-step programming of the parameters required for commissioning the unit. Below are the steps included in the first start-up procedure (some steps may not be displayed, if not necessary):

1/9 - Choose the model (only if the replacement controller is not configured);

2/9 - Water hardness: auto or user; defined;

3/9 - Manual water hardness setting;

4/9 - Control type;

5/9 – Select the main room probe type:

6/9 – Select the limit probe type;

7/9 - Set the limits for active probes;

8/9 - Set the drain to dilute cycles: auto or user-defined;

9/9 - Set manual drain to dilute cycles.

Index	Parameter	Description
Eg01	Wizard	Start the wizard for initial unit configuration Set whether to display the wizard when next restarting Default: wizard enabled: YES

For further information, see the paragraph "Activation sequence".

7.8.2 Set the language (Service and Installer menu)

The first time that the unit is powered on, the menu language needs to be selected. To change the language, go to screen Eg02. Press "ENTER" and UP/DOWN to set, ESC to exit without changes.

Index	Parameter	Description
Eg02	Language	Set the language. Default: English Possible settings: 1. English 2. Italiano 3. Deutsch 4. Français 5. Español

7.8.3 Set the unit of measure (Installer menu)

Screen Eg03 is used to select the units of measure: International (°C, kg/h) or Imperial (°F, lb/h). In addition, the language selection screen when starting can be disabled.

Index	Parameter	Description
Eg03	Unit of measure and language when starting	Select the units of measure. Show change language when starting. Default: units of measure: depending on unit P/N show change language when starting: YES Possible settings: units of measure: International, Imperial show change language when starting: YES, NO

7.8.4 Set and change passwords (Service and Installer menu)

Screen Eg04 is used to change and/or set the Service and Installer passwords.
The passwords are 4-digit numbers.
The default Installer password is: 0077
The default Service password is: 0044

Index	Parameter	Description
Eg04	Change passwords	Set and change the passwords Default: Installer: 0077 Service: 0044

7.8.5 Factory reset (Service and Installer menu)

To reset all parameters to the default values, use screen Eg06.

Index	Parameter	Description
Eg06	Inst. default	Unit factory reset. Remarks: follow the directions shown on the display Confirmation will be requested before resetting the values

⚠ **Important:** when performing a reset, all the control settings will be lost and the values will return to the factory defaults.

7.8.6 Software update from USB pen drive

The update package can be downloaded from ksa.carel.com. The unit's software can be updated using a USB pen drive plugged directly into the c.pHC controller. In the pen drive's root directory, create an UPGRADE directory and copy the software update file to this directory. Once having plugged the pen drive into the controller, go to screen Eg07 and set the "Update unit software" parameter to "YES".

Index	Parameter	Description
Eg07	Software update	Start software update from USB pen drive
		Default: No

The steps to update the SW/OS (software/operating system) are:

- Save user parameters;
- SW/OS upgrade;
- Restore default parameter values;
- Write user parameters.

Once the update is complete, remove the USB pen drive from the controller.

After updating, the main user parameters do not need to be configured as the update automatically restores them.

7.8.7 Export parameters to USB pen drive

The unit's parameter configurations can be exported to USB pen drive, and subsequently loaded onto a second unit, making installation and setup even faster. To save the parameters, plug the USB pen drive into the c.pHC and then go to screen Eg08 and set the "Export?" parameter to "YES".

Index	Parameter	Description
Eg08	Export parameters	Start exporting configuration from unit to USB pen drive
		Default: No

The exported file will automatically be saved in the pen drive's root directory and will be called: "UG4cgg.txt"

7.8.8 Import parameters onto the unit

The unit's parameter configurations can be imported from a USB pen drive. To import the parameters, make sure that the exported file is in the pen drive's root directory and is called: "UG4cgg.txt". Then plug the USB pen drive into the c.pHC and go to screen Eg09, and set the "Import?" parameter to "YES".

Index	Parameter	Description
Eg09	Import parameters	Start importing configuration from USB pen drive to unit
		Default: No

7.9 Menu E. Settings - h. Supervision

7.9.1 Supervisor network address setting (Installer menu)

Supervision can be enabled on the Ethernet network or BMS serial port. The following protocols are available to be set by the user:

Port	Protocol
BMS	Carel, ModBus, BACnet
Ethernet	ModBus, BACnet

On screen Eh01, the port and supervisor protocol can be set, as well as the unit's supervision address for the BMS port.

Index	Parameter	Description
Eh01	Unit supervision address on BMS port	Set the unit's supervision address and protocol for BMS port.
		Default: Address: 1; Protocol: ModBus; On/Off from SV: No; Control from SV: No;
		Possible settings: Address: 1-247 Protocol: Modbus, BACnet, Carel

Address is the device's supervision address on the BMS port. Enabling or disabling the parameters "On/Off from SV" and "Control from SV" activates or deactivates the response to the corresponding signals from the supervisor. For other supervisor protocols, select Carel protocol and use the Carel external gateway (supernode for humidification).

7.9.2 BMS port (Installer menu)

Screen Eh02 is used to set supervisor communication on the BMS port.

Index	Parameter	Description
Eh02	BMS port configuration settings	Set the communication parameters for supervision via BMS: Baud rate, stop bits and parity
		Default: Baudrate: 19200 Stop bits: 2 Parity: None

7.9.3 BACnet MS/TP supervisor settings (Installer menu)

To set the address, maximum number of master units and maximum number of frames for BACnet MS/TP supervision systems, use screen Eh03. The screen is only displayed when supervision is configured on the BMS port with the BACnet protocol.

Index	Parameter	Description
Eh03	BACnet MS/TP config	Set the address, maximum number of masters and maximum number of frames
		Default: Address: 0 Max masters: 127 Max frames: 10

For BACNet MS/TP, in addition to setting the parameters in screen Eh03, it is also necessary to set the parameter on screen Eh06.

Index	Parameter	Description
Eh06	Device inst.	Default: 77000

7.9.4 Ethernet network (Installer menu)

To connect the unit to the local Ethernet network for operation in Master/Slave mode, software Backup/Rotation or use the webserver, set the DHCP, IP, subnet mask, gateway and DNS parameters.

⚠ **Important:** these values must be provided by the local network administrator.

Index	Parameter	Description
Eh04	Ethernet network settings	Set the unit's IP address for the Ethernet connection. Set DHCP, subnet mask, gateway, DNS.
		Default:

Index	Parameter	Description
		DHCP: Off IP: 192.168.0.1 (unit's IP address) Mask: 255.255.255.0 (subnet mask) GW: 192.168.0.1 (gateway) DNS: 0.0.0.0 Update ?: NO

The following values are set by default on each unit:
DHCP: Off
unit's IP address: 192.168.0.1
subnet mask: 255.255.255.0
gateway: 192.168.0.1
DNS: 0.0.0.0

After having modified the parameters, set the "Update?" parameter to "YES" to update the IP address. To start the update, set the "Update?" parameter to "YES".

Important: the controller is not accessible directly on the internet as a firewall guarantees remote access only via a secure connection (Carel tERA cloud or encrypted VPN connection).

7.9.5 Supervisor settings for ModBus or BACnet on TCP/IP (Ethernet port) (Installer menu)

Both Modbus and BACnet are available on the Ethernet port; to enable the protocol go to screen Eh06:

Index	Parameter	Description
Eh05	Settings of supervisor from Ethernet port	Select the protocol on the Ethernet port En. ModBus TCP/IP: Enabled/Disabled En. BACnet TCP/IP: Enabled/Disabled Default: En. ModBus TCP/IP: Disabled En. BACnet TCP/IP: Enabled

If using the BACnet protocol, the address must be set on screen Eh04.

Note: only one instance of the BACnet protocol is available, therefore it is not possible to activate BACnet on the BMS port and on the Ethernet port at the same time.

7.9.6 Supervisor port

Go to screen Eh07 to set which serial port the supervisor is connected to: BMS or Ethernet.

Index	Parameter	Description
Eh07	Regulation from SV	Select the supervisor port Default: Disabled Possible settings: None, Ethernet (BACnet), BMS port

Important: after changing the protocol, restart the controller.
Any changes made to the communication protocols will require a controller

To restart, simply press the "ENTER" button when the following flashing message is shown:



Fig.7.m

7.10 Menu E. Settings - i. Logout

7.10.1 Logout from the settings menu (Installer and Service menu)

To logout, use the icon available in the “Settings” menu

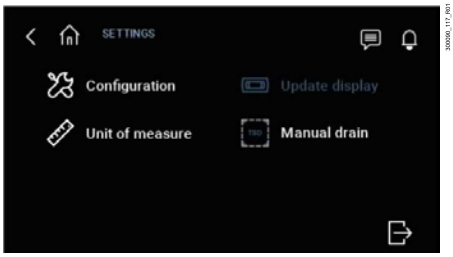


Fig.7.n

In “character terminal” mode, screen EI01 screen is used to exit the Settings menu and logout. The screen also describes the type of login performed (installer or service). When accessing this screen, the following information will be shown (in the currently selected language). “To log out, press "ENTER"”: the main screen will then be shown again. Press “ESC” to exit if not wanting to end the session.

7.11 Touch display: software update

To update the touch display, simply enter the system menu (from the home page), Settings menu and enter the password. The following screen will be displayed:

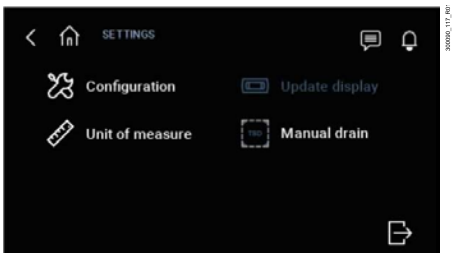


Fig.7.o

Plug the USB key into the display (the mini USB port on the display is accessible from inside the electrical panel (outdoor model) or from inside the panel (indoor model), the “update display” menu will be shown. Click this menu and follow the steps described on the display to update the software.

8. Master/Slave system

8.1 Description of the system

To obtain steam production higher than that provided by one single unit, several humidifiers can be connected together in a Master/Slave system.

A maximum of 19 Slave units can be connected to each Master, meaning a total of 20 humidifiers.

The Master and Slave units are connected using a local Ethernet network, and in the case of just two units (one Master and Slave) involves a direct connection between the two controllers via Ethernet RJ45 Category 5 cable.

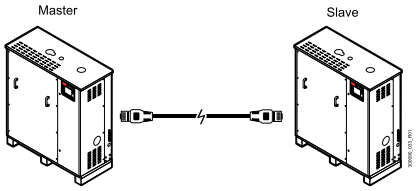


Fig.8.a

If the Master/Slave system comprises three or more units, a network switch is required.

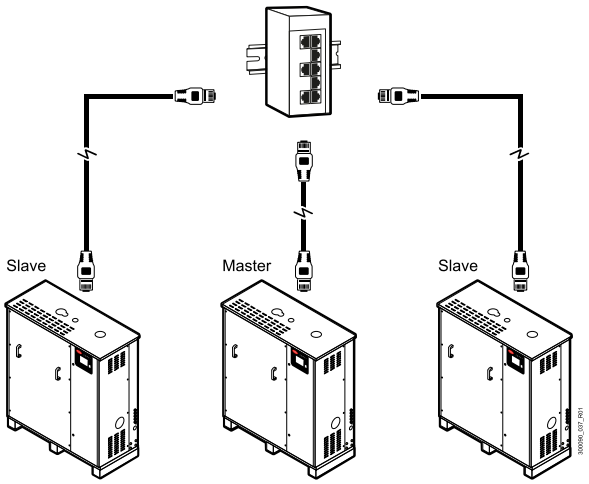


Fig.8.b

The Ethernet port is available on the humidifier's c.pHC controller:

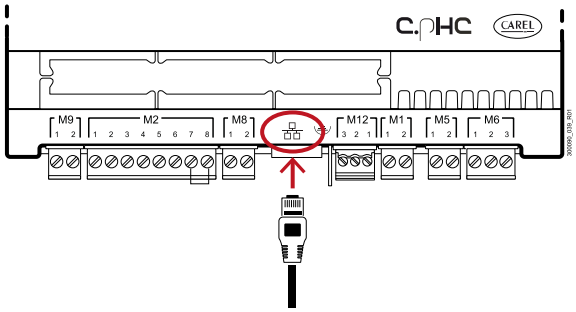


Fig.8.c

Note: use CAT-5 STP Ethernet cable with a maximum length of 100 m. To connect the shield, use the earth connector provided on the controller.

8.2 Switch for connection

The Master/Slave connection of more than two units requires the use of an industrial grade switch. Carel supplies a switch (P/N: KITSE08000) that can connect a maximum of eight units (8 Ethernet ports). If necessary, use several switches with a cascaded arrangement.

Technical specifications:

Number of ports	8
Installation	DIN rail
Operating temperature	-10 to 60°C (14 to 140°F)
Power supply	12/24/ 48 Vdc
	18-30 Vac (47-63 Hz)
Current @ 24 Vdc	0.13 A
Protection	IP30

Tab.8.a

8.3 Type of installation

The Master/Slave system features one main unit (Master), which manages operation of the secondary units (Slaves). The external signal or probe connections, depending on the set control type, are made to just one of the humidifiers in the system. The unit that the signal is connected to will automatically be identified as the Master. Consequently, the Master does not need to be especially configured.

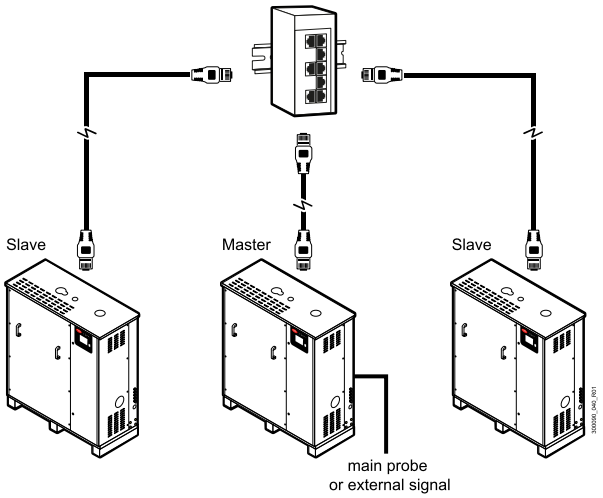


Fig.8.d

As long as the Master unit is powered, the system will operate even if the Master itself has a malfunction (e.g. alarms, production stopped, ...), as the controller on this unit will send all the necessary data to the Slaves. Obviously, if system redundancy is not provided for, the amount of steam produced will be lower than actual requirements. If the Master is completely switched off, the Master/Slave system cannot read the signals/control probes. Therefore, it is recommended to connect at least two (or all) of the humidifiers in the system to the external signal or to fit them with distinct probes.

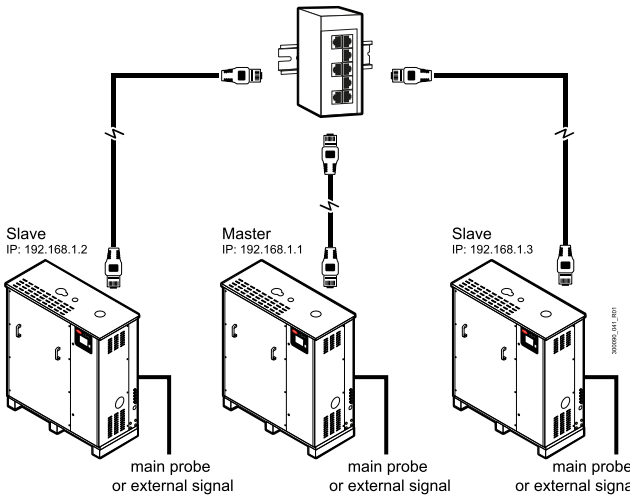


Fig.8.e

Systems set up in this way will be able to cover the humidification requirement. In this specific case, the Master will always be the unit with the lowest IP address of those connected to the signal/probes. If necessary, an additional humidifier (backup) can be installed to cover request in the event of malfunctions on one of the units in the system.

8.4 System configuration

To configure the Master/Slave system, proceed as follows:

1. Connect the probes or the external signal to the unit and complete the configuration (control type, type of signal, maximum production …);
2. Set the IP addresses of the individual units so that they belong to the same subnetwork (subnet mask); the screen index for this configuration is Eh02 (E. Settings – h. Supervision). The IP address is set on the display on each unit, assigning a different address to each unit in the same subnetwork. If necessary, contact the local network administrator. Remember that the default address for each unit is 192.168.0.1, and the default subnet mask is 255.255.255.0.
3. Connect the units making up the Master/Slave system to the local Ethernet network via a switch. If using just two units, an RJ45 category 5 cable can be used, connected directly to the Ethernet ports on the two c.pHC controllers.
4. Configure the Master/Slave system by enabling the units one at a time (this operation can be performed on the display on any of the units):
   I. Display screen index Ed01 and then access configuration mode by pressing "PRG";
   II. Enter the IP address for "Unit 1" and confirm by pressing "Enter";
   III. Repeat the operations described above (1 and 2) for all the other units in the Master/Slave system. (The units will join the Master/Slave system (online status) immediately after being connected to the network.)

Note:

- the Master unit will always (automatically) be the one with the lowest IP address out of the units connected to the probes or external signal;
- it may take a few seconds (max 10 s) for the Master to start sending the production request to the Slave/Slaves. This is also true when, automatically, the Master unit is changed (for example, in the event of malfunctions).

A cascaded connection of several units can also be made by simply using terminals 11U and 12U, which provide the production request (0-10 V).

8.4.1 Maximum production

In the same way as for the individual unit configuration, for the Master/Slave system the maximum capacity can also be set.

Procedure:

Enter menu Ed07 (E. Settings - d. Network), press PRG and scroll with UP/DOWN until reaching menu Ed03. The "Capacity" parameter identifies the maximum production request for the Master/Slave system, and can therefore be set by the user. The "Maximum capacity" parameter (read-only), on the other hand, indicates the sum of the sizes of each unit added to the system; this value is therefore the maximum effectively available to the Master/Slave system. Consequently, "Capacity" will always be ≤ "Maximum capacity". In any case, the maximum production can be defined for each individual unit in the system by limiting production from its maximum according to the size. In this case, "Maximum capacity" will be updated keeping in consideration these reductions.

8.4.2 Rotation logic

The activation logic of the units in the Master/Slave system can be set, choosing between "Grouped" or "Balanced" (screens Ed03 and Ed04). On screen Ed07 (E. Settings - d. Network), scroll with UP/DOWN until reaching Ed03 or Ed04 (NB: Ed04 is only visible if rotation is enabled on Ed03).

Grouped:

- the units are activated in series, one after the other, according to the request.

If the steam request does not require operation of all of the units, rotation can also be divided between the units with the lowest operating hours (auto-rotation function), so as to ensure uniform operation of the units over time (same number of operating hours). To enable and configure the rotation times, the "auto-rotation time" parameter must be set on screen Ed04.

Note: if the "auto-rotation time" parameter is = 0, the auto-rotation function is disabled.

Balanced:

- the units are activated in parallel at the same time, splitting the total production request between the number of units.

8.4.3 Advanced preheating

Note: the function has meaning only in the case of "Grouped" rotation.

By activating the advanced preheating it is possible to automatically pre-activate the Slave units that are currently in standby. When the request reaches 90% of production (of the units currently in production), the preheating of the remaining units is activated. The function is enabled/disabled on screen Ed04.

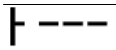
8.4.4 Disconnecting a unit

To remove a unit from the Master/Slave system, thus reducing the number of units available in the system, use the "Disconnect unit" function on screen Ed06. This can be done on any of the humidifiers in the system.

Note: once the unit has been disconnected, this will no longer be visible in the Master/Slave system, as its IP address will be removed from the list. If a unit is disconnected erroneously, the system can be restored on screen Ed01 (entering its IP). This must be done from the display on a unit that is already connected to the system.

8.4.5 Display the system

To display a summary of the Master/Slave system, go to screen Ed08. On screen Ed07 (E. Settings - d. Network), press DOWN to access menu Ed08. This has a total of five pages that display all of the units (01, 02, ..., 20), the status of each unit and current production as a percentage. The following table lists the Master/Slave network unit status indications:

Symbol	Unit status
	Current unit (unit being accessed, display or web server)
	Online
	Offline
	Not configured and not included in the Master/Slave system

Tab.8.b

The units in the Master/Slave system can also be selected one by one, displaying maximum production, unit status, operating hours, current production request and any alarms. To enter this display, from screen Ed08, select the desired unit and press ENTER, thus accessing screen Ed09. Scroll using the UP/DOWN buttons to display the details of all the units.

8.4.6 Software backup

Master/Slave mode can also be used to manage the software backup function; if one or more units in the Master/Slave system is affected by a malfunction, the system automatically restores humidification production by activating the backup units. The lost production in relation to the request is thus compensated by increasing production on the individual units (where possible) and/or starting any units in standby. Even if not strictly necessary, to ensure the backup function, the external control signal must be sent to all the units in the Master/Slave system; if using probes on the other hand, each unit must be fitted with a probe. Only in this way can complete operation be guaranteed in the event of malfunctions.

Note: if a unit goes offline due to a malfunction or shutdown, it will temporarily be excluded from the system, and when next restarted it may take 15 seconds or more to automatically come online again.

8.4.7 Software backup for maintenance

During maintenance or cleaning on a unit in the Master/Slave system, the backup for maintenance function can be activated. This allows production to start on a unit in standby, set as backup, before switching off the humidifier in question for maintenance. In this way, service continuity can be guaranteed in applications where required humidity control is very precise and continuous over time.

Procedure:

1. Access screen Ed07 (Network);
2. Press DOWN to display the list of units (Ed08);
3. Scroll to the unit on which maintenance is required (Unit 1, Unit 2, …) and press ENTER to confirm (screen Ed09).
4. Press PRG to access screen Ed10 and set "Request switch unit off" = YES. Wait for the display to show the message: "The unit can now be switched off for maintenance" and then switch the unit off.

Once maintenance has been completed, simply switch the humidifier on again, and it will automatically come back online.

Note: to activate the advanced software backup functions for maintenance, the backup unit must be connected to the probes or external signal.

9. Web server

9.1 Integrated web server

The integrated web server is used to configure and monitor the main unit parameters directly from a PC. In fact, using the Ethernet port on the humidifier controller, the unit can be accessed via a local network by simply entering its IP address in the browser.



Fig.9.a

To log in and access the various menu items, use the installer or service password (see the chapter "Start-up and user interface").

9.2 Connecting to the integrated web server

For the physical connection to the controller from a PC, must use the RJ45 Ethernet port on the c.pHC and an Ethernet category 5 cable.

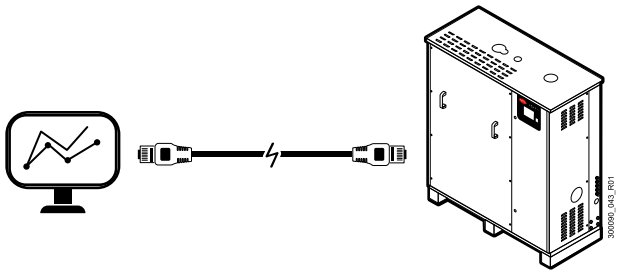


Fig.9.b

Location of the Ethernet port on the c.pHC controller.

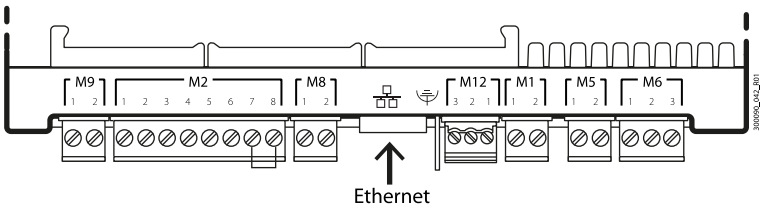


Fig.9.c

The PC and the humidifier controller (or all controllers, for Master/Slave systems) must be part of the same subnetwork.

The default network configurations for the c.pHC are:

- unit's IP address: 192.168.0.1;
- subnet mask: 255.255.255.0;
- gateway: 192.168.0.1.

For example, the PC's network settings can be changed as follows:

- unit's IP address: 192.168.0.2
- subnet mask: 255.255.255.0
- gateway: 192.168.0.1

to do this, access the "Network and sharing center" on the PC and go to "Local network connection". Then modify the addresses, as shown above, for Internet protocol version 4.

Each c.pHC controller can obviously also be added to the Ethernet local network, allowing the web server to be accessed from any point in the network and for each humidifier.

Important: the controller is not accessible directly on the internet as a firewall guarantees remote access only via a secure connection (Carel tERA cloud or encrypted VPN connection).

9.3 Web server functions

The web server "Home" page provides access to the display, so as to carry out all the configuration operations in the same way as if working directly on the unit. In addition, there will be an immediate response from the unit on its display.

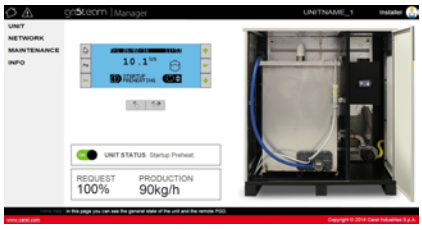


Fig.9.d

The main menus are:

- UNIT
- NETWORK
- MAINTENANCE
- INFO

Unit menu

Probes: information and configuration of the main and limit probe. Select the type of signal and define the minimum and maximum values.

Wireless: enable and associate each wireless probe to the main probe group or limit probe group. Read the humidity and/or temperature, level and battery status.

Control: select the control type. Set the set point, differential, minimum and maximum.

Configuration: set the time and date. Configure the main alarms and variation in percentages of fill and drain times.

Scheduler: enable and set the daily and weekly time bands.

Network menu

Display a summary of the status of the units in the advanced Master/Slave system.

Maintenance

Timers: display boiler and unit operating hours. Display the time remaining until maintenance is required and set the maintenance pre-alert.

Logs: display the log of main variables (production, set point, drain pump status, fill valve status, request, unit status).

Live: real time display of the main variables (production, set point, drain pump status, fill valve status, request, unit status).

Info

Unit info: information on the unit model and software version. Select language and unit of measure.

Resources: useful links (Carel website, gaSteam manuals and pages on Carel website).

Guide & FAQ: general information on using the web server.

Note: to avoid incorrect settings, some of the main unit operating parameters can only be modified via the web server when the unit is off (off from keypad).

10. Unit hardware backup

For applications that require continuous humidity control, a second backup unit may be required, which is automatically activated in the event of a malfunction on the first. The gaSteam controller features a dedicated digital input and output for the backup connection, so as to guarantee, via the normally-open contact, activation of the second unit.

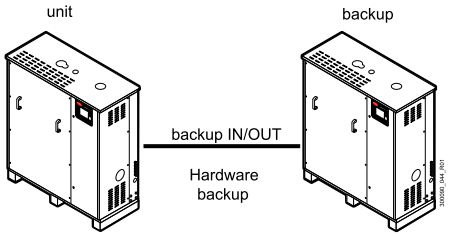


Fig.10.a

Connection between two units for the hardware backup function:

Terminal	Function
9U	Backup and rotation contact common
10U	Backup and rotation contact (NO)
2U	GND
4U	Digital input for backup/rotation function

Tab.10.a

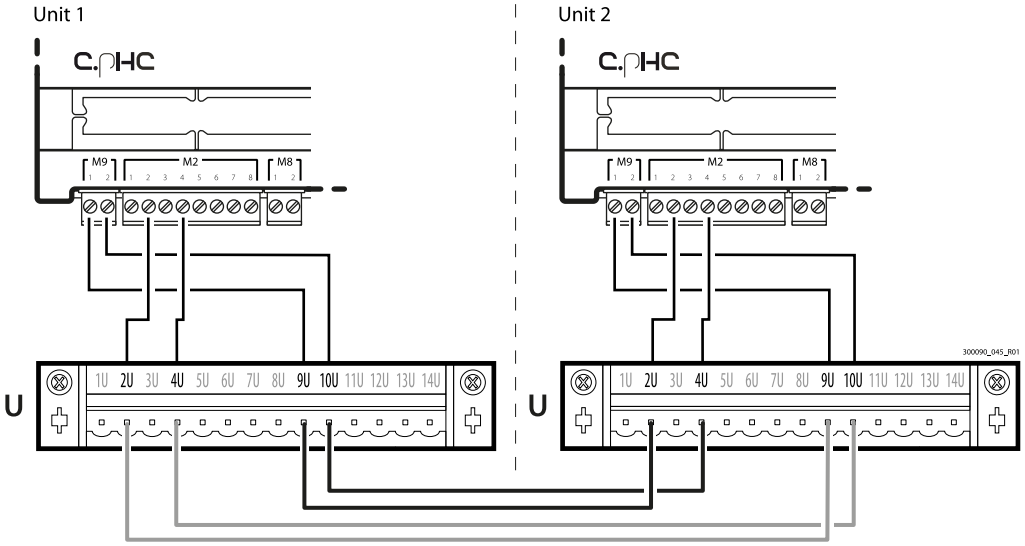


Fig.10.b

The hardware backup function must be activated on the menu in screen Ee01. Screen Ee02 is used to set the unit with higher priority in the event of simultaneous activation. To ensure service continuity, the external signal or probes must be connected to both units; in this way, both the main unit and the backup unit will be completely independent. If using autonomous control with probes, it is recommended to use dedicated probes for each unit (main probe and limit probe), in other words, do not share the probes between the units. This will guarantee service even in the event of probe malfunctions.

11. Supervisor network

11.1 Protocols and configuration

The humidifier can be connected to a supervisor via serial (BMS) or Ethernet network. The Carel, ModBus and BACnet protocols are supported as standard by the units.

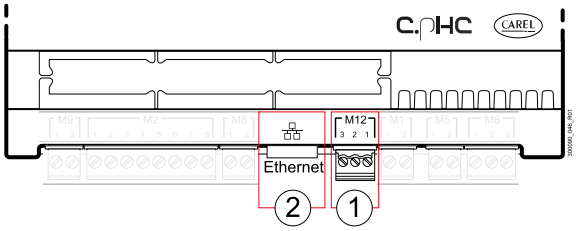


Fig.11.a

Port	Terminal on c.pHC controller	Protocol
BMS (1)	M12.1, M12.2, M12.3	Carel, ModBus, BACnet
Ethernet (2)		ModBus, BACnet

Tab.11.a

To configure supervision on the BMS port (1), go to "SETTINGS – SUPERVISION", specifically screens Eh01, Eh02 and Eh03. If connecting the supervisor to the Ethernet port (2), set the network parameters (DHCP, IP address, subnet mask, gateway, DNS) on screen Eh04. For other supervisor protocols, select the Carel protocol on the BMS serial port and use the Carel external gateway (supernode for humidification: SNU0000EM0), combined with the desired supervision card (depending on the protocol).

Note:

- use CAT-5 STP Ethernet cable, max length = 100 m. To connect the shield, use the earth connector provided on the controller;
- for BMS and Fieldbus, use shielded cables with the shield connected to earth.

12. Wireless probes

12.1 Installation and electrical connections

For installations where standard wired probes can not be used, (for example, modifications to existing systems), wireless probes can be used. These are connected via a device called an access point (Carel P/N: WS01AB2M20), for up to four wireless probes. The Carel wireless probes available are for room installation (WS01G01M00) or industrial applications (WS01F01M00), both of which measure humidity and temperature. Installation diagram (with four wireless room probes):

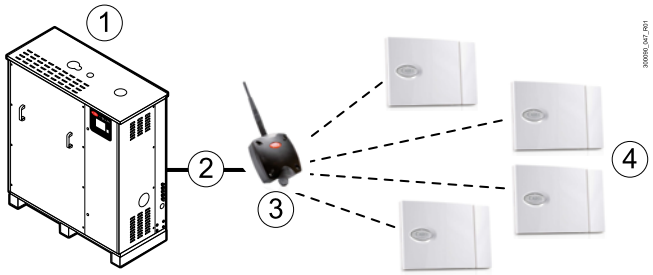


Fig.12.a

Key:

1. gaSteam humidifier;
2. humidifier/access point connection;
3. Access point;
4. Wireless probes for measuring temperature and humidity;

gaSteam/access point connection:

To connect the humidifier to the access point, use the Fieldbus connection on terminal M3 (M3.1: Tx/Rx-, M3.2: Tx/Rx +, M3.3: GND):

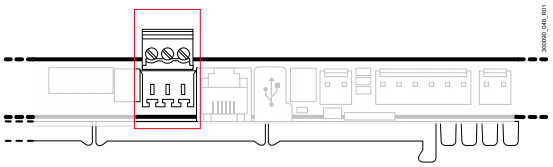


Fig.12.b

Note: the devices have a range of around a hundred metres in a free space, i.e. without any obstacles. In a closed space, the range varies considerably depending on the type of environment and the surrounding objects (shelves, metal partition walls etc.).

If installing multiple wireless probes, the controller will calculate the weighted average between the various probe readings, depending on the settings made by the user and the defined probe groups. Also see the paragraph "Wireless probes". The controller also displays (locally only) the battery and signal level for each probe (screens D05 to D08). The table below shows the part numbers and descriptions of the Carel devices that can be used:

P/N		Model	Characteristics	Power supply
WS01F01M00		SI sensor	Temperature/humidity for industrial use	Battery

P/N		Model	Characteristics	Power supply
WS01G01M00		SA sensor	Room temperature/humidity	Battery
WS01AB2M20		Access Point	ZigBee™ radio gateway - RS485 ModbusR	12-24 Vac/dc ± 10% 100 mA; 50/60Hz; use a Class II safety transformer with a minimum power rating of 2 VA. A 12 Vac transformer is recommended

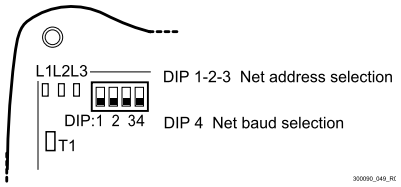
Tab.12.a

12.2 Wireless probe installation

The main steps for installing the wireless devices are:

- power up the access point (12/24 Vac/dc ±10%m, 100 mA) and complete the initialisation procedure, creating the network and selecting the channel;
- after having opened the domain on the access point, complete the binding procedure so as to uniquely identify each probe.

The address to be used by the access point is set using the dipswitches on the device, and is the following:



Dipswitch				Access point add.	Serial port baud rate (bit/s) N82
1	2	3	4	2	19200
OFF	ON	OFF	ON		

Tab.12.b

Fig.12.c

This assigns the address 2 to the access point, with a baud rate (bit/sec) of 19200 (N82). The addresses of the four serial probes, on the other hand, must be set as shown in the table below:

	Address	Dipswitch							
		1	2	3	4	5	6	7	8
Probe 1	16	0	0	0	0	1	0	0	0
Probe 2	17	1	0	0	0	1	0	0	0
Probe 3	18	0	1	0	0	1	0	0	0
Probe 4	19	1	1	0	0	1	0	0	0

Tab.12.c

Remember to verify the quality of the wireless signal between the access point and each wireless probe. For a complete description of the installation procedure, see the Carel manual on the corresponding probes and access point. To configure the probes, go the screens: Ec03, Ec04, Ec05, Ec06 and Ec07, described in the paragraph "Wireless probes". In particular, screen Ec03 is used to enable the wireless probes connected to the access point, knowing that addresses 16, 17, 18 and 19 are Probe 1, Probe 2, Probe 3 and Probe 4 respectively.

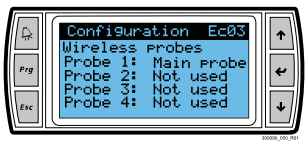


Fig.12.d

13. Alarm table

Any active alarms will be shown on the corresponding screen with direct access from the display.

The Alarm icon starts flashing; pressing the Alarm button once displays the type of alarm. In the event of potentially dangerous alarms, the humidifier automatically stops steam production. For certain alarm events, as well as the alarm signal, the alarm relay is activated (see the Alarm table). When the cause of the alarm is no longer present:

- the alarm is reset automatically or manually: the humidifier starts again and the alarm relay is deactivated;
- the message shown on the display can only be cleared manually.

Even if no longer active, the alarm status continues to be indicated until "reset display" button is pressed. Alarms that are still active cannot be reset. If there is more than one alarm, the display shows the entire list. From the alarm screen, the alarm log can be displayed ("Enter" on displays with keypad).

Code	Alarm	Cause	Possible solution	Reset	Action
ALC01	Autotest failed (unit blocked)	Probable problems with: Feedwater, level control, fill solenoid valve Note: check the alarm log for any warnings relating to the events that generated the alarm (autotest warning, level sensor warning, low production warning)	Make sure that the unit receives water Switch the unit off and clean the level control and the fill valve	Restart required	Total shutdown
ALW03	Warning: High conductivity	High water conductivity warning	Check feedwater conductivity If necessary, feature a suitable water treatment system The problem cannot be solved by softening the feedwater	Manually reset the warning	Signal only
ALC02	High conductivity (unit blocked)	High water conductivity alarm	Switch the unit off and clean the electrodes that measure the water conductivity If the problem persists, change the origin of the feedwater or feature a suitable treatment system (demineralisation, even partial) The problem cannot be solved by softening the feedwater		Closed
ALC03	Level sensor blocked (unit blocked)	The level sensor may not be working correctly	Switch the unit off and clean the boiler, the level sensor and the fill solenoid valve Check correct water supply to the boiler;	Restart required	Total shutdown
ALB01	Missing water	No feedwater	Check that the supply pipe to the humidifier and the internal tubing are not blocked or choked and that there is enough pressure (0.1-0.8 MPa, 1-8 bars) Check operation of the fill solenoid valve Check that there is not excessive back-pressure in the steam outlet, preventing water from entering the boiler due to gravity Check that the steam outlet hose is not choked or that there are no pockets of condensate Check that the level sensor is working correctly, clean if necessary	Manual/ Automatic	Stop production
ALB02	Low production	Input power not available; With the unit on, there is no steam production or water preheating; flue gas outlet or air intake blocked; gas shut-off valve closed, burner and/or controller malfunction	check the gas pressure and that the gas shut-off valve is open; make sure that the air intake is not blocked; check the fan power connection; check operation of the brushless fan, the flue gas outlet and air intake; burner flame malfunction; check the connections between the flame control board and the burner, and between the electrical panel and the flame control; check the connections between the unit controller and the intermediate terminal block in the electrical panel	Manual	Stop production
ALA01	Main probe broken or	Main room probe not connected or damaged	Check probe connection and the type of control selected	Manual	Stop production

Code	Alarm	Cause	Possible solution	Reset	Action
	not connected				
ALA02	Limit probe broken or not connected	Limit probe or second probe not connected or damaged	Check probe connection and the type of control selected	Manual	Stop production
ALA03	Preheat. probe broken or not connected	NTC water temperature probe not connected or damaged	Check preheating operation and the settings of the parameters on screen Eb02;	Automatic	Stop production
ALH01	High humid/temp (regulation probe)	High humidity in the room (high temperature with temperature control)	Check operation of the probe and the limits set on screen c01	Automatic	Signal only
ALH02	Low humid/temp (regulation probe)	Low humidity in the room	Check operation of the probe and the limits set on screen c01	Automatic	Signal only
ALH03	High humid/temp (limit probe)	High humidity at outlet	Check the operation of the outlet limit probe	Automatic	Signal only
ALW01	Foam presence	Entrainment of foam in the boiler during boiling	The entrainment of foam is generally due to the presence of surfactants in the water (lubricants, solvents, detergents, water treatment or softening agents) or an excessive concentration of dissolved salts. Purge the feedwater lines Clean the boiler	Automatic	Signal only
ALT01	Maintenance required	Planned maintenance reminder	Stop the unit and carry out complete maintenance on the humidifier, then reset the "boiler operating hours" counter (screen Ea07)	Automatic (reset operating hours)	Signal only Unit shutdown in relation to water hardness.
ALW02	Cylinder full	Boiler full of water to the high level sensor on the cover, without humidification request	Check the fill valve for leaks; Check whether the high level sensor is dirty; Check that there is no condensate return along the steam hose;	Automatic	Signal only
ALA04	Conductivity sensor	Conductivity meter not connected or damaged	Check the connection of the conductivity meter for measuring feedwater conductivity	Manual	Stop production
ALR01	Retain memory error	Problem in the electronic controller	Replace the controller	Automatic	Signal only
ALP01..4	Wireless probe 1..4 offline	No communication with probe 1..4	Check binding between probe and access point, check the battery. Check the probe signal level.	Automatic	Signal only
ALM01	No model selected	Model not set	Set a model	Automatic when the model is selected	Stop production
ALN01..20	Network: problem with unit 1..20	Network unit alarm (see the specified unit for details)	Carry out the checks based on the alarm displayed on the unit in question	Automatic	Signal only
ALP05..8	Warning: low battery wireless probe 1..4	Battery discharged on wireless probe 1..4	Check battery charge, replace if necessary;	Automatic	Signal only
ALA05	Cannot read main probe Wireless probe	All the main wireless probes are not working	Check probe connections and binding to the access point	Stop - automatic	Stop production

Code	Alarm	Cause	Possible solution	Reset	Action
ALA06	Cannot read lim. probe Wireless probe	All the limit wireless probes are not working	Check probe connections and binding to the access point	Stop - automatic	Stop production
ALP09	High boiler temp.	The boiler temperature read by the NTC probe is too high > 110°C	Check that there is water in the boiler; check that the level sensor is working and/or clean it	Stop with manual reset	Stop production
ALA07..9	Klixon fan 1..3	thermal protector on fan 1..3. Safety thermostat activated due to abnormal overheating of the boiler from operation without water	stop the unit and perform complete maintenance on the boiler	Stop with manual reset	Stop production
ALP10..12	Flue gas probe 1..3 broken	flue gas temperature probe 1 broken. NTC probe for measuring the flue gas temperature not connected or not working	Check the probe connections and/or operation	Stop with manual reset	Stop production
ALP13..15	Warning flue gas temp. 1..3	flue gas temperature probe 1..3 warning > 175°C. Extremely high flue gas temperature, boiler full of scale	switch the unit off, clean the heat exchanger, check burner calibration.		
ALP16..18	Alarm flue gas temp. 1..3	flue gas temperature probe 1..3 alarm > 180 °C. Extremely high flue gas temperature, boiler full of scale	switch the unit off, clean the heat exchanger, check burner calibration	Stop with manual reset	Stop production
ALA10..12	Alarm burner 1..3 missing flame	ignition failure on burner 1..3. No flame with production request. • when the unit is activated there is no steam production; • flue gas outlet or air intake blocked; • gas shut-off valve closed, burner and/or controller malfunction; • No gas supply	• check the gas pressure and that the gas shut-off valve is open; • make sure that the air intake is not blocked; • check the fan power connection; • check operation of the brushless fan, the flue gas outlet and air intake; • burner flame malfunction; • check the connections between the flame control board and the burner, and between the electrical panel and the flame control; • check the connections between the unit controller and the intermediate terminal block in the electrical panel	Stop with manual reset	Stop production
ALA13	AntiFreeze check components	temperature below 5°C, water drained using the pump to protect the humidifier	check the temperature inside and outside of the humidifier. Increase the temperature inside the humidifier by fitting additional heaters where necessary.	Stop with manual reset	Stop production
ALA20..22	Speed fault burner 1..3	Speed fault on burner 1..3	check the electrical connections between the control board and the fan. if necessary, remove the brushless fan and check operation.	Stop with manual reset	Stop production
ALP24	Missing model Slave	model not set on slave unit	Set the model on the slave	Stop with manual reset	Stop production
ALP25	Missing water Slave	No feedwater	Check that the supply pipe to the humidifier and the internal tubing are not blocked or choked and that there is enough pressure (0.1-0.8 MPa, 1-8 bars) Check operation of the fill solenoid valve Check that there is not excessive back-pressure in the steam outlet, preventing water from entering the boiler due to gravity Check that the steam outlet hose is not choked or that there are no pockets of condensate Check that the level sensor is working correctly, clean if necessary	Stop with manual reset	Stop production
ALM21	Aarm foam slave	Entrainment of foam in the boiler during boiling	The entrainment of foam is generally due to the presence of surfactants in the water (lubricants, solvents, detergents, water treatment or softening agents) or an excessive concentration of dissolved salts. Purge the feedwater lines Clean the boiler	Stop with manual reset	Stop production
ALB03	Low production Slave	Input power not available; With the unit on, there is no steam production or water preheating; flue gas outlet or air intake blocked; gas shut-off valve closed, burner and/or	check the gas pressure and that the gas shut-off valve is open; make sure that the air intake is not blocked; check the fan power connection; check operation of the brushless fan, the flue gas outlet and air intake; burner flame malfunction; check the connections between the flame control board and the burner, and between the		

Code	Alarm	Cause	Possible solution	Reset	Action
		controller malfunction	electrical panel and the flame control; check the connections between the unit controller and the intermediate terminal block in the electrical panel		
ALC05	Lev. sensor Slave	The level sensor may not be workingcorrectly	Switch the unit off and clean the boiler, the level sensor and the fill solenoid valve. Check correct water supply to the boiler;	Stop with manual reset	Stop production
ALC06	Alarm autotest Slave	Probable problems with: Feedwater, level control, fill solenoid valve Note: check the alarm log for any warnings relating to the events that generated the alarm (autotest warning, level sensor warning, low production warning)	Make sure that the unit receives water Switch the unit off and clean the level control and the fill valve		
ALW12	Alarm cylinder Slave	Boiler full of water to the high level sensor on the cover, without humidification request	Check the fill valve for leaks; Check whether the high level sensor is dirty; Check that there is no condensate return along the steam hose;		
ALC07	High Conductivity Slave	High feedwater conductivity alarm	Switch the unit off and clean the electrodes that measure the water conductivity If the problem persists, change the origin of the feedwater or feature a suitable treatment system (demineralisation, even partial) The problem cannot be solved by softening the feedwater	Stop with manual reset	Stop production
ALA18	PreHeat probe Slave	NTC water temperature probe not connected or damaged	Check preheating operation and the settings of the parameters on screen Eb02;	Stop with manual reset	Stop production
ALA19	Conduct probe Slave	Conductivity meter not connected or damaged	Check the connection of the conductivity meter for measuring feedwater conductivity	Stop with manual reset	Stop production
ALR02	Retain memory Slave	Problem in the electronic controller	Replace the controller		
ALA23	AntiFreeze Slave drain pump forced	temperature below 5°C, water drained using the pump to protect the humidifier	check the temperature inside and outside of the humidifier. Increase the temperature inside the humidifier by fitting additional heaters where necessary.	Stop with manual reset	Stop production
ALW04	Warning: Autotest	Probable problems with: Feedwater, level control, fill solenoid valve Note: check the alarm log for any warnings relating to the events that generated the alarm (autotest warning, level sensor warning, low production warning)	Make sure that the unit receives water Switch the unit off and clean the level control and the fill valve		Signal only
ALW05	Warning: Level sensor malfunct.	The level sensor may not be working correctly	Switch the unit off and clean the boiler, the level sensor and the fill solenoid valve Check correct water supply to the boiler;		Signal only
ALW06	Warning: Low production	Input power not available; With the unit on, there is no steam production or water preheating; flue gas outlet or air intake blocked; gas shut-off valve closed, burner and/or controller malfunction	check the gas pressure and that the gas shut-off valve is open; make sure that the air intake is not blocked; check the fan power connection; check operation of the brushless fan, the flue gas outlet and air intake; burner flame malfunction; check the connections between the flame control board and the burner, and between the electrical panel and the flame control; check the connections between the unit controller and the intermediate terminal block in the electrical panel		Solo segnalaz.
ALA14	Unit Slave Offline	Slave unit Offline			
ALA15	Master Offline	Master unit Offline			
ALA16	AntiFreeze	AntiFreeze. Temperature below 10°C,	check the temperature inside and outside of the humidifier. Increase		

Code	Alarm	Cause	Possible solution	Reset	Action
	force pre-heating	preheating activation to protect the humidifier	the temperature inside the humidifier by fitting additional heaters where necessary.		
ALA17	AntiFreeze Slave force pre-heating	AntiFreeze in Slave unit. Temperature below 10°C, preheating activation to protect the humidifier	check the temperature inside and outside of the humidifier. Increase the temperature inside the humidifier by fitting additional heaters where necessary.		
ALP21	SV Offline	Supervisor Offline			
ALW07	Warning Autotest Slave	Autotest warning in Slave unit. Probable problems with: Feedwater, level control, fill solenoid valve Note: check the alarm log for any warnings relating to the events that generated the alarm (autotest warning, level sensor warning, low production warning)	Make sure that the unit receives water. Switch the unit off and clean the level control and the fill valve		Signal only
ALW09	Alarm Autotest Slave	Autotest alarm in Slave unit. Probable problems with: Feedwater, level control, fill solenoid valve. Note: check the alarm log for any warnings relating to the events that generated the alarm (autotest warning, level sensor warning, low production warning)	Make sure that the unit receives water. Switch the unit off and clean the level control and the fill valve		
ALW10	Warning Sensor level Slave	Warning level sensor in Slave unit. The level sensor may not be working correctly	Switch the unit off and clean the boiler, the level sensor and the fill valve. Check correct filling of water in boiler		Signal only
ALW11	Warning Low production Slave	Warning low production in Slave unit. Input power not available; With the unit on, there is no steam production or water preheating; flue gas outlet or air intake blocked; gas shut-off valve closed, burner and/or controller malfunction	check the gas pressure and that the gas shut-off valve is open; make sure that the air intake is not blocked; check the fan power connection; check operation of the brushless fan, the flue gas outlet and air intake; burner flame malfunction; check the connections between the flame control board and the burner, and between the electrical panel and the flame control; check the connections between the unit controller and the intermediate terminal block in the electrical panel		Signal only

Tab.13.a

14. Spare parts and maintenance

Before all operations:

- disconnect the appliance from the mains power supply;
- close the water supply and gas shut-off valves;
- drain the water circuit using the manual electric pump function or the drain valve provided, after connecting a pipe to drain the water outside of the unit and avoid flooding.

⚠ Important:

- do not use detergents or solvents to clean the plastic components.
- descaling can be performed using a 20% acetic acid solution, then rinsing with plenty of water.

14.1 Boiler maintenance

Access the boiler as described in the chapter "Introduction and installation".

Remove panels A, B and C as follows (see the figure):

- detach the steam hose from the boiler;
- undo screws V and V1;
- undo the internal and external screws that hold secure panel B;
- remove panels A, B and C.

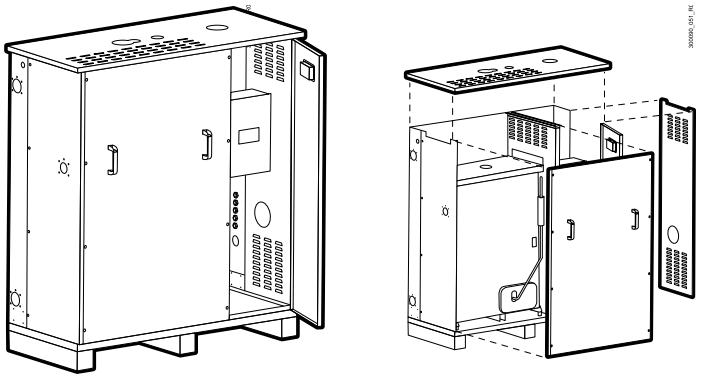


Fig.14.a

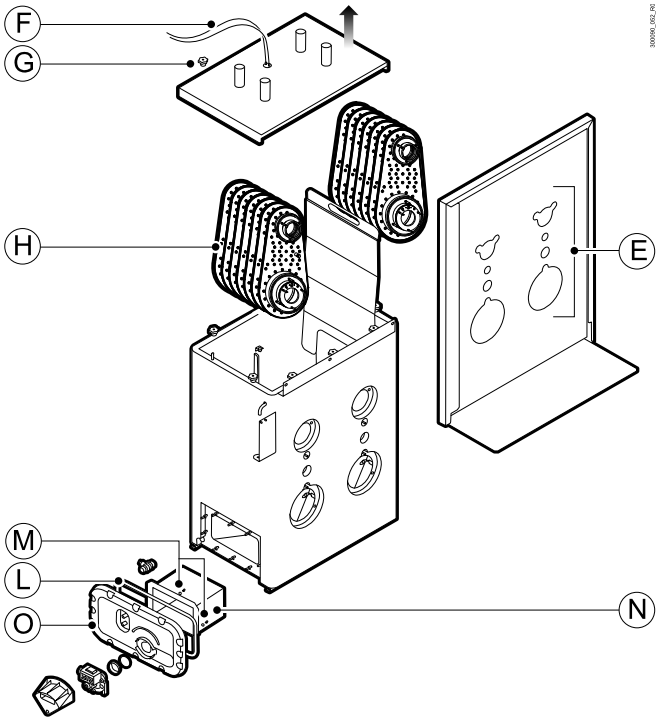


Fig.14.b

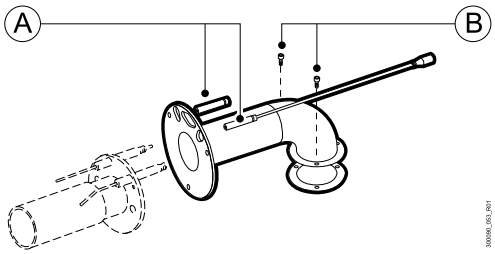


Fig.14.c

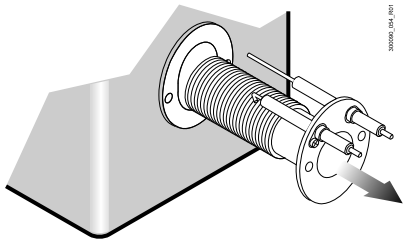


Fig.14.d

To remove the heat exchanger, proceed as follows:

- disconnect the cables from the burner electrodes (the detection electrode must be disconnected from the burner control board, see the figure, pos. "**A**");
- remove the fan manifold by undoing the screws **B** (Fig. 14.c) and remove the burner combustion head (Fig. 14.d);
- disconnect the cables from the foam detection electrode F (Fig. 14.b);
- unscrew and remove the locking nuts **G**;
- remove the boiler cover;

- remove the nuts **E** from the side of the burner;
- take out the heat exchanger **H** and wash it using a 20% acetic acid solution, removing scale using implements that do not scratch the lining (e.g. wood or plastic). At the end of the process, rinse thoroughly.
- disconnect the power supply cable and all the pipes/hoses connected to the electric pump and the cover **O**;
- undo the panel fastening nuts and remove it, making sure not to damage the gasket **L**;
- unscrew and remove the screws **M** to release the filter **N** and wash it in a 20% acetic acid solution;
- using a wooden or plastic scraper, scrape the inside of the vaporiser chamber.

14.2 Cleaning the burner

The burner must be checked by authorised and qualified personnel once or twice a year, according to use.

Before performing any maintenance on the burner, check its general condition, carrying out the operations listed below:

- remove the burner head as described previously;
- using a brush, clean the inside of the burner head; make sure not to crush the metal mesh (Fig. 14.e);
- detach all gas and electrical connections from the burner assembly;
- check for dust deposits on the fan and if necessary remove the parts required to clean it (Fig. 14.f).
- clean the fan using a brush (Fig. 14.g).

⚠ **Important:** to avoid damaging the fan, never use a jet of compressed air when cleaning it.

When reassembling the parts, check:

- the condition of the gaskets (replace if necessary);
- that the position of the electrodes corresponds to the figure.

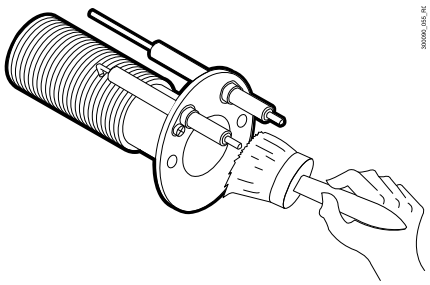


Fig.14.e

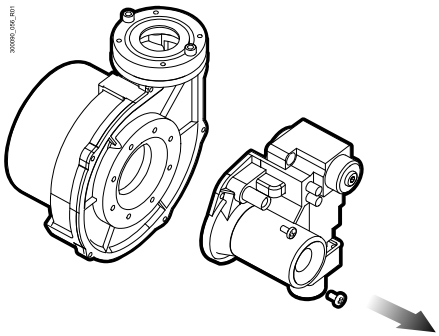


Fig.14.f

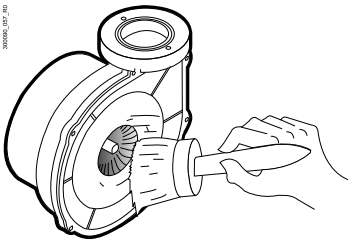


Fig.14.g

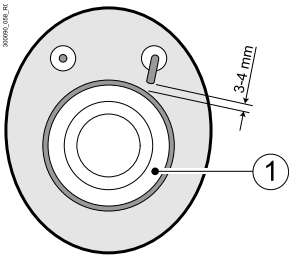


Fig.14.h

14.3 Checking ionisation current

Ionisation current is checked by placing a microammeter set to 10 µA full scale (direct current) in series with the flame detection electrode. Incorrect positioning of the electrode may lead to a decrease in the ionisation current and burner shutdown due to incorrect flame detection. In this case, check the correct position and condition of the electrode, its electrical connections, and the earth connection of the burner. The ionisation current is normally 5 µA.

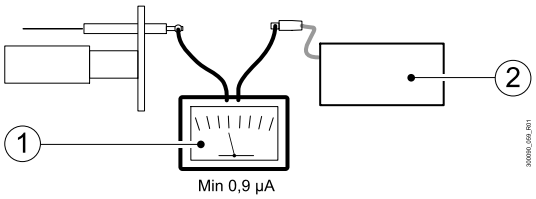


Fig.14.i

1	microammeter with 10 µA full scale
2	test equipment

14.4 Flue gas temperature sensor

The flue gas temperature sensor is located in the flue and does not require periodical maintenance. If the sensor needs to be replaced due to a fault, proceed as follows:

- stop the humidifier by turning the switch to 0 (OFF) and checking that the display is off;
- open the electrical compartment to access the panel;
- loosen terminals 10T and 11T (or 10Z and 11Z, see the wiring diagram) on the terminal block in the bottom of the electrical panel and the probe holder nut on the adapter joint and remove the probe and the cable.

Then replace the sensor by following the same procedure in reverse.

14.5 Water temperature sensor

This does not require periodical maintenance.

To replace the sensor, proceed as follows:

- stop the humidifier by turning the switch to 0 (OFF) and checking that the display is off;
- open the panels to access the water circuit (Fig. 14.a);
- remove the sensor from the stainless steel housing in the level sensor support at the front;
- loosen terminals 12T and 13T (see the wiring diagram) on the terminal block at the bottom and remove the probe.

Then replace the sensor by following the same procedure in reverse.

14.6 Fuses

Fuses 1, 2, 3 measure 10.3x38 mm and are contained inside the fuse carrier; while fuse 4 measures 6.3x20 mm; to check the fuses, test continuity using a tester. Use fuses with the ratings indicated in the table:

model	**UG045, UG090, UG150, UG180, UG300, UG450**
fuses 1 and 2	2 A T
fuse 3	1 A F
fuse 4	3.15 A T

Tab.14.a

14.7 Cooling fan

The cooling fan starts when the unit is switched on, and is used to keep the operating temperature of the electrical panel and the electronics within the design limits. If the fan is faulty, after having removed panel B (Fig. 14.a):

- detach the electrical connections;
- replace the fan after having unscrewed the fastening screws;

Note: being thermally protected, the fan may switch off temporarily if it overheats, and then will start again after having cooled down.

15. Wiring diagrams

15.1 Electrical panel

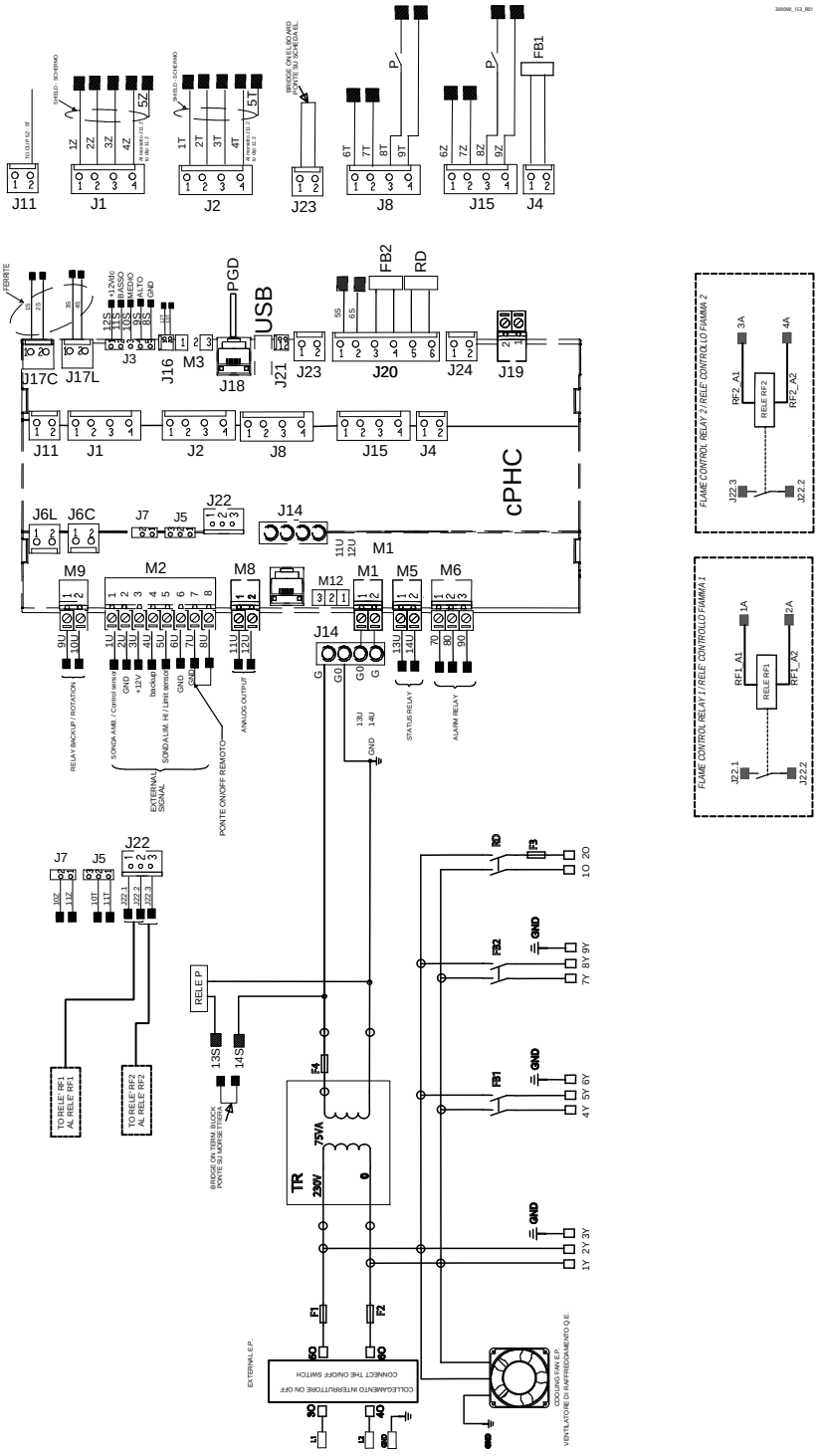
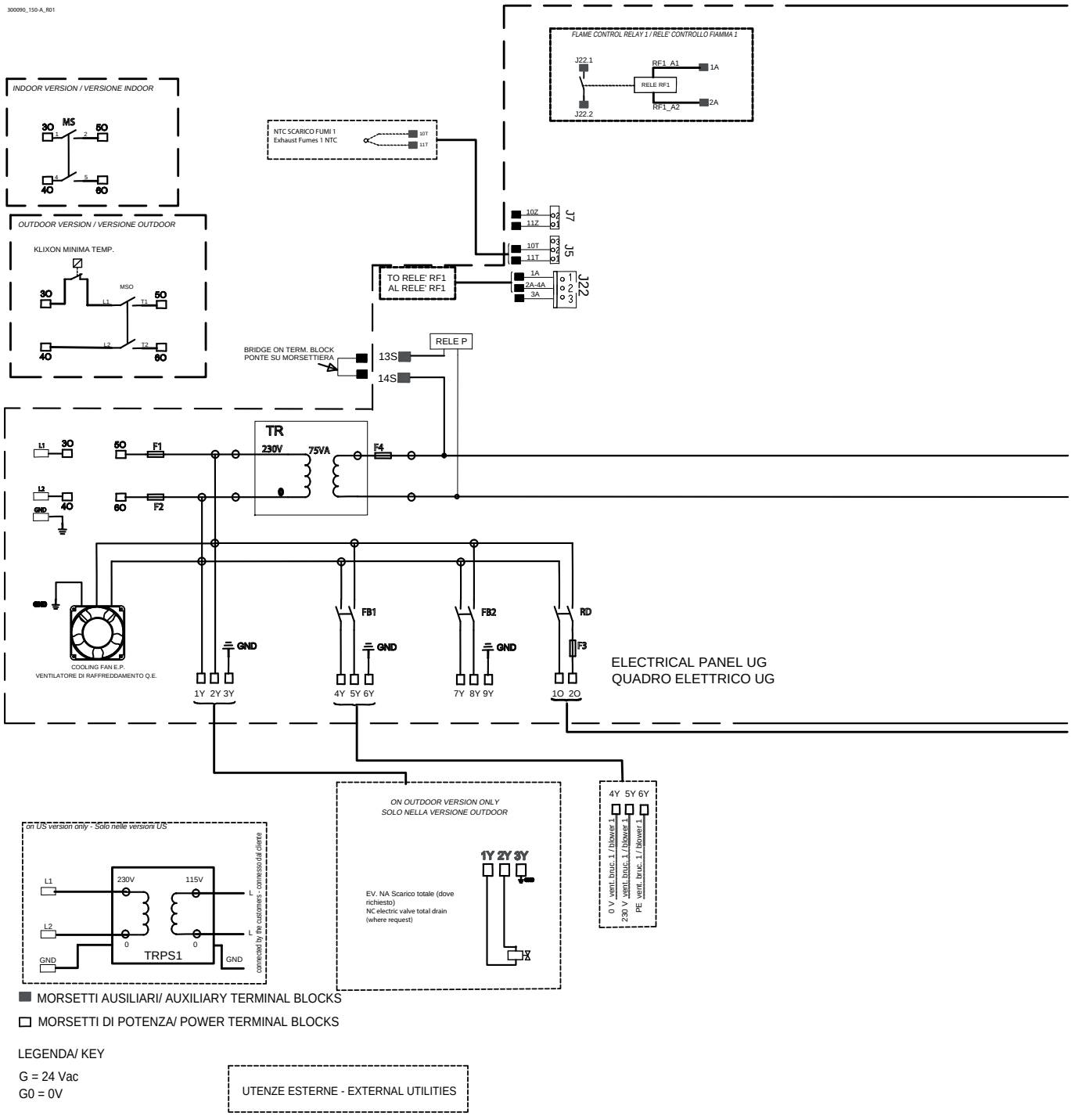


Fig.15.a

15.2 Wiring diagram for UG045 - UG090 - UG150



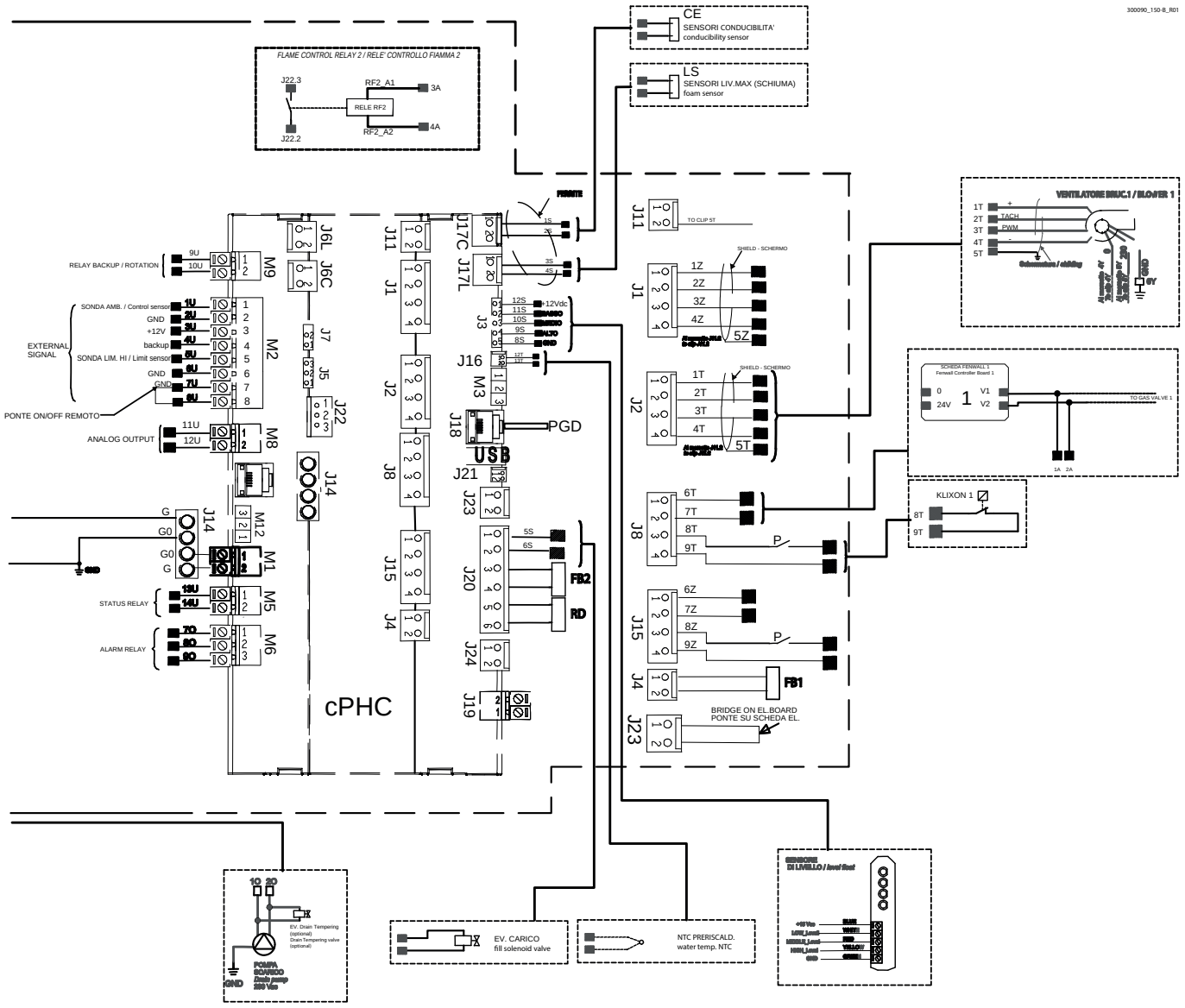
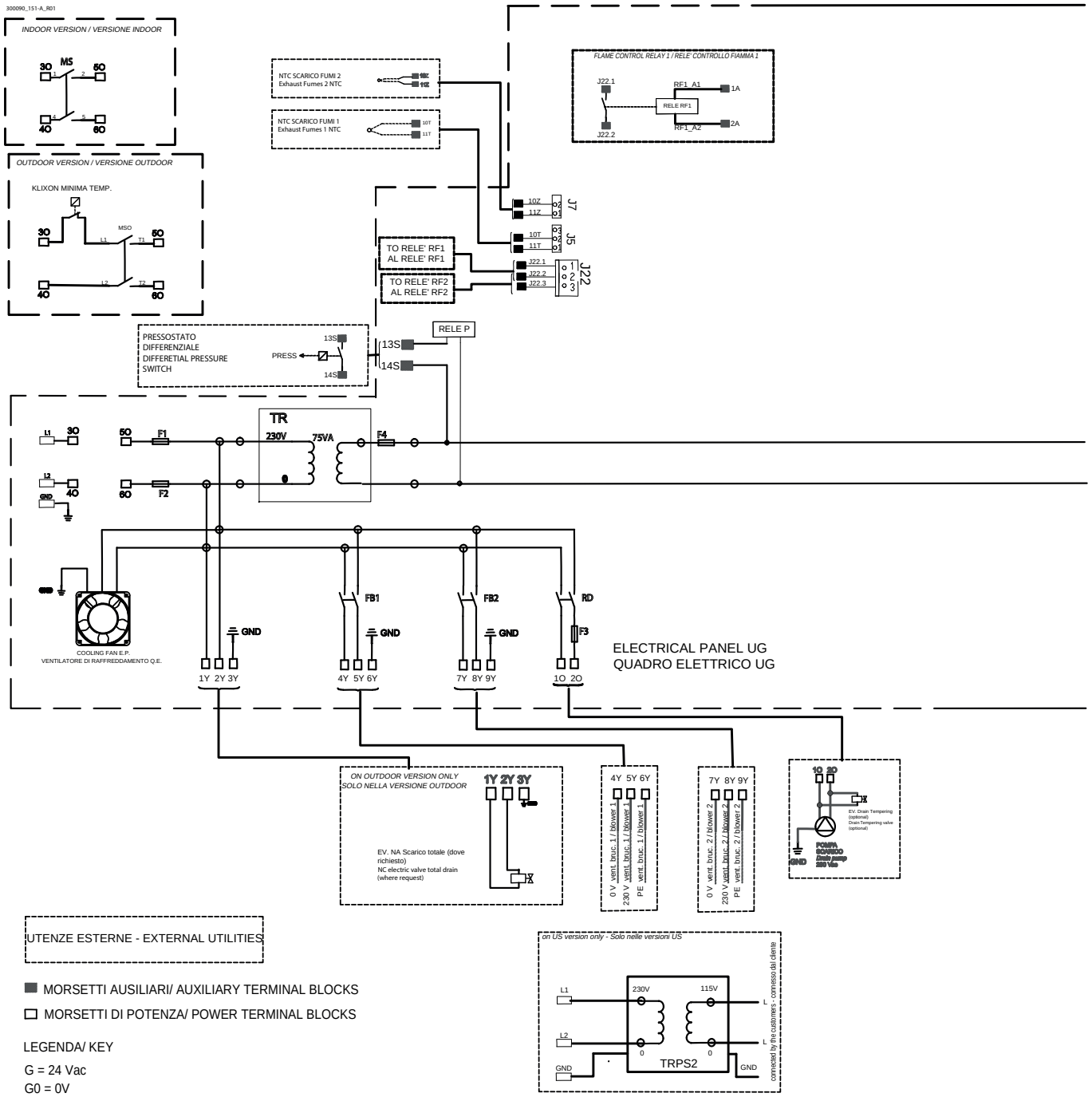

CE
SENSORI CONDUCIBILITA'
conducibility sensor
LS
SENSORI LIV.MAX (SCHIUMA)
foam sensor
FLAME CONTROL RELAY 2 / RELE' CONTROLLO FIAMMA 2
RELE RF2
cPHC
PGD
USB
EV. CARICO
fill solenoid valve
NTC PRERISCALD.
water temp. NTC
BRIDGE ON EL.BOARD
PONTE SU SCHEDA EL.
KLIXON 1
TO GAS VALVE 1
RELAY BACKUP / ROTATION
EXTERNAL SIGNAL
PONTE ON/OFF REMOTO
ANALOG OUTPUT
STATUS RELAY
ALARM RELAY

Fig.15.b

15.3 Wiring diagram for UG180 - UG300



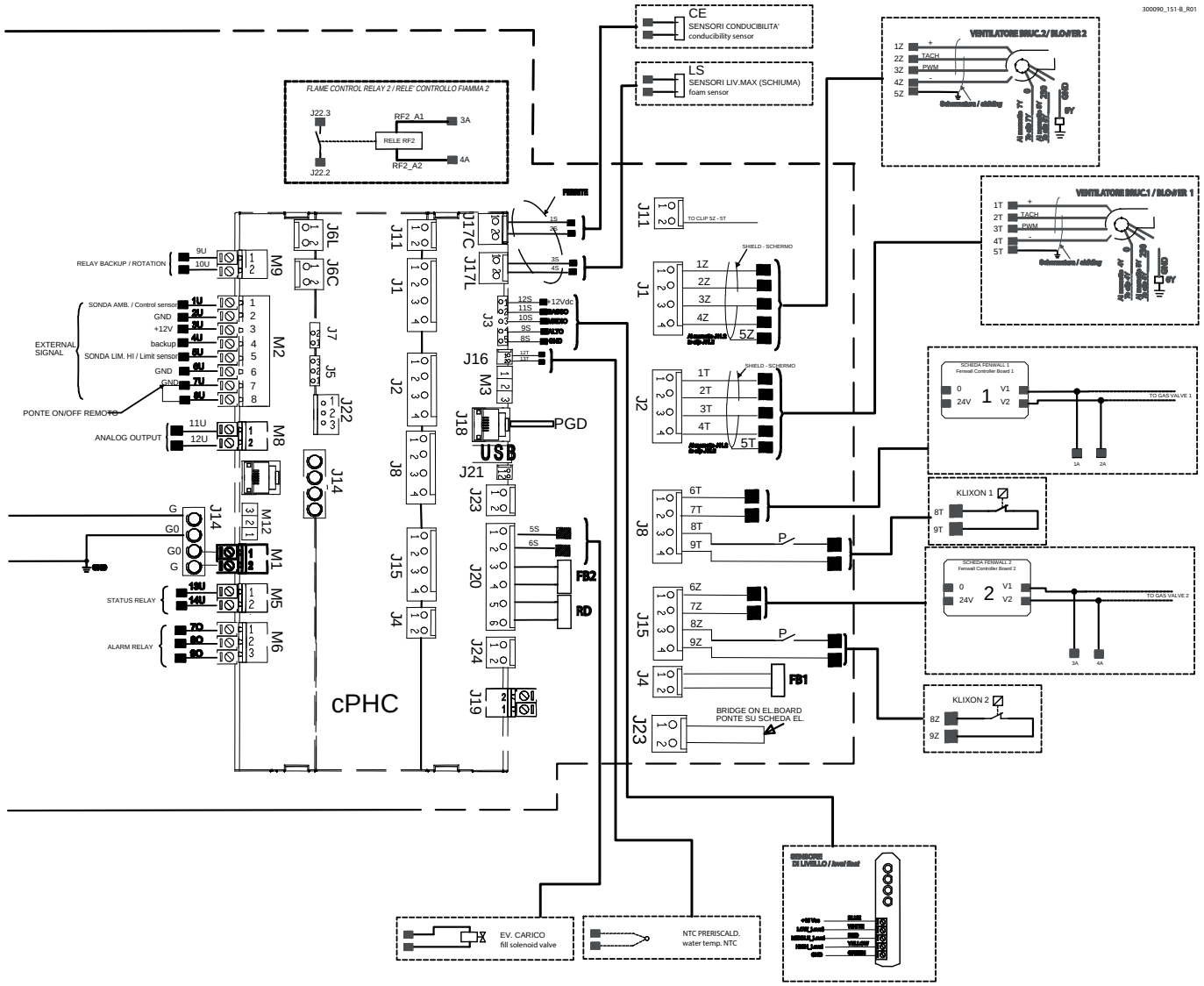
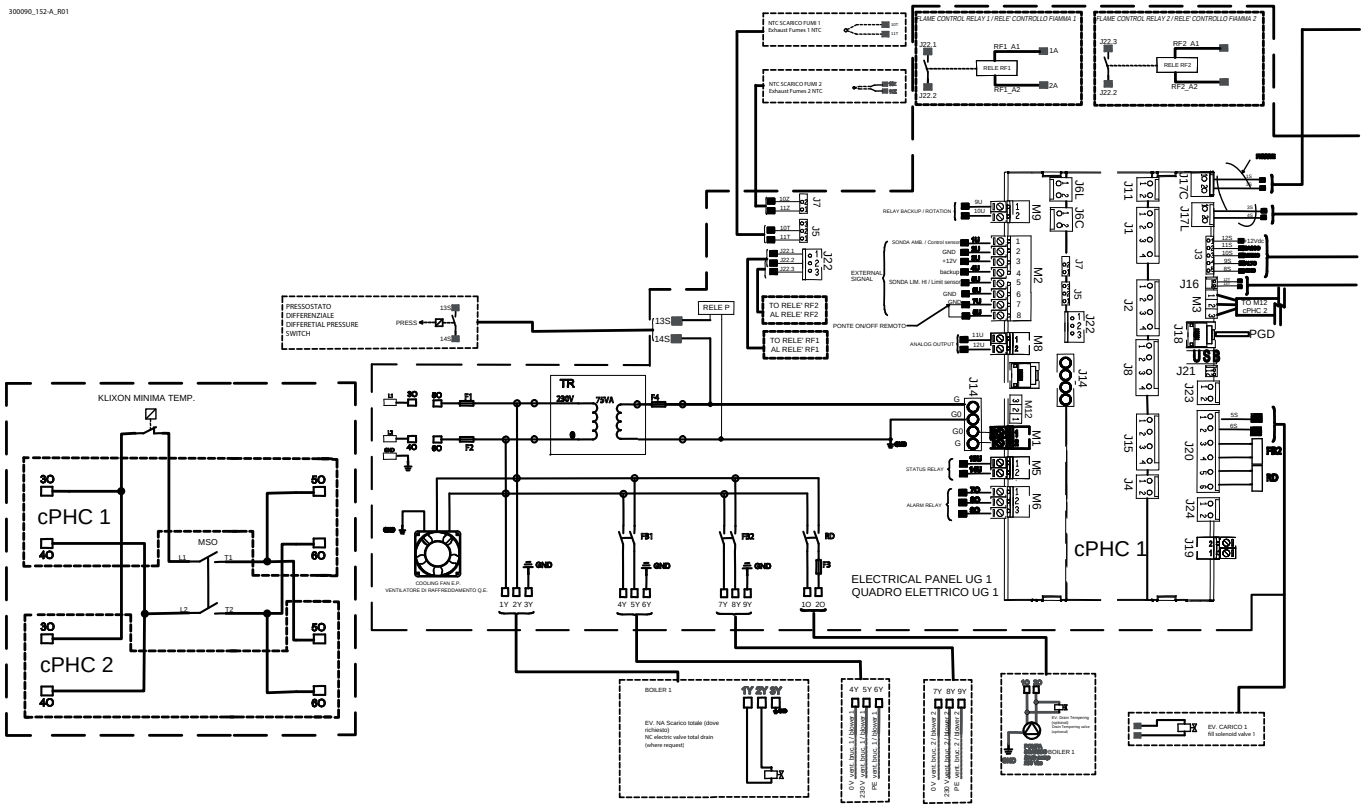
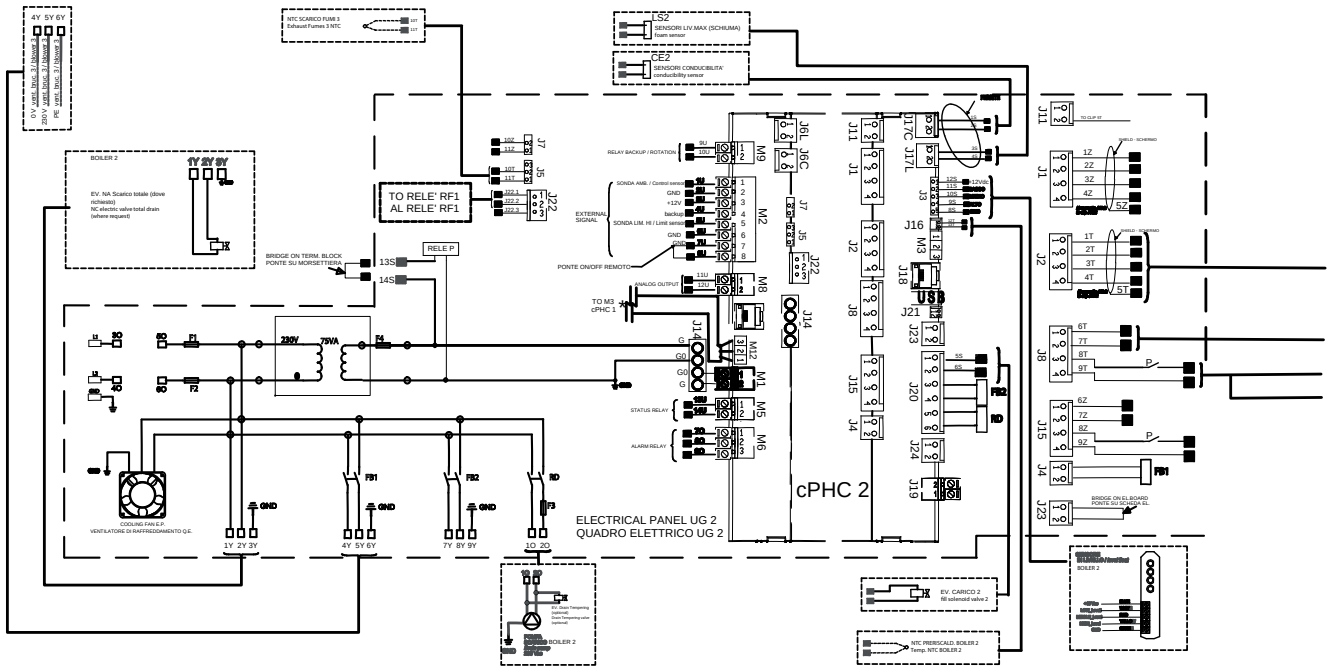
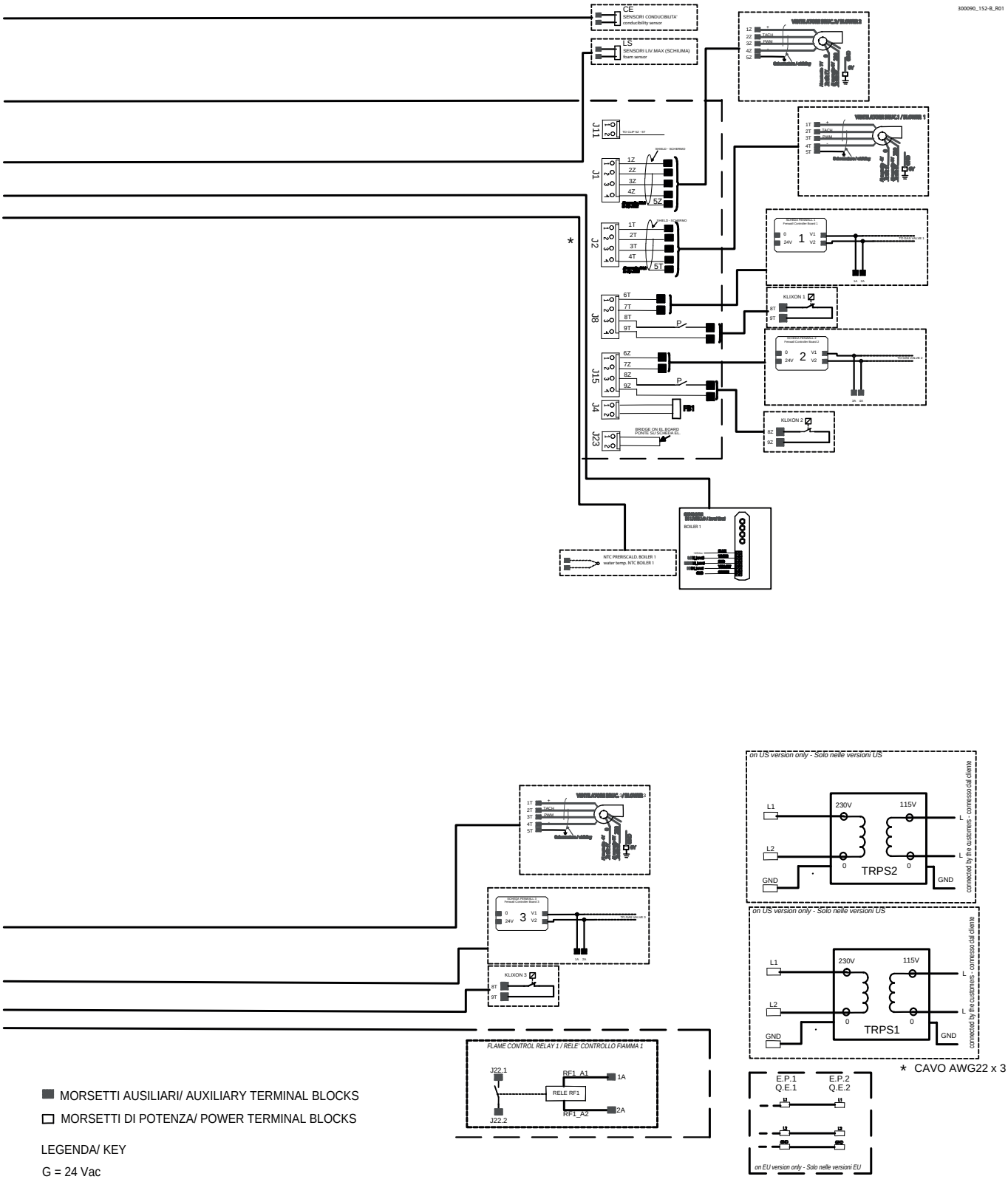

CE
SENSORI CONDUCIBILITA'
conducibility sensor
LS
SENSORI LIV.MAX (SCHIUMA)
foam sensor
FLAME CONTROL RELAY 2 / RELE' CONTROLLO FIAMMA 2
RELE RF2
RF2_A1
RF2_A2
J22.3
J22.2
3A
4A
RELAY BACKUP / ROTATION
SONDA AMB. / Control sensor
GND
+12V
backup
SONDA LIM. HI / Limit sensor
GND
EXTERNAL SIGNAL
PONTE ON/OFF REMOTO
ANALOG OUTPUT
STATUS RELAY
ALARM RELAY
M1
M2
M5
M6
M8
M9
M12
J1
J2
J3
J4
J5
J6L
J6C
J7
J8
J11
J14
J15
J16
J17C
J17L
J18
J19
J20
J21
J22
J23
J24
PGD
USB
FB1
FB2
RD
cPHC
TO CLIP 5Z - 5T
SHIELD - SCHERMO
5Z
5T
KLIXON 1
KLIXON 2
BRIDGE ON EL.BOARD
PONTE SU SCHEDA EL.
TO GAS VALVE 1
TO GAS VALVE 2
EV. CARICO
fill solenoid valve
NTC PRERISCALD.
water temp. NTC
FERRITE

Fig.15.c

15.4 Wiring diagram for UG450







■ MORSETTI AUSILIARI/ AUXILIARY TERMINAL BLOCKS

□ MORSETTI DI POTENZA/ POWER TERMINAL BLOCKS

LEGENDA/ KEY

G = 24 Vac

G0 = 0V

Fig.15.d

Wiring diagram key

Ref.	Description
Base	Selectable fuse carrier
cPHC	Main board + expansion card
P	Differential pressure switch (180/300/450 kg/h)
AL	3-pin through connector
A	4-pin through connector
Klixon 1/2/3	Safety thermostat
TRPS1	115-230 V transformer (UG45-90-150)
U-T-S	14-pin through connector
Z	14-pin through connector
Y-O	4-pin through connector
MSO	ON/OFF switch (outdoor)
MS	Two-pole ON/OFF switch
K.MINIMA	Safety thermostat (outdoor only)
TRPS2	115-230 V transformer (UG180-300)
F1-F2	Power fuses
F3	Pump fuse
RD	Pump control relay
FB1-FB2	Burner fan control relay
P	Pressure switch control relay
RF1-RF2	Flame control relay 1-2

Tab.15.a

15.5 Electrical panel

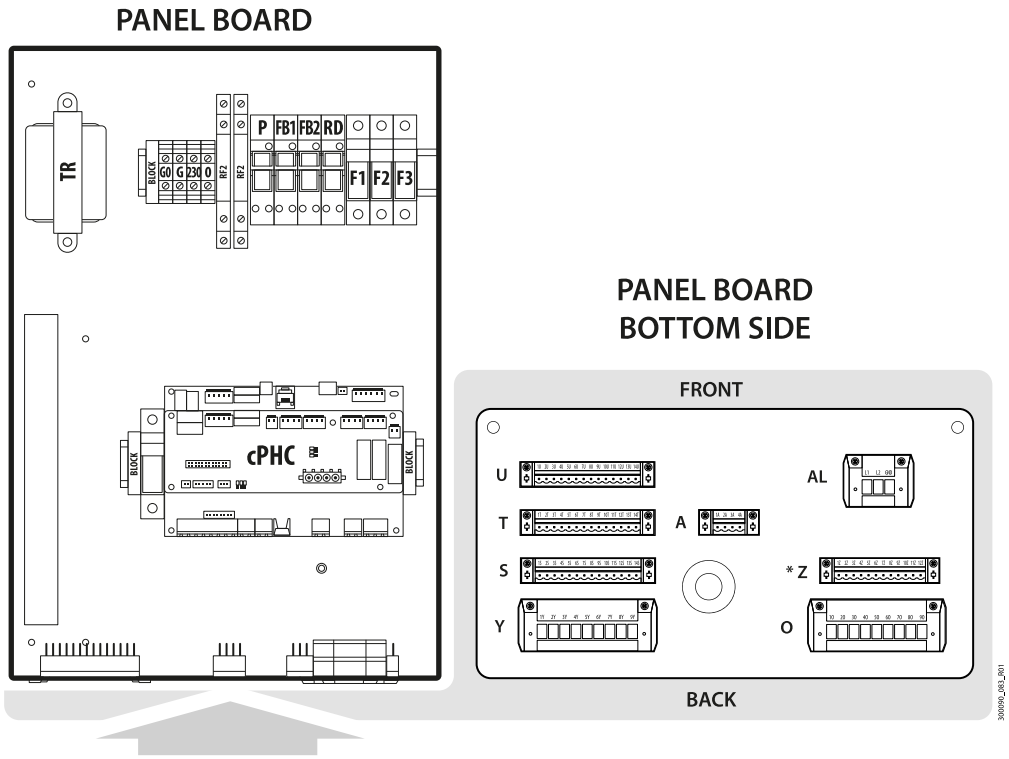


Fig.15.e

16. General features and models

Models for indoor installation:

INDOOR models	UG045	UG090	UG150	UG180	UG300
rated supply voltage (Vac)	230 V 50 Hz (ver. UG***HD004) / 115V 60 Hz (ver. UG***HD1104)				
steam connection (Ø mm)	2x40	2x40	1x80	4x40	2x80
steam outlet pressure limits (Pa)	0-2000 (0-0.30 PSI)				
feedwater					
connection	3/4"				
temperature limits °C (°F)	1-40 (33.8-104)				
pressure limits (MPa)	0.1-0.8 (1-8 bars) (14.5-166 PSI)				
fill water instant flow-rate (l/min)	18 (4.76 US gall/min)				
drain water					
connection ø mm (in)	50 (1.97)				
typical temperature °C (°F)	≤100 (212)				
drain water instant flow-rate (l/min)	32 (6.60 gall/min)				
operating conditions	1T40°C (33T104°F); 10-90% rH non cond.				
storage conditions	-10T70°C (14T158°F), 5-95% rH				
protection rating	IP20				
auxiliary voltage / frequency (V - Hz)	24 / 50-60				
instant steam production (kg/h / lbs/h) (1)	45 (100)	90 (200)	150 (330)	180 (400)	300 (660)
power consumption at rated voltage (W)	180	250	260	385	400

Tab.16.a

(1) average steam production is affected by factors, such as: ambient temperature, water quality, steam distribution system.

Models for outdoor installation:

OUTDOOR models	UG045	UG090	UG150	UG180	UG300	UG450
rated supply voltage (Vac)	230 V 50 Hz (ver. UG***YD004) / 115 V 60 Hz (version UG***Y1104)					
steam connection (Ø mm)	2x40	2x40	1x80	4x40	2x80	3x80
steam outlet pressure limits (Pa)	0-2000 (0-0.30 PSI)					
feedwater						
connection	3/4"					
temperature limits °C (°F)	-20 to + 45 (33.8-113) with heaters					
pressure limits (MPa)	0.1-0.8 (1-8 bars) (14.5-166 PSI)					
fill water instant flow-rate (l/min)	18 (4.76 US gall/min)					
drain water						
connection ø mm (in)	50 (1.97)					
typical temperature °C (°F)	≤100 (212)					
drain water instant flow-rate (l/min)	32 (6.60 gall/min)					
operating conditions	1T45°C (33T104°F); 10-90% rH non cond.					
storage conditions	-10T70°C (14T158°F), 5-95% rH					
protection rating	outdoor IAS 12-94					
auxiliary voltage / frequency (V - Hz)	24 / 50-60					
instant steam production (kg/h / lbs/h) (1)	45 (100)	90 (200)	150 (330)	180 (400)	300 (660)	450 (1000)
power consumption at rated voltage (W)	180	250	260	385	400	660

Tab.16.b

(1) average steam production is affected by factors, such as: ambient temperature, water quality, steam distribution system.

controller specifications:

type	c.pHC
auxiliary voltage / frequency (V - Hz)	24 / 50-60
maximum auxiliary power (VA)	90
probe inputs (general specifications)	selectable for signals: 0-1 Vdc, 0-10 Vdc, 2-10 Vdc, 0-20 mA, 4-20 mA
	input impedance: 60 kΩ with signals: 0-1 Vdc, 0-10 Vdc, 2-10 Vdc
	50 Ω with signals: 0-20 mA, 4-20 mA
power supply for active probes (general specifications)	24 Vdc (24 Vac rectified), Imax= 250 mA
	12 Vdc 5%, Imax= 50 mA
alarm relay outputs (general specifications)	250 V 2 A (2 A)
	type of micro-switching action 1C
remote enabling input (general specifications)	voltage-free contact; max. resistance 50 Ω; Vmax = 24 Vdc; Imax = 5 mA
serial communication	three-wire RS485 / Fieldbus / BMS
USB port	USB type A
Ethernet port	Ethernet

Tab.16.c

16.1 Heating-water circuit specifications

		UOM	UG045	UG090	UG150	UG180	UG300	UG450
rated heat output	rated	kW (BTU/h)	33 (112763)	62.5 (213449)	105 (358275)	125 (426518)	210 (716550)	315 (1074825)
	minimum		7.8 (26614)	14.7 (50158)	24.7 (84280)	14.7 (50158)	24.7 (84280)	24.7 (84280)
rated heat	input		34.8 (118742)	65 (221789)	108 (368511)	130 (443578)	216 (737023)	324 (1105534)
	minimum		8.7 (29686)	16.25 (55447)	27 (92128)	16.25 (55.5)	27 (92128)	27 (92128)
rated steam production	rated	kg/h (lbs/h)	45 (100)	90 (200)	150 (330)	180 (400)	300 (660)	450 (990)
	minimum		11.25 (25)	22.5 (50)	37.5 ()	22.5 (50)	37.5 ()	37.5 ()
max steam temp.		°C (°F)	105 (221)	105 (221)	105 (221)	105 (221)	105 (221)	105 (221)
water content in steady operation		l	120	120	200	200	415	615
NOx emissions		class	5(<70 mg/kW/h	5(<70 mg/kW/h	5(<70 mg/kW/h	4 (<100 mg/kWh)	4 (<100 mg/kWh)	4 (<100 mg/kWh)
CO2	natural gas (G20)	% vol	9.4	9.4	9.4	9.4	9.4	9.4
	natural gas (G25)		9.3	9.3	9.3	9.3	9.3	9.3
	propane (G31)		11.2	11.2	11.2	11.2	11.2	11.2
	butane (G30)		11.6	11.6	11.6	12	12	12
CO		mg/kWh	* <25	* <60	* <60	* <60	* <60	* <60
D. flue gas pipe***		mm	80 (3")	80 (3")	80 (3")	2xØ80 (3")	2xØ80 (3")	3xØ80 (3")
D. air intake pipe***			80 (3")	80 (3")	80 (3")	2xØ80 (3")	2xØ80 (3")	3xØ80 (3")
D. steam hose			2x Ø40	2x Ø40	1x Ø80	4x Ø40	2x Ø80	3x Ø80
natural gas combustion flow-rate (G20)	rated	m^3 St/h**	3.68	6.87	11.45	13.4	22.7	34.4
	minimum		0.9	1.75	2.91	1.67	2.85	2.85
natural gas combustion flow-rate (G25)	rated		4.2	8.7	14.6	17.5	29.2	43.8
	minimum		1.02	1.98	3.3	1.98	3.3	3.3
propane combustion flow-rate (G31)	rated		1.43	2.68	4.47	5.36	8.94	13.41
	minimum		0.48	0.68	1.13	0.68	1.13	1.13
butane combustion flow-rate (G30)	rated		1.1	2.06	3.43	4.12	6.86	10.29
	minimum		0.37	0.545	0.91	0.545	0.91	0.91
gas supply pressure	natural gas (G20)	Pa/mbar/PSI	2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9
	natural gas (G25)		2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9	2000/20/0.9
	propane (G31)		3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44
	butane (G30)		3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44	3000/30/0.44
max allowable pressure drop for		Pa/mbar/PSI	90/0.90/0,013	82/0.82/0,012	82/0.82/0,012	95/0.95/0,014	95/0.95/0,014	95/0.95/0,014

		UOM	UG045	UG090	UG150	UG180	UG300	UG450
air intake and flue gas exhaust								

Tab.16.d

* value refers to combustion of natural gas (G20); Tab. 8.b

** m ^{3} St = dry gas at 15°C and 1013.25 mbar atmospheric pressure;

***: using the specific KITNSTALL for USA.

16.2 Technical values of the flue gas according to useful heat input

fuel type	**natural gas (G20)**			**natural gas (G25)**			**propane (G31)**			**butane (G30)**		
	UG045	UG090	UG180	UG045	UG090	UG180	UG045	UG090	UG180	UG045	UG090	UG180
rated heat input	34.76/	65.00/	130.0/	34.76/	65.00/	130.0/	34.76/	65.00/	130.0/	34.76/	65.00/	130.0/
(kW/l/h/BTU)	118.712	221.987	443.973	118.712	221.987	443.973	118.712	221.987	443.973	118.712	221.987	443.973
flue gas flow-rate (kg/s)	0.0163	0.0303	0.0606	0.0167	0.03115	0.0623	0.0154	0.0283	0.0566	0.0147	0.0276	0.0551
flue gas temperature °C (°F)	135 (253)	170 (338)	165 (329)	123 (253)	175 (347)	163 (325)	123 (253)	175 (347)	165 (329)	123 (253)	175 (347)	163 (325)
CO2 percentage in flue gas (%)	9.4	9.4	9.4	9.3	9.3	9.3	11.2	11.4	11.2	11.6	11.6	12

Tab.16.e

fuel type	**natural gas (G20)**			**natural gas (G25)**		
	UG150	UG300	UG450	UG150	UG300	UG450
rated heat input	108/	216/	324/	108/	216/	324/
(kW/l/h/BTU)	368.511	737.023	1,106,000	368.511	737.023	1,106,000
flue gas flow-rate (kg/s)	0.048	0.096	0.144	0.0167	0.03115	0.0623
flue gas temperature °C (°F)	172 (342)	168 (334)	165 (329)	172 (342)	168 (334)	165 (329)
CO2 percentage in flue gas (%)	9.4	9.4	9.4	9.3	9.3	9.3

Tab.16.f

16.3 Steam hose models

humidifier model:	**UG045**	**UG090**	**UG150**	**UG180**	**UG300**	**UG450**
steam outlet ø mm (in):	2x 40 (1.6")	2x 40 (1.6")	80 (3.1")	4x 40 (1.6")	2x 80 (3.1")	3x 80 (3.1")
maximum capacity kg/h (lb/h):	45 (99)	90 (198)	150 (330)	180 (396)	300 (661)	450 992()
1312367AXX	√	√		√		
1312461AXX			√		√	√
NORTH AMERICAN MARKET ONLY						
SHOSE00040	√	√		√		
1312461AXX			√		√	√

Tab.16.g

16.4 Linear distributor models and typical installations

				gaSteam					
humidifier outlet ø mm				2x 40 (2x 1.6")		80 (3.1")	4x 40 (4x 1.6")	2x 80 (2x 3.1")	3x 80 (3x 3.1")
humidifier capacity kg/h				45	90	150	180	300	450
distributor inlet (øA) mm (in)	max. distributor capacity kg/h (lbs/h)	minimum width of duct/AHU mm (in)	P/N	UG045	UG090	UG150	UG180	UG300	UG450

40 (1.6")	25 (55)	850 (33.5")	DP085D40RH	2	(4)**			See Figs. 16.b and 16.c
40 (1.6")	35 (77)	1050 (41.3")	DP105D40RH	2	(4)**			
40 (1.6")	45 (99)	1250 (49.2")	DP125D40RH	2	2	4	4	
40 (1.6")	45 (99)	1650 (65")	DP165D40RH	2	2	4	4	
40 (1.6")	45 (99)	2050 (80.7")	DP205D40RH		2	4	4	
40 (1.6")	25 (55)	850 (33.5")	DP085D40R0	2	(4)**			
40 (1.6")	35 (77)	1050 (41.3")	DP105D40R0	2	(4)**			
40 (1.6")	45 (99)	1250 (49.2")	DP125D40R0	2	2	4	4	
40 (1.6")	45 (99)	1650 (65")	DP165D40R0	2	2	4	4	
40 (1.6")	45 (99)	2050 (80.7")	DP205D40R0		2	4	4	
40 (1.6")	45 (99)	600 (23.6")	DP060D40RU	2	2	4	4	

Tab.16.h

if the duct does not feature the required width for the distributor, two shorter distributors (numbers indicated in brackets) can be used, branching the steam hose.

*: use Carel "Y" kit UEKY000000, inlet 40 mm (1.6") and 2 outlets x 30 mm (1.2")

**: use Carel "Y" kit UEKY40X400, inlet 40 mm (1.6") and 2 outlets x 40 mm (1.6")

For the typical installations of linear distributors, see the figure below:

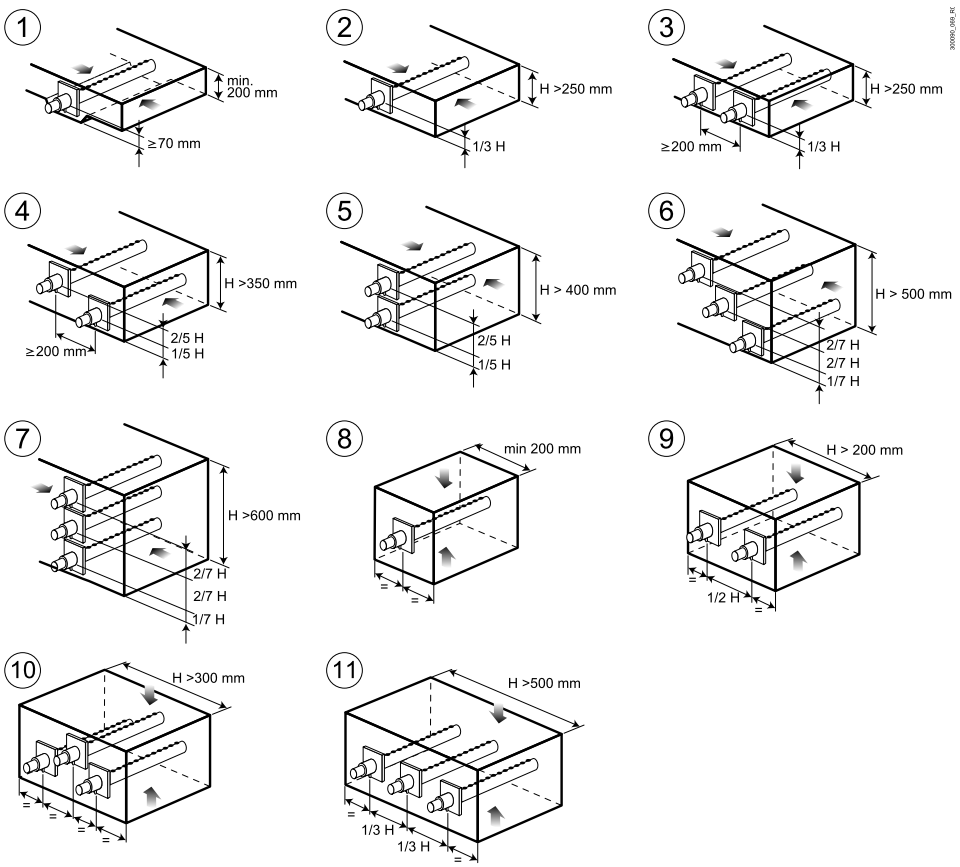


Fig.16.a

Typical installations of gaSteam series gas-fired humidifiers:

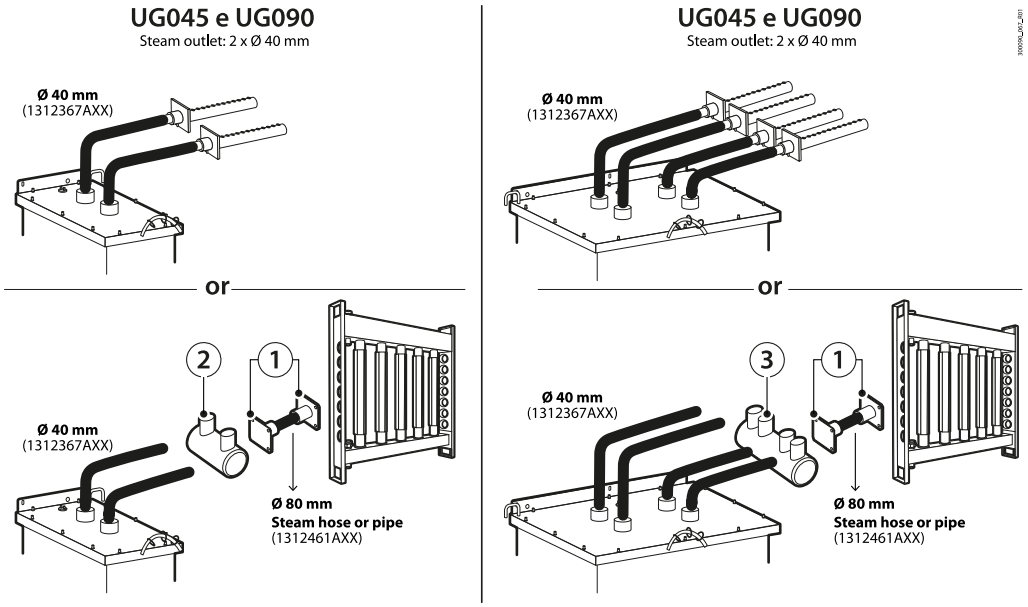
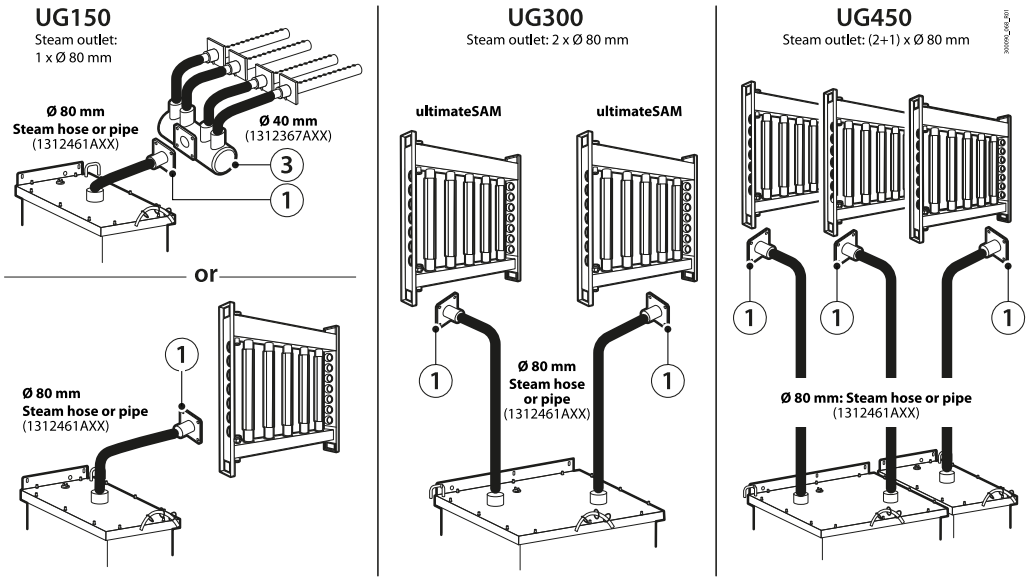


Fig.16.b



Key:

Ref	P/N
1	SAKIT80100
2	SAKIT40200
3	SAKIT40400

Tab.16.i

CAREL can accept no responsibility for possible errors in this manual.
CAREL reserves the right to modify its products without notice

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